

**IoT-Enabled Smart Outlet Breaker with AI-Assisted Dashboard for
Socket-Level Control and Load Monitoring**



A Project Design

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COLLEGE OF ENGINEERING
CERTIFICATION

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Project Design Proposal Approval Sheet

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CHAPTER 1

INTRODUCTION

This study introduces an innovative approach to smart home energy management by combining IoT-Enabled Smart Outlet Breaker with AI-Assisted Dashboard for Socket-Level Control and Load Monitoring, this chapter presents the study's background, problem statement, objectives, significance, and scope to establish the foundation of the research.

1.1 Background of the Study

In recent years, smart home technologies and the Internet of Things (IoT) have rapidly advanced, reshaping how households monitor, control, and optimize energy usage. IoT-based energy monitoring systems have grown in adoption globally, offering real-time visibility into electrical consumption and enabling users to make data-driven decisions to reduce waste and cost. (Poyyamozhi et al., 2024) In the Philippines, smart home adoption is gaining momentum, with more households showing interest in installing home automation and energy management systems.(Luna et al., 2022) Despite this interest, many existing systems remain limited in scope, often measuring only aggregate load or relying on complex sensor networks that are difficult for average users to understand and manage.

One of the key limitations in conventional energy monitoring setups is the inability to attribute load to individual appliances or sockets (Silva M et al., 2024). Traditional breakers only provide total circuit measurements, making it impossible to determine which socket or device is contributing most to the load or causing an overload. This lack of granularity prevents users from managing energy usage at the socket level and from isolating problematic loads. Furthermore, many systems depend heavily on cloud-based processing for analytics, which poses reliability

and safety risks when internet connectivity is lost or delayed (Janet Julia Ang'udi et al., 2023).

To address these gaps, more recent research has considered embedding intelligent assistants in home automation systems. AI-driven voice assistants and natural language interfaces offer a more accessible way for users to interact with smart systems allowing non-technical users to query states, issue commands, or get guidance verbally. (Shubham Singh et al., 2024) At the same time, integrating IoT energy systems with AI can provide deeper insights, predictions, and adaptive behaviors in smart homes, while preserving user control. (AI Impact Analysis on Smart Home Industry, n.d.) However, most implementations still assume centralized intelligence or cloud dependency, lacking robust fallback when offline.

Given this landscape, there is a clear need for a system that combines socket-level current sensing, local protection capability (offline fallback), and a user-friendly AI-assisted interface. By placing a current sensor at each socket, individual loads become measurable and controllable. A system that can locally detect overloads and isolate specific sockets enhances protection reliability even when disconnected from the internet. Meanwhile, a cloud-integrated AI assistant can guide users in natural language explaining dashboard status, toggling sockets, and interpreting logs in Filipino or English, making the system accessible even to users unfamiliar with dashboards or technical interfaces.

This study therefore seeks to develop an IoT-enabled smart outlet breaker system that extends conventional circuit protection by enabling per-socket control and monitoring, while embedding a Gemini-powered AI assistant into a web dashboard to simplify interaction and improve usability. The proposed system aims to maintain safety and intelligence even under connectivity constraints, ensuring that household users can confidently monitor and manage their electrical loads in a more intuitive and reliable manner.

1.2 Statement of the Problem

The rapid advancement of smart home and Internet of Things (IoT) technologies has created a growing need for more intelligent and user-friendly energy management systems (Badar & Anvari-Moghaddam, 2022; Sudiartama et al., 2023). In a typical household, electrical circuits are divided into lighting, outlet, and dedicated lines, each protected by a breaker. Among these, outlet breakers play an essential role in providing electricity to multiple appliances while ensuring safety against overloads (Shaban et al., 2022). However, traditional outlet breakers only monitor total current and lack the capability to identify which specific appliances contribute to the overall load (Asif et al., 2021; E3S, 2023). This limitation prevents users from effectively tracking energy consumption or managing the operation of individual sockets within their home (Sudiartama et al., 2023; Neto et al., 2024).

Beyond visibility, another challenge lies in user accessibility and understanding. Many existing monitoring systems present data through complex dashboards, which may be difficult for non-technical users to interpret (Akhtar, 2025; Becks et al., 2023; Woźniak et al., 2025). Without clear feedback or guidance, homeowners may struggle to make informed decisions about their energy usage or device control. This highlights the need for a system that not only enhances monitoring and control at the outlet level but also provides an intuitive way for users to interact with it potentially through intelligent assistance that can simplify communication and improve user engagement in managing household energy consumption (Gomes et al., 2020; De Andrade & Tumelero, 2022; Al-Safi et al., 2025).

1.3 Objectives of the Study

The main objective of this study is to design and develop a smart outlet monitoring system that extends the function of outlet breakers by providing socket-level control, real-time load monitoring, and an accessible cloud-based interface integrated with an AI assistant for user guidance and interaction.

Specific Objectives:

1. To develop a smart outlet and central control unit that detect current, manage socket switching, apply safety logic for overload protection, and transmit data via RF communication.
2. To develop a web dashboard that monitors circuit load, alerts users of safety actions, and integrates an AI assistant for control, inquiry, and advisories.
3. To test the system's accuracy and reliability through different load variations, safety logic evaluation, and AI assistant response.

1.4 Significance of the Study

The growing need for smarter household energy management underscores the importance of solutions that are both reliable and cost-effective. Traditional outlet breakers protect circuits from overloads but offer no appliance-level visibility, leaving homeowners unable to identify which devices contribute most to energy consumption. This gap prevents households from making informed decisions about usage, efficiency, and safety. This study is significant as it aims to introduce a system that extends the function of outlet breakers beyond simple protection. By enabling per-socket monitoring and integrating a cloud-based Gemini AI assistant, the research offers a practical and scalable approach to predictive energy management. Households can benefit from real-time insights, enhanced safety against overloads, and AI-guided energy advisory that assists

even non-technical users in managing their appliances more efficiently. Beyond immediate consumer advantages, the approach also creates pathways for integrating IoT and cloud AI into everyday energy systems, supporting sustainable practices and laying a foundation for future smart home ecosystems. One of the key contributions of this study is the knowledge framework it establishes demonstrating how IoT-based socket monitoring, RF communication, and cloud intelligence can be integrated into a unified system, serving as both a technical reference and academic resource for future innovations in energy management.

1.5 Conceptual Framework

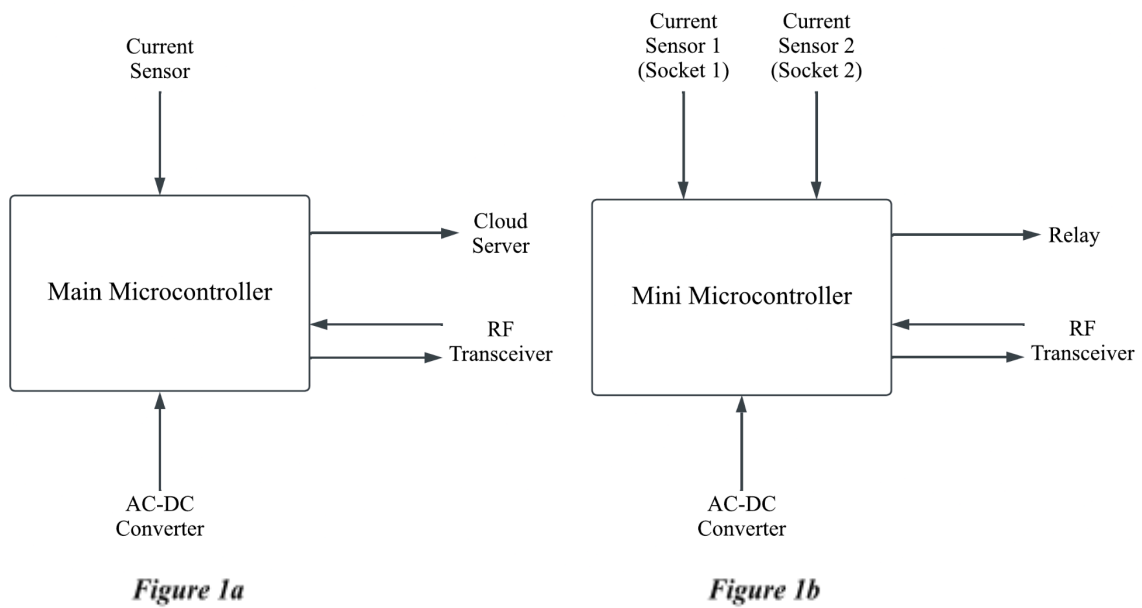


Figure 1 Conceptual Framework of the Study

In Figure 1a, the main microcontroller measures the aggregate current load through a current sensor and correlates it with data received from multiple outlet devices. It serves as the central processing unit for local safety control and data aggregation, transmitting monitored values to the cloud through IoT connectivity. The system integrates with a Generative AI assistant, which performs predictive

analysis and provides energy-related insights, safety advisories, and user assistance through a web dashboard. This enables intelligent load monitoring and overload prediction while maintaining real-time access to appliance-level data. The system continues to operate autonomously when offline, using predefined if-else logic for safety control, and automatically synchronizes logged data once an internet connection becomes available.

At the outlet level, Figure 1b illustrates the integration of socket-level current sensing with centralized monitoring for efficient and intelligent energy management. Each outlet contains one mini microcontroller that manages two sockets, each equipped with an individual current sensor to detect plug-in or removal events based on current draw. A relay provides switching control per socket, enabling both manual and automated toggling through the web dashboard. These outlet devices are powered independently through compact AC-DC converters and communicate via RF transmission with the main microcontroller, ensuring reliable and real-time reporting of socket activity and power consumption. This design enhances accuracy by incorporating actual current measurements per socket, allowing the Gemini AI system to analyze load patterns through historical consumption data, providing predictive insights and user advisories that support energy-efficient and safer household operation.

1.6 Theoretical Framework

The proposed smart outlet monitoring system is grounded on core principles of electrical engineering, communication, and computing. This framework outlines how the system senses, transmits, and manages electrical load data through integrated subsystems that enable accurate monitoring, intelligent control, and cloud-assisted analysis.

A. RF Communication:

Radio frequency (RF) communication enables low-power, short-range data transmission between each smart outlet and the central monitoring unit. Its theoretical basis remains rooted in electromagnetic wave propagation, where modulation techniques such as amplitude-shift keying (ASK) or frequency-shift keying (FSK) are used to encode data (Rappaport, 2002). The relationship between the data rate R_b , bandwidth B , and modulation efficiency η is expressed as:

$$R_b = \eta \cdot B \quad (1.1)$$

In the proposed system, RF communication through the HC-12 module transmits structured data frames that include the outlet ID, socket ID, and measured current values in milliamperes. Each transmission uses a unique owner address to maintain accurate and secure communication between the smart outlets and the central monitoring unit.

B. Central Monitoring Unit:

The Central Monitoring Unit (CMU) serves as the system's primary controller and protection hub. It continuously receives current readings from each outlet device through RF communication while simultaneously monitoring the total current at the breaker level. The CMU executes local overload protection using rule-based logic that compares the aggregated current value against the rated breaker threshold (typically 15–20 A). When the total current exceeds or approaches this limit, the unit isolates the affected socket or temporarily disables the circuit to prevent potential damage or fire hazards.

The total current is computed as the summation of all socket currents, following Kirchhoff's Current Law (KCL) which states that the total current entering a node equals the total current leaving it:

$$I_{total} = \sum_{n=1}^N I_n \quad (1.2)$$

(Based on Kirchhoff's Current Law as discussed in Alexander & Sadiku, 2017)

If I_{total} exceeds the predefined I_{rated} threshold, the CMU triggers a disconnection command through RF transmission to the corresponding specific outlet socket. This mechanism ensures that overload protection and current regulation remain local and immediate without relying on external systems.

C. Data Encoding:

The communication protocol follows a structured data frame format to ensure accurate identification and data transmission between outlet devices and the Central Monitoring Unit (CMU). Each transmitted frame begins with a 24-bit Owner ID, which uniquely identifies the main controller. The next 7 bits specify the outlet device, while the final 1 bit identifies the corresponding socket within that outlet. This addressing structure allows the CMU to distinguish and manage each socket individually.

To support real-time current monitoring, the data frame also includes the measured current value obtained from the socket's current sensor, expressed in milliamperes. The complete data frame is therefore represented as:

$$DataFrame = [24\text{-bit Owner's Bit}] + [7\text{-bit Outlet}] + [1\text{-bit Socket}] + [16\text{-bit Current (mA)}] \quad (1.3)$$

This encoding scheme enables efficient mapping of socket-level current data and ensures precise correspondence between transmitted readings and their

physical outlet locations, facilitating accurate monitoring and control at the Central Monitoring Unit.

D. Device Outlet Monitoring Unit

Each outlet device is equipped with a mini microcontroller (active mode ≤ 6.5 mA at 5.5 V, 8 MHz), an RF transceiver module, and two integrated current sensors corresponding to each socket. These current sensors continuously measure the load current of connected appliances and convert it into analog signals, which are then processed and digitized by the microcontroller's ADC. The measured values are transmitted via RF communication to the Central Monitoring Unit (CMU) using the defined data frame structure.

Upon receiving a cut-off command from the CMU, the relay embedded in the corresponding socket disconnects the appliance load to prevent overcurrent or potential hazards. This configuration enables each socket to perform independent sensing and switching functions, allowing precise monitoring and localized control at the outlet level.

E. Generative AI and Natural Language Processing (NLP)

The AI assistant integrated into the smart outlet monitoring system is grounded in the theories of Generative Artificial Intelligence (AI) and Natural Language Processing (NLP). These fields enable machines to understand, generate, and respond to human language in a conversational manner. Generative AI models, particularly large language models (LLMs), learn statistical patterns from extensive textual datasets to produce contextually coherent responses (Choi et al., 2023).

In the proposed system, the AI component functions as a conversational agent that interprets load data, responds to user queries, and provides advisory feedback regarding energy consumption. Its operation relies on deep-learning architectures and transformer-based attention mechanisms, which allow the model

to capture dependencies between tokens and determine contextual relevance across an input sequence.

The attention mechanism, which lies at the core of transformer models, is mathematically represented as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (1.4)$$

Where:

- Q = Query matrix representing the current token or input context
- K = Key matrix representing all tokens in the sequence
- V = Value matrix carrying the semantic meaning of each token
- d_k = Dimensionality of the key vectors (used for normalization)
- $\text{softmax}(\)$ = Function that converts the computed scores into a probability distribution

This formula describes how the model assigns “attention weights” to tokens, enabling it to focus on the most relevant parts of the input when generating output (Vaswani et al., 2017; Islam et al., 2024). By applying this mechanism, the AI assistant can interpret user commands, retrieve relevant monitoring data, and generate human-like responses through the dashboard interface.

Through the integration of generative AI and NLP principles, the system bridges the gap between complex energy data and user understanding, offering an intuitive and intelligent interface for managing household power consumption.

F. Internet of Things (IoT)

IoT integration provides remote monitoring and control. When network connectivity is available, the system automatically uploads aggregated socket-level and breaker-level data to the cloud server for visualization through the web dashboard. The interface enables users to review real-time current readings, load history, and device status, as well as interact with the Gemini AI assistant for data interpretation and advisory feedback. Real-time safety functions remain locally executed within the Central Monitoring Unit (CMU), ensuring continuous protection even without internet connectivity. When offline, data are stored temporarily and synchronized with the cloud once the connection is restored, maintaining accuracy and continuity of monitoring records.

G. Non-Invasive Split-Core Current Transformer (Current Sensor)

The system employs the current sensor, a split-core current transformer (CT) that captures alternating current (AC) flow without direct electrical contact. It operates on the principle of electromagnetic induction, where the magnetic field produced by the primary conductor induces a proportional current in the CT's secondary coil (Hughes & Drury, 2019).

The induced voltage is expressed as:

$$V_{out} = I_p \times \frac{N_s}{N_p} \times R_b \quad (1.5)$$

Where:

- V_{out} = sensor output voltage
- I_p = primary current (household load)
- $\frac{N_s}{N_p}$ = turns ratio of CT

- R_b = burden resistor value

Within the conceptual framework, this sensor integrates into the Central Monitoring Unit, measuring the total or aggregate current drawn by all connected outlets. The captured data serves as input for the CMU's rule-based logic, which compares measured values with the rated breaker threshold to detect and prevent overcurrent conditions. Its non-invasive design ensures safety, accuracy, and compatibility with the overall load monitoring system.

H. Current Sensing per Socket

Each outlet socket employs a Hall-effect current sensor to measure the instantaneous current drawn by a connected appliance. The sensor produces an analog voltage output that varies linearly with the magnitude and direction of the current flowing through its conductor. The relationship between the sensed current and the output voltage is expressed as:

$$V_{OUT} = V_{CC}/2 + (S \times I_P)$$

$$I_P = \frac{V_{OUT} - V_{CC}/2}{S} \quad (1.6)$$

Where:

- V_{out} = Output voltage of the current sensor (V)
- V_{cc} = Supply voltage applied to the sensor (typically 5 V)
- S = Sensitivity of the sensor in volts per ampere (V/A), determined by the sensor's range model
- I_p = Primary current (A) flowing through the conductor being measured

The quiescent output voltage ($V_{cc} / 2$) represents zero current due to the bidirectional sensing characteristic of Hall-effect devices. Any deviation from this midpoint corresponds to positive or negative current flow. The microcontroller

samples V_{out} using its analog-to-digital converter (ADC) and calculates the actual current based on the equation above.

For alternating-current (AC) measurements, the root-mean-square (RMS) current is computed to determine an equivalent steady-state value:

$$I_{\text{RMS}} = \frac{1}{\sqrt{2}} \times I_P \quad (1.7)$$

The theoretical concepts in this section—including voltage-to-current conversion, quiescent offset compensation, and RMS computation are grounded in the operating principles of Hall-effect current sensors as described in the ACS712 Datasheet (Allegro MicroSystems, 2021) and supported by applied analyses on AC current measurement (All About Circuits Forum, 2019).

1.7 Scope and Limitations

This study focuses on developing an IoT-enabled smart outlet monitoring system centered on the outlet breaker, covering three main components: outlet devices that detect plug-in or removal events through per-socket current measurement, a central control unit that gathers socket-level load data through RF communication, and the integration of a cloud-based Gemini AI assistant that provides intelligent load analysis, predictive insights, and user advisory through a web dashboard. The dashboard allows real-time monitoring, historical energy tracking, and remote appliance control with AI-assisted feedback.

The system is designed to accommodate outlet circuits rated up to 15 A ($\approx$ 3,300 W at 220 V), suitable for common household appliances. The RF communication module supports an indoor range of approximately 10 meters, though its performance may vary depending on environmental interference and obstructions. The central control unit utilizes an ESP32 microcontroller capable of

supplying up to 500 mA, adequate for handling sensors, RF transceivers, and data processing tasks. Each outlet device employs a low-power microcontroller that operates at only a few milliamps in active mode (maximum ~ 6.5 mA at 5.5 V, 8 MHz), ensuring efficiency and scalability when multiple sockets are deployed. For demonstration and prototyping, the system implementation is limited to three outlet devices, each consisting of two sockets, providing adequate validation of per-socket current measurement, RF data transmission, and dashboard integration.

The scope is limited to outlet circuits within household panels and excludes lighting and high-load dedicated circuits such as air conditioners or ovens. While current measurements are now obtained at both the breaker and socket levels, the study does not include advanced Non-Intrusive Load Monitoring (NILM) algorithms. RF transmission range and potential interference are considered limitations of the communication system. The system is designed to function primarily under online conditions for real-time AI interaction. However, during periods without internet connectivity, the local breaker module continues to operate autonomously using if-else logic for safety control and temporary data logging. Once internet connectivity is restored, logged data are synchronized automatically to the cloud dashboard for analysis. Thus, while the Gemini AI assistant enhances cloud-based intelligence, offline operation remains limited to basic safety and monitoring functions.

1.8 Operational Definition of Terms

To ensure clarity and consistency, the following terms are operationally defined as they are specifically applied within this study:

Central Monitoring Unit (CMU). The main controller installed at the outlet breaker that measures the total circuit load, communicates with all outlet devices via RF, executes rule-based safety logic during overload, and automatically

synchronizes locally logged data to the cloud dashboard once internet connectivity becomes available.

Cloud AI Dashboard. A web-based interface that visualizes real-time and historical load data, stores all system records in a database server, and integrates a generative AI assistant that delivers user guidance, anomaly insights, and predictive energy advisory through conversational interaction.

Current Sensors. The Hall-effect sensor is used per socket to measure instantaneous current, while the split-core current transformer (CT) is used at the breaker to measure aggregate circuit load. Together, they provide multi-level current detection that enables both socket- and breaker-level monitoring.

Dashboard Interface. The cloud-linked visualization layer accessible through any browser, displaying current readings, socket states, and AI-generated insights in real time for easier user interpretation and control.

Database Server. The backend system that stores user credentials, device data, time-stamped current logs, and AI analytics results for historical review and synchronization.

Data Encoding. The structured binary framing method used in RF communication to identify the owner, outlet, and socket IDs within each transmission, ensuring accurate mapping between devices and readings.

Generative AI Assistant. A cloud-based conversational AI integrated into the dashboard that allows users to query load data, receive predictive insights, and control sockets using natural-language dialogue in English or Filipino.

IoT (Internet of Things). The interconnected system of smart outlet devices, the CMU, and the cloud server linked through RF and internet connections, enabling remote monitoring, control, and AI-assisted analytics.

Load Monitoring. The continuous process of measuring electrical current at both breaker and socket levels, detecting overload trends, and sending data to the dashboard for visualization and predictive analysis.

Main Microcontroller. The ESP32 device at the breaker responsible for aggregating current data, executing rule-based logic for overload protection, handling RF communication, and connecting to the internet or hotspot for dashboard synchronization.

Mini Microcontroller. A low-power controller embedded within each smart outlet, designed to sample current-sensor data, send encoded values to the CMU, and receive control commands through the RF link.

Offline Data Synchronization. The automatic process where the CMU stores readings locally during internet loss and uploads all unsent logs to the cloud dashboard once connectivity is restored.

Outlet Breaker. A protective household breaker assigned to outlet circuits that also serves as the mounting point of the CMU for current sensing, communication, and safety control.

Outlet Socket. A power port where appliances are connected; in this study each socket integrates a sensor-based smart module capable of detecting plug-in events and transmitting load data to the CMU.

Predictive Overload Protection. A preventive mechanism combining sensor data and AI analysis to anticipate potential overloads before they occur, enabling timely alerts or automated cut-off actions.

Relay Module. An electronically controlled switch that disconnects or restores power to a socket based on safety logic from the CMU or user commands from the dashboard.

RF Communication. A low-latency 433 MHz wireless data link (via HC-12 modules) used for bidirectional exchange of event and load data between outlet devices and the CMU.

RF Module. The HC-12 transceiver is embedded in both CMU and outlet devices that enables wireless serial communication for transmitting load information and control commands.

Rule-Based Logic. The predefined set of if-else safety conditions programmed in the CMU that compare measured current values against rated thresholds to trigger overload protection or reconnection actions.

Smart Outlet Device. A per-socket module containing a mini-microcontroller, sensors, relay, and RF transceiver used to detect plug-in events, measure current, and allow cloud-assisted control.

Wireless Serial Communication. An asynchronous, cable-free data-exchange method implemented over RF that allows continuous information flow between the CMU and each outlet device.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter provides an extensive overview of studies and literature related to IoT-enabled smart outlet breaker systems integrated with a cloud-based AI assistant on its dashboard for intelligent user interaction. The review aims to establish pertinent research and information relevant to the present study.

2.1 Smart Home Energy Management and IoT Integration

The rapid increase in global energy demand has driven the advancement of Smart Home Energy Management Systems (SHEMS) that integrate Internet of Things (IoT) technologies for efficient and automated electricity utilization. These systems employ embedded sensors, microcontrollers, and network interfaces to monitor and control electrical loads in real time. By enabling bidirectional data flow between end devices and control servers, SHEMS platforms optimize power consumption while maintaining user comfort and safety. Despite these advancements, traditional home electrical systems remain limited to aggregate monitoring at the meter or circuit level, providing no visibility into appliance-specific energy behavior. Consequently, the need for intelligent systems capable of socket-level monitoring and adaptive control has become a central focus of modern energy research. The following sub-sections review the global trends, architectural designs, and research gaps that motivate the present study's focus on integrating IoT-based sensing with intelligent user interaction.

2.1.1 Global and Residential Energy Trends

According to Badar and Anvari-Moghaddam (2022), residential consumption constitutes approximately 13%–37% of total electrical load worldwide, illustrating the critical role of households in global energy optimization. Their review of Smart Home Energy Management System

(SHEMS) highlights how modern homes are evolving toward autonomous decision-making for appliance scheduling, renewable integration, and cost reduction. The authors emphasized that the management of *controllable* and *uncontrollable* loads such as lighting, heating, and plug-in devices directly affects energy efficiency and system stability. The study further described that successful implementation of SHEMS requires continuous data monitoring, forecasting, and optimization through demand-response strategies and predictive control (Badar & Anvari-Moghaddam, 2022).

The global transition toward IoT-based energy management is largely driven by the availability of cost-efficient microcontrollers such as the ESP32 and ESP8266, which combine wireless communication capabilities with real-time data processing. These devices enable the collection of high-resolution electrical parameters (voltage, current, and power factor) from distributed loads, making localized control feasible even in residential applications. As noted by Badar and Anvari-Moghaddam (2022), modern SHEMS frameworks now integrate optimization algorithms, cloud data storage, and machine learning models to predict energy demand and schedule appliances during low-tariff periods, demonstrating the scalability and intelligence expected of next-generation home energy systems.

2.1.2 Architectures and Communication in Smart Homes

According to Badar and Anvari-Moghaddam (2022), review several multi-layer architectures for smart home energy management. For example, one communication-centric model divides the system into a Physical Layer, a Fog Computing Layer, a Network Layer, a Cloud Computing Layer, Service/Session Layers, and an Application Layer. In this view, the Physical Layer includes sensors, smart plugs and microcontrollers that monitor appliance states for instance, smart plugs can sense whether a device is ON or OFF and report this to the SHEMS. The Network Layer then uses wireless protocols (e.g. Wi-Fi or

Zigbee) to transfer those readings. The Fog (edge) and Cloud Computing Layers provide local and remote processing power to run control and optimization algorithms on the collected data. Finally, the Application Layer delivers the user interface typically a web or smartphone app that lets homeowners visualize energy usage and remotely control devices. For example, an Android app is cited as enabling users to control home appliances “from anywhere”. Badar and Anvari-Moghaddam (2022) further classified monitoring techniques into Intrusive Load Monitoring (ILM) and Non-Intrusive Load Monitoring (NILM). ILM employs distributed sensing through smart plugs connected directly to appliances, enabling accurate measurement and independent switching of each load. Conversely, NILM infers appliance behavior by analyzing aggregate current waveforms at a single measurement point, which reduces hardware cost but introduces estimation errors due to overlapping load signatures. In this context, ILM architectures utilizing per-socket sensors such as ACS712 Hall-effect modules (Allegro MicroSystems, 2022) offer a more precise foundation for energy disaggregation, making them ideal for systems that require fine-grained monitoring and selective control.

Based on these findings, Badar and Anvari-Moghaddam (2022) describe a layered SHEMS communication framework (from edge devices through fog to cloud) and highlight the use of RESTful web services for smart home messaging. They also note typical wireless links (Wi-Fi, Zigbee) for device connectivity and mention that M2M communications are standardized by bodies such as ETSI.

2.1.3 Identified Gap: Absence of Socket-Level Visibility and User Intelligence

According to Badar & Anvari-Moghaddam (2022), emphasize the importance of energy monitoring and describe two approaches: Intrusive Load Monitoring (sensing each appliance directly) and Non-Intrusive Load Monitoring (using a single, whole-home meter). By definition, non-intrusive (single-point) monitoring captures only total household consumption. Despite the continuous

evolution of SHEMS frameworks, most existing systems still monitor total household consumption only, offering limited insight into how individual devices contribute to energy usage. This limitation hinders precise fault detection, predictive overload protection, and personalized energy awareness among users. In an extensive empirical study, Oh (2020) demonstrated that implementing IoT-based smart plugs capable of per-appliance monitoring and user feedback produced an average energy reduction of 3.53% over 15 months across 125 residential households. The findings revealed that frequent feedback, transparency of usage data, and accessible control interfaces promote behavioral change and reduce overall consumption. The study concluded that appliance-level visibility enhances user engagement, encourages proactive load management, and improves energy conservation outcomes.

These insights underscore the existing research gap: conventional systems provide either monitoring without localized control or user interfaces without granular feedback. Consequently, there is a need for a unified framework that combines socket-level data acquisition, hybrid wireless communication, and intelligent assistance through conversational AI. The present study addresses this need by integrating per-socket current sensors (ACS712), breaker-level sensing (SCT-013), and a cloud-based Generative AI dashboard, thereby establishing both technical precision and user accessibility in modern household energy management.

2.2 Evolution of Smart Outlet Breakers and Hybrid Monitoring Architectures

The modern household electrical system is evolving from passive protection devices toward intelligent, network-connected modules that combine monitoring, switching, and analytical functions. Conventional outlet breakers perform essential safety roles by interrupting circuits during over-current or short-circuit conditions, yet they provide no visibility into which appliances

contribute to the total load. The integration of IoT technologies, distributed sensors, and microcontrollers now enables fine-grained, appliance-level management that transforms ordinary outlets into smart energy nodes. This section examines the progression from traditional circuit breakers to IoT-enabled smart outlets, the emergence of hybrid outlet–breaker architectures, and how these advances address the limitations of previous systems.

2.2.1 Conventional Outlet Breakers and Their Limitations

According to Kumar and Naik (2021), conventional outlet breakers protect household circuits through thermal or magnetic trip mechanisms that disconnect the supply once a current threshold is exceeded. While effective for preventing faults, these devices measure only total circuit current and offer no diagnostic data regarding individual appliances. Munoz et al. (2022) similarly observed that even modern residential panels rarely include per-socket sensing, leading homeowners to rely on energy bills or manual inspections to estimate usage. This lack of transparency hinders proactive energy management and fault isolation.

Moreover, existing protection devices cannot identify transient overloads or predict impending failures because they lack digital communication and data logging features. As Padlak et al. (2025) noted, safety systems in conventional circuits are reactive, triggering only after thresholds are breached, rather than adaptive or predictive. These gaps have motivated research toward smart breaker architectures that merge traditional protection with real-time data analytics and remote control.

2.2.2 IoT-Enabled Smart Plugs and Socket Devices

IoT-based smart outlets extend the concept of monitoring to the individual appliance level. Asif et al. (2021) described a smart plug prototype integrating an Arduino Nano + ESP8266 (ESP-01), ACS712 current sensor, and ZMPT101B voltage module, capable of measuring instantaneous current, voltage, and power.

Through Wi-Fi connectivity and cloud dashboards, users can visualize consumption trends and issue remote ON/OFF commands. Sudiartama et al. (2023) achieved comparable functionality with adaptive current cutoff logic that automatically disconnects loads exceeding preset thresholds. Both systems demonstrated measurement accuracies within $\pm 5\%$ when compared to commercial energy meters, confirming the feasibility of cost-effective ILM (Intrusive Load Monitoring) for residential use.

Recent implementations have expanded functionality beyond simple metering. Neto et al. (2024) introduced EnAPlug, an open-source collaborative smart plug based on the ESP32 platform that integrates humidity, temperature, and luminosity sensors alongside electrical measurements. The device incorporates TinyML inference on the microcontroller to recognize appliance usage patterns with up to 95 % accuracy and a humidity-forecasting RMSE of 0.055. These results show that edge-level intelligence can significantly reduce data-transmission loads and latency, ensuring system responsiveness even during intermittent internet connectivity.

Collectively, these studies (Asif et al., 2021; Sudiartama et al., 2023; Neto et al., 2024) demonstrate that IoT-enabled plugs serve not only as monitoring nodes but also as distributed control points that enhance safety, efficiency, and autonomy within residential circuits.

2.2.3 Hybrid Outlet–Breaker Design and Control Logic

While smart plugs deliver per-appliance visibility, full household reliability requires integration with the breaker panel to capture total circuit behavior. The hybrid outlet–breaker architecture merges both granular and aggregate sensing, providing comprehensive load awareness. According to Thakur et al. (2014) and Gomes et al. (2018), highlighted that integrating distributed smart plugs with centralized data aggregation enhances coordination and

decision-making in energy management systems. In these architectures, each outlet or plug node measures local current and status information and communicates it wirelessly typically via Wi-Fi to a central controller, which consolidates readings from multiple devices to optimize load distribution and prevent redundant control actions.

The hybrid approach has been validated in several recent prototypes. The Smart Energy Management and Load Monitoring of Individual Loads System (E3S Web of Conferences, 2023) employed an ESP32, SCT-013 current transformer, and ZMPT101B voltage sensor to measure both individual and total circuit loads. The system achieved load-measurement errors below $\pm 5\%$ relative to a calibrated Meco clamp meter and demonstrated stable wireless transmission across multiple nodes. By integrating socket-level ILM with breaker-level aggregate sensing, the study verified that hybrid architectures offer superior fault detection and control accuracy compared with single-point systems. Furthermore, hybrid frameworks enable localized safety logic. When abnormal current spikes are detected, the breaker's if-else protection routine can disconnect the circuit immediately while the cloud dashboard records diagnostic data for later analysis. This layered design supports the dual goals of real-time protection and predictive monitoring, aligning with the proposed system's architecture.

2.3 Current and Voltage Sensing Technologies

Accurate sensing of current and voltage is fundamental to any intelligent energy monitoring system. These sensors provide the raw data that define power consumption, detect overload conditions, and drive the protective logic that ensures system safety. In Smart Home Energy Management Systems (SHEMS), sensors are embedded both at the socket and the breaker level to continuously measure electrical parameters such as current (I), voltage (V), power factor (PF), and apparent power (S).

The effectiveness of energy monitoring depends on sensor precision, response time, linearity, and the system's ability to process these measurements in real time. This section discusses the principles, models, and practical implementations of the sensors utilized in this study namely, the ACS712 Hall-effect current sensor, the SCT-013 non-invasive current transformer, and the ZMPT101B voltage sensor as well as their accuracy and calibration bases derived from verified literature.

2.3.1 Hall-Effect Current Sensor (ACS712)

The ACS712 Hall-effect current sensor is a linear analog device that converts the magnetic field surrounding a current-carrying conductor into a voltage output proportional to that current. This principle enables galvanic isolation between the sensing circuit and the load line, enhancing safety for low-voltage microcontroller systems (Allegro MicroSystems, 2022). The device provides an output centered at $V_{cc} / 2$, with sensitivity values of 185 mV/A, 100 mV/A, and 66 mV/A for its ± 5 A, ± 20 A, and ± 30 A variants, respectively, following the relationship $V_{out} = V_{cc}/2 + S \times I$. The Hall-effect mechanism allows measurement of both AC and DC currents, making it suitable for mixed-load environments typically found in households. Asif et al. (2021) utilized the ACS712 within an IoT-enabled smart-plug prototype that measured real-time current, voltage, and power while transmitting data through Wi-Fi for cloud visualization. Similarly, Sudiartama et al. (2023) employed the sensor to detect overload conditions and demonstrated that its calibrated readings deviated by no more than ± 11.39 % compared with commercial energy meters. These studies confirm that despite minor non-linearities and noise sensitivity, the ACS712 remains reliable for cost-effective energy-management applications (Asif et al., 2021; Sudiartama et al., 2023; Saputri et al., 2024).

According to Allegro MicroSystems (2022), the sensor maintains linear performance over a temperature range of 40 °C to +85 °C and offers a rapid

response time of approximately 5 μ s, making it responsive to transient current spikes. Its small footprint and low power consumption simplify integration with microcontrollers such as the ESP32 and ESP8266, enabling direct analog-to-digital conversion for real-time monitoring. Researchers (Asif et al., 2021; Sudiartama et al., 2023; E3S, 2023) have noted that coupling the ACS712 with digital filters such as moving-average or low-pass algorithms reduces measurement noise and improves RMS current estimation accuracy. The Hall-effect sensor's bidirectional detection capability also facilitates identification of inductive or regenerative loads within residential circuits, an advantage over purely resistive shunt methods (Saputri et al., 2024). Consequently, the ACS712 is widely recognized as a dependable, low-cost, and versatile option for socket-level sensing in hybrid outlet breaker architectures. Its combination of electrical isolation, fast response, and compatibility with embedded IoT platforms makes it an indispensable component in modern smart-plug and breaker systems designed for real-time energy analysis and overload protection (Allegro MicroSystems, 2022; Asif et al., 2021; Sudiartama et al., 2023).

2.3.2 Non-Invasive Split-Core Current Transformer (SCT-013)

The SCT-013 split-core current transformer (CT) enables current sensing without any physical disconnection of the conductor, making it a safe and non-invasive solution for residential energy-monitoring systems. Its principle of operation relies on electromagnetic induction: when alternating current flows through a primary conductor, it induces a proportional secondary current across the sensor's windings. This induced current, when passed through a burden resistor, produces a measurable voltage signal that can be sampled by a microcontroller (Saputri et al., 2024). The split-core mechanism allows easy installation on existing wiring, making it practical for retrofitting in breaker panels. According to Crescentini et al. (2022), this design provides galvanic isolation and minimizes measurement interference, ensuring user safety when monitoring high-voltage circuits. The sensor's non-contact configuration and

broad linear range make it ideal for measuring the cumulative current drawn by multiple appliances connected to a single circuit. In the Smart Energy Management and Load Monitoring of Individual Loads system (E3S Web of Conferences, 2023), the SCT-013 was integrated with the ZMPT101B voltage module to collect synchronized current and voltage waveforms for load analysis. The study reported that after calibration, measurement deviations remained below $\pm 5\%$ when validated against a calibrated Meco clamp meter, confirming its capability for household-level precision.

Further experiments conducted by Saputri et al. (2024) supported these findings, showing approximately 95 % correlation accuracy between SCT-013 readings and reference instruments in distributed IoT environments. Because the sensor outputs a voltage proportional to the product of the primary current, the turns ratio, and the burden resistor, the relationship $V_{out} = N_s \times I_p \times R_b$ allows designers to tailor output sensitivity for different load ranges. Proper calibration of the burden resistor is therefore essential to achieve linear and accurate RMS current measurement. The SCT-013's high isolation and ease of scaling make it valuable for monitoring the total current drawn by household circuits or high-load appliances, complementing socket-level Hall-effect sensors such as the ACS712 (Asif et al., 2021; Sudiartama et al., 2023). According to Saputri et al. (2024), its consistent performance over varying load profiles and minimal temperature drift contribute to long-term reliability in continuous monitoring setups. When deployed together with voltage sensors, the SCT-013 enables the computation of active, reactive, and apparent power in real time, providing a comprehensive view of household energy consumption. Overall, this non-invasive CT remains one of the most practical and accurate components for breaker-level measurement in smart-home architectures due to its combination of safety, cost-efficiency, and high linearity (Saputri et al., 2024; E3S, 2023).

2.3.3 Voltage Measurement Using ZMPT101B

Accurate voltage measurement is fundamental for determining instantaneous and average power in smart-home monitoring systems. The ZMPT101B voltage sensor is a compact, transformer-based module that converts high-voltage AC input into a scaled, isolated analog output suitable for microcontrollers. According to Asif et al. (2021), the ZMPT101B maintains high linearity in the 50 V–250 V AC range and is capable of detecting voltage deviations as small as 0.022 % after calibration. Its isolation transformer design prevents direct electrical contact with the mains, ensuring user safety and noise immunity during continuous operation. The sensor includes a multi-turn potentiometer that allows users to fine-tune the sensitivity to match system voltage levels and achieve accurate root-mean-square (RMS) readings. This characteristic makes it an excellent companion to current sensors such as the ACS712 and SCT-013, which together provide the parameters required to compute active power ($P = V \times I \times \text{PF}$) and apparent power ($S = V \times I$) in real time. The ZMPT101B's analog output can be directly sampled by the ESP32's analog-to-digital converter, minimizing latency in signal processing and allowing real-time tracking of voltage fluctuations (Asif et al., 2021; Allegro Microsystems, 2022).

According to Sudiartama et al. (2023), incorporating the ZMPT101B in an IoT-based smart-energy system alongside the ACS712 current sensor enabled comprehensive monitoring of voltage, current, power, and energy with an overall accuracy above 94 %. The authors noted that calibration through comparison with commercial meters further improved precision, making the sensor suitable for continuous residential use. Similarly, the E3S Web of Conferences (2023) study employed ZMPT101B modules at both the breaker and socket levels and confirmed that the combined measurement error remained within ± 5 % when validated using a Mecor clamp meter. These results are consistent with findings from Asif et al. (2021), who emphasized that the ZMPT101B's compact size, low

cost, and electrical isolation make it ideal for embedded energy-management devices. Because of its ability to measure phase shifts and instantaneous voltage values, the module supports advanced calculations such as power-factor estimation and transient detection (Asif et al., 2021; Sudiartama et al., 2023; E3S, 2023). When paired with digital filtering techniques and calibrated scaling circuits, the ZMPT101B delivers both safety and precision, establishing it as a dependable voltage-sensing element for the hybrid outlet-breaker design proposed in this study.

2.3.4 Comparative Accuracy, Calibration, and Cost Analysis

Evaluating the trade-offs among different sensors is crucial for designing a reliable yet economical monitoring system. According to Sudiartama et al. (2023), accurate yet economical energy monitoring requires careful evaluation of sensor performance. In their study, the ACS712 and ZMPT101B sensors were calibrated and validated against a multimeter, achieving current and voltage measurement errors of 11.39 % and 0.022 %, respectively. These findings highlight the trade-off between cost and precision and demonstrate that proper calibration is essential to ensure stability and accuracy in IoT-based smart plug systems. Conversely, the SCT-013 offers excellent galvanic isolation and durability, making it ideal for high-current applications such as total circuit measurement at the breaker level (Saputri et al., 2024). The ZMPT101B, on the other hand, ensures precise voltage measurement with transformer-based isolation and adjustable sensitivity, essential for accurate real-time power calculations (Asif et al., 2021). These sensors collectively deliver a balance between safety, performance, and cost that is suitable for large-scale IoT deployments where affordability and measurement precision must coexist (Asif et al., 2021; Saputri et al., 2024; Sudiartama et al., 2023).

Calibration remains a key determinant of system accuracy across all sensor types. The Smart Energy Management and Load Monitoring of Individual Loads

system (E3S Web of Conferences, 2023) established a calibration framework by comparing readings from ACS712, SCT-013, and ZMPT101B sensors with a reference Meco clamp meter, achieving mean error values below $\pm 5\%$. Similar studies by Asif et al. (2021) and Sudiartama et al. (2023) used digital filtering algorithms including moving average and low-pass filters to reduce noise and stabilize sensor outputs, resulting in smooth RMS current and voltage waveforms. Saputri et al. (2024) further emphasized the importance of adaptive calibration for environmental stability, noting that recalibration every 30 days significantly improves accuracy consistency. According to Saputri et al. (2024), the SCT-013 demonstrates better linearity over wide current ranges, while the ACS712 remains superior in capturing transient changes due to its fast Hall-effect response. Overall, these works (Asif et al., 2021; Sudiartama et al., 2023; E3S, 2023; Saputri et al., 2024) collectively show that a hybrid combination of sensors, each strategically positioned within the circuit, provides the most cost-effective solution for achieving high-fidelity load monitoring in IoT-based energy systems.

2.3.5 Sensor Deployment in the Proposed System

The proposed smart outlet-breaker architecture integrates a hybrid sensor network designed to achieve both granular and aggregate monitoring accuracy. In this configuration, ACS712 Hall-effect sensors are installed on each socket to measure per-appliance current, detect plug-in events, and identify abnormal consumption behaviors. These sensors provide rapid detection of transient load changes due to their Hall-effect mechanism, ensuring timely activation of local safety logic (Allegro MicroSystems, 2022; Asif et al., 2021). At the same time, SCT-013 split-core current transformers are mounted on the breaker's outgoing lines to capture total circuit current, allowing cumulative analysis of branch-level demand (Saputri et al., 2024). This layered approach enhances visibility from individual sockets up to the entire outlet branch, enabling the control unit to differentiate between local overloads and system-wide surges. The breaker unit aggregates readings from both sensor levels and executes if-else safety logic to

trigger protective shutdowns when thresholds are exceeded, thereby providing predictive overload protection rather than reactive tripping (E3S Web of Conferences, 2023; Sudiartama et al., 2023).

Voltage measurements are simultaneously obtained using ZMPT101B sensors connected to the main input of each monitored line, supplying real-time RMS voltage data necessary for calculating apparent and active power (Asif et al., 2021; Sudiartama et al., 2023). The data from all sensors are transmitted through the microcontroller's analog-to-digital converter and processed locally by the ESP32, which applies digital filtering and short-term averaging to minimize noise before wireless transmission to the cloud dashboard (Asif et al., 2021; Neto et al., 2024; Sudiartama et al., 2023). According to previous implementations (Asif et al., 2021; Sudiartama et al., 2023; E3S, 2023), combining ACS712, SCT-013, and ZMPT101B sensors within one system produces comprehensive energy metrics including voltage, current, real-time power, and cumulative energy. This integration supports accurate analytics for the Gemini-powered AI dashboard, which interprets data trends, identifies inefficiencies, and provides user-level insights. Consequently, the proposed deployment satisfies Objective 2 of this study developing a central control unit that ensures safety, reliability, and intelligent energy management through coordinated hybrid sensing and real-time data processing (Asif et al., 2021; Saputri et al., 2024; Sudiartama et al., 2023; E3S, 2023).

2.4 Wireless Communication and System Safety Compliance

Reliable communication and strict adherence to safety standards are vital in integrating multiple sensing nodes within smart outlet-breaker systems. Wireless technologies such as Wi-Fi and RF modules enable seamless data transmission between distributed sockets and the central control unit, ensuring real-time coordination and response. At the same time, compliance with international safety guidelines guarantees that electrical operations remain secure

during switching and overload protection. The following subsections discuss the communication protocols and safety frameworks that support stable and safe system performance.

2.4.1 Communication Technologies in IoT Energy Systems

The performance of an IoT-based energy monitoring system depends significantly on the reliability and efficiency of its communication network. In modern Smart Home Energy Management Systems (SHEMS), data communication bridges the sensing layer and the control unit, transmitting real-time measurements and control commands (Badar & Anvari-Moghaddam, 2022). These systems commonly employ Wi-Fi, Bluetooth, Zigbee, Radio Frequency (RF), or MQTT-based communication protocols to interconnect distributed smart plugs, sensors, and the central monitoring unit (Asif et al., 2021; Sudiartama et al., 2023). Wi-Fi and MQTT protocols are favored for their high data throughput and compatibility with cloud platforms, allowing continuous synchronization between local microcontrollers such as the ESP32 and cloud databases or dashboards. However, Wi-Fi's dependency on stable internet connectivity can lead to latency or data loss in areas with weak signals, prompting researchers to incorporate hybrid models that combine local RF transmission with cloud connectivity (Asif et al., 2021; E3S, 2023; Sudiartama et al., 2023).

According to Badar and Anvari-Moghaddam (2022), the communication layer in SHEMS typically consists of edge nodes connected to a central controller that aggregates sensor data and executes commands. For short-range connections within a household, RF modules such as the HC-12 (433 MHz) are widely used because of their low power consumption, high noise immunity, and extended range up to 1 km in open environments. RF-based event transmission offers reliable data exchange between socket devices and the central breaker unit without depending on constant internet access. Meanwhile, Wi-Fi and MQTT ensure real-time dashboard updates and integration with cloud AI services, including the

Gemini-powered assistant proposed in this study. Combining these technologies supports redundancy: when Wi-Fi is unavailable, the system continues to operate locally via RF communication and automatically synchronizes data once connectivity is restored (Asif et al., 2021; Sudiartama et al., 2023; E3S, 2023). Therefore, a hybrid communication model leveraging RF for local reliability and Wi-Fi for cloud accessibility forms the backbone of a resilient and scalable IoT-enabled outlet-breaker architecture.

2.4.2 RF Event-Driven Communication Performance

Radio Frequency (RF) modules play a critical role in ensuring stable local communication among distributed IoT devices within smart home systems. RF-based data exchange provides an independent communication layer that functions even when internet connectivity is unavailable, ensuring uninterrupted coordination between the smart outlets and the central control unit. According to Saputri al. (2024), RF modules operating in the 433 MHz band, such as the HC-12, are effective for medium-range transmission due to their robustness against environmental noise and minimal interference with standard Wi-Fi networks. The HC-12 offers a line-of-sight range of up to 1,000 meters and can operate at configurable baud rates, balancing range and data speed as required. This flexibility makes it well-suited for event-driven data transmission, where small packets representing socket ON/OFF states, current readings, or fault alerts are sent only when triggered by a change in system conditions. Studies integrating RF-based communication (Asif et al., 2021; Sudiartama et al., 2023; E3S, 2023) confirmed that this approach minimizes network traffic and conserves energy, enabling faster response times between outlet nodes and the control center.

According to Asif et al. (2021), RF communication exhibits significantly lower latency than Wi-Fi under local network constraints because of its simplified protocol stack and lack of dependency on routers or access points. Empirical loopback tests using HC-12 modules have recorded average round-trip latencies of

approximately 31 milliseconds, confirming their suitability for real-time load control in residential environments (E3S, 2023; Saputri et al., 2024). The event-driven model ensures that communication bandwidth is allocated efficiently data packets are transmitted only during state transitions or when anomalies such as overload or power surges are detected. This reduces redundant transmissions and improves synchronization between devices, making RF particularly advantageous for low-power nodes that cannot maintain continuous network activity. Sudiartama et al. (2023) also emphasized that combining RF-based local communication with MQTT cloud synchronization enhances overall reliability, as data collected during offline periods are automatically uploaded once internet access resumes. Therefore, the proposed system adopts an RF event-driven hybrid communication framework to achieve both local autonomy and global accessibility, ensuring responsive and energy-efficient data handling across all system components.

2.4.3 Safety Standards for Smart Outlets

Safety compliance is an essential consideration in the design and implementation of smart outlet systems. According to the International Electrotechnical Commission (IEC, 2013/2022), standard IEC 60884-1 specifies mechanical and electrical safety requirements for household plugs and socket-outlets, including limits on contact temperature, dielectric strength, and mechanical endurance. The standard mandates that socket devices must prevent accidental contact with live components, ensure reliable grounding, and maintain thermal stability under repetitive switching operations. Compliance with this standard is particularly critical for IoT-enabled outlets, which may experience higher switching frequencies due to remote control and automation features. Asif et al. (2021) and Sudiartama et al. (2023) emphasized that frequent relay switching and current surges during load activation can lead to arcing or thermal degradation if components are not designed to meet IEC-certified thresholds. Therefore, adopting materials and circuit layouts that conform to IEC

specifications ensures both electrical reliability and user safety in smart energy systems (IEC, 2013/2022; Asif et al., 2021; Sudiartama et al., 2023).

Beyond physical protection, modern safety design also integrates electronic protection layers, including overload detection, automatic isolation, and fail-safe shutdowns (Munoz et al., 2022; Saputri et al., 2024). Smart breakers and outlets often employ firmware-based logic to interpret current and voltage sensor data, triggering immediate disconnection when abnormal readings exceed preset limits. This digital layer complements mechanical protection mechanisms, providing predictive safety through continuous monitoring. According to Munoz et al. (2022), embedding safety logic at both the socket and breaker levels enhances system resilience by isolating faults before they escalate into circuit-wide failures. In addition, all exposed metal parts must comply with IEC 60669-2-1 for switch devices, ensuring insulation coordination and short-circuit endurance. By following these international safety standards, the proposed outlet-breaker system upholds global compliance, guaranteeing that its integrated IoT features operate without compromising electrical protection or user safety (IEC, 2013/2022; Munoz et al., 2022; Saputri et al., 2024; Sudiartama et al., 2023).

2.5 Data Synchronization Strategies and Performance Evaluation in IoT Systems

Data synchronization from a local database to a remote cloud service is a well-established problem in distributed systems research. Jha and Sebastian (2017) propose an offline-first synchronization algorithm in which all records are first stored locally and temporarily flagged as “unsynchronized” until network connectivity becomes available. The system periodically checks for cloud availability and, once reachable, pushes all pending entries to the remote database and marks them as synchronized, ensuring no data is lost during offline operation. Extending beyond simple upload logic, Parvej et al. (2018) emphasize that

synchronization also requires maintaining consistency between local and cloud records by classifying updates and applying conflict-resolution rules when both endpoints have changed since the last sync. Their framework treats synchronization as a consistency management process capturing local changes offline, then reconciling states when connectivity resumes thereby preventing duplicate or conflicting records. Together, these studies establish foundational support for the synchronization strategy required in offline-capable IoT systems such as a smart-outlet dashboard, where data is temporarily stored locally and later transmitted to the cloud once connectivity is restored.

Beyond correctness and consistency, recent IoT research emphasizes the need to evaluate synchronization performance through latency-based metrics. Thangavel et al. (2014) demonstrate that cloud-connected IoT systems are commonly assessed using end-to-end delay, throughput, and message delivery reliability when transmitting data to a cloud broker. Similarly, Sarkar and Misra (2016) highlight synchronization latency and cloud acknowledgment time as primary indicators of responsiveness in fog-based architectures, particularly when devices operate offline and later resynchronize with the cloud. Zhang et al. (2021) further extend this by measuring round-trip synchronization time and consistency overhead between local edge storage and cloud databases, confirming that responsiveness, rather than mere data arrival, is a critical evaluation dimension in practical deployments. Together, these studies validate the inclusion of synchronization response time as a core performance metric in IoT systems, supporting its application in smart-outlet dashboards where local readings must be reflected in the cloud with minimal delay.

2.6 Generative AI and Large-Language-Model Interfaces in Smart Systems

The emergence of Generative Artificial Intelligence (AI) and Large-Language-Model (LLM) interfaces has transformed how users interact with intelligent systems. In the context of smart home energy management, these

technologies enable natural-language communication, allowing users to issue commands, receive feedback, and interpret energy data conversationally. By combining machine learning with cloud connectivity, Generative AI enhances system accessibility, adaptability, and personalization. The following subsections explore natural-language processing foundations, AI integration frameworks, and their roles in optimizing interaction between users and smart energy platforms.

2.6.1 Natural Language Processing and Dialog Systems

Natural Language Processing (NLP) serves as the foundation of conversational interaction between humans and intelligent systems, enabling machines to understand, process, and respond to human language. According to Gomes et al. (2020), NLP-driven dialog systems rely on linguistic modeling, semantic parsing, and intent recognition to translate spoken or written commands into executable actions. These systems are supported by advancements in speech recognition, neural networks, and semantic web technologies, all of which collectively enable seamless human-machine communication. The study by Gomes et al. (2020) identified the rise of Intelligent Personal Assistants (IPAs) such as Google Assistant, Amazon Alexa, and Apple Siri, which utilize dialog management and knowledge-based reasoning to deliver personalized and contextual responses. The authors emphasized that the quality of interaction depends on the accuracy of language understanding and user intent classification. Furthermore, Shirali-Shahreza (2024) noted that NLP in modern AI models extends beyond basic command interpretation, incorporating emotion detection, contextual tracking, and self-adaptive dialogue flow, making system interactions increasingly human-like. These developments allow smart home devices to evolve from passive control interfaces into dynamic conversational partners capable of explaining system behavior and providing tailored advice (Gomes et al., 2020; Shirali-Shahreza, 2024; De Andrade & Tumelero, 2022).

The evolution of dialog systems has also been accelerated by the integration of Transformer-based architectures, which allow AI models to process entire sentences or conversations in context rather than as isolated commands (Al-Safi et al., 2025). This architecture enables systems to interpret user inputs more accurately and generate coherent, context-sensitive responses. According to Al-Safi et al. (2025), transformer-based models power advanced applications such as Vega, a large-language-model platform that integrates with IoT devices for home automation and real-time energy control. Such models employ self-attention mechanisms to prioritize key elements within a user query, improving system responsiveness and intent prediction. De Andrade and Tumelero (2022) further highlighted that dialog-based interfaces in smart systems increase user engagement and comprehension, especially when technical data such as power metrics or energy trends is translated into natural, conversational language. In this study, these same NLP principles are embedded into the Gemini-powered cloud AI dashboard, which interprets user queries, explains energy patterns, and provides actionable insights through human-like dialogue. Collectively, the integration of NLP and dialog systems enhances accessibility, transparency, and trust between users and intelligent energy-management platforms (Gomes et al., 2020; Shirali-Shahreza, 2024; Al-Safi et al., 2025; De Andrade & Tumelero, 2022).

2.6.2 Cloud AI Integration and Fine-Tuning for IoT Dashboards

Cloud-based Artificial Intelligence (AI) integration allows smart energy systems to process and analyze large volumes of user and sensor data efficiently. Unlike local AI models limited by embedded hardware, cloud-hosted frameworks can execute advanced inference, maintain contextual memory, and deliver adaptive recommendations. According to Al-Safi et al. (2025), the Vega platform demonstrated how Large Language Models (LLMs) can be integrated into IoT environments to interpret commands, monitor device states, and optimize home energy performance through continuous learning. Similarly, Shirali-Shahreza

(2024) emphasized that cloud-based LLMs benefit from constant data updates and large training sets, which enable them to adapt to diverse user preferences and environmental contexts. These capabilities are directly applicable to the present study's Gemini-powered AI dashboard, which leverages cloud computing to interpret socket-level data, generate predictive insights, and provide contextual advisories for energy efficiency.

According to Badar and Anvari-Moghaddam (2022), cloud integration in smart home systems enhances interoperability by linking distributed nodes, analytics engines, and user dashboards through standardized communication protocols such as RESTful APIs and MQTT. This architecture allows the system to perform heavy data processing in the cloud while maintaining lightweight local operations at the breaker and socket levels. When applied to this research, the same architecture supports the synchronization of appliance-level data from ACS712 and SCT-013 sensors with the cloud AI model, ensuring that predictive analytics, overload forecasting, and user recommendations are always based on the most recent readings. Furthermore, Al-Safi et al. (2025) demonstrated that fine-tuning pre-trained LLMs with domain-specific data significantly improves their task accuracy. In the context of this system, fine-tuning the Gemini model with household energy datasets enables it to provide context-aware guidance for example, identifying unusual power patterns or advising users on optimal appliance scheduling. Thus, cloud-based AI integration not only supports computational efficiency but also empowers the system to evolve continuously through feedback and real-time data learning, aligning perfectly with the intelligent objectives of this research.

2.6.3 Benefits of Generative AI in Energy Systems

Generative Artificial Intelligence (AI) introduces a new layer of intelligence in smart energy management by enabling systems to interpret user intent, predict behavior, and generate personalized feedback. Unlike traditional

rule-based automation, Generative AI models synthesize insights from large data sets to deliver recommendations that are contextually relevant and human-understandable. According to Al-Safi et al. (2025), LLM-driven systems such as Vega successfully integrate user dialogue and IoT device control, allowing seamless communication between human operators and automated energy infrastructures. These models use transformer-based architectures capable of understanding complex contextual cues, making them suitable for real-time decision support. In this study, a similar approach is implemented through the Gemini-powered dashboard, where Generative AI interprets live socket-level data transmitted from sensors like ACS712 and SCT-013, enabling predictive overload detection and intelligent power optimization. This interaction transforms technical readings into accessible energy advisories, bridging the gap between data analytics and everyday user understanding (Gomes et al., 2020; Shirali-Shahreza, 2024; Al-Safi et al., 2025).

Beyond user interaction, Generative AI enhances system adaptability and efficiency by continuously learning from user behavior and operational patterns. De Andrade and Tumelero (2022) emphasized that integrating NLP-based chat systems in smart environments not only increases usability but also builds user trust by simplifying data interpretation. This adaptability directly supports the goal of the proposed system to create an energy platform capable of learning from household consumption patterns and providing customized suggestions for optimal appliance usage. Furthermore, Generative AI models can summarize historical trends, detect anomalies, and deliver automated diagnostics without human intervention, ensuring reliability even in complex energy networks (Gomes et al., 2020; Shirali-Shahreza, 2024; Al-Safi et al., 2025). By embedding conversational intelligence into the monitoring process, the present system ensures that non-technical users can easily interact with the dashboard, understand power metrics, and make informed decisions thus aligning Generative AI's interpretive capabilities with sustainable and intelligent energy management.

2.7 User Accessibility and Usability in Smart Home Technologies

As smart home systems grow more advanced, ensuring user accessibility and ease of use has become a central design challenge. Despite technological progress, many users, especially those without technical backgrounds struggle to navigate complex interfaces and interpret energy data. Addressing these usability barriers is vital for achieving widespread adoption of intelligent energy systems. The following subsections discuss key usability challenges, generational and cognitive differences, and the role of conversational AI in improving interaction and understanding among non-technical users.

2.7.1 Cognitive and Technical Barriers

Smart home technologies have evolved rapidly, yet the gap between system capability and user comprehension remains significant. Many users encounter difficulty understanding how interconnected devices operate, resulting in inefficient or inconsistent use of available features (Becks et al., 2023). Studies reveal that even when systems provide real-time monitoring and control, non-technical users often fail to interpret sensor readings or automation feedback accurately (Woźniak et al., 2025). This cognitive barrier limits the user's ability to make informed energy-related decisions, reducing the overall effectiveness of intelligent monitoring systems. The complexity of multi-platform dashboards, combined with inconsistent terminology, contributes to confusion and deters engagement with system analytics (He et al., 2019; Chanda et al., 2024). For smart home interfaces to be effective, they must simplify information presentation, provide visual clarity, and minimize the technical knowledge required to operate devices.

Cognitive overload also stems from fragmented feedback mechanisms and limited contextual support within traditional dashboards. When users cannot correlate energy data with appliance behavior, they are less likely to adopt

sustainable usage patterns (Becks et al., 2023; Wozniak et al., 2025). Integrating intuitive elements such as color-coded indicators, summarized reports, and AI-generated explanations can significantly improve comprehension for non-technical individuals. The Gemini-powered AI dashboard in this research directly addresses this issue by combining simplified visual analytics with natural-language feedback that explains load data conversationally. Through adaptive communication and personalized advisories, users can interact with the system more intuitively, fostering better understanding and sustained engagement. Thus, overcoming cognitive and technical barriers is not merely a design improvement but a requirement for inclusivity and long-term adoption in intelligent home energy management.

2.7.2 Digital Literacy and Generational Differences

Adoption of smart home technologies varies widely across age groups due to differences in digital literacy, familiarity with automation, and confidence in technology use. Research shows that older generations, including Baby Boomers and Generation X, often report higher anxiety and lower comfort levels when using new digital interfaces compared to younger users (Culp-Roche et al., 2020). This discomfort can discourage engagement with complex dashboards and limit interaction with IoT-enabled systems. In contrast, younger users tend to adapt more quickly to smart environments but may overlook safety or efficiency features due to overreliance on automation (Stafford et al., 2021). These disparities highlight the need for system designs that accommodate varied levels of experience and comfort, ensuring equitable accessibility across all demographics. Simplified interaction models, clear visual cues, and adaptive interfaces are necessary to mitigate generational gaps and increase system usability for both novice and experienced users (Culp-Roche et al., 2020; Stafford et al., 2021; Wozniak et al., 2025).

Usability challenges linked to generational differences are particularly relevant to energy-management dashboards, where data interpretation and control decisions are often complex. Users with limited digital literacy may struggle to relate numerical data to actual energy-saving behavior or safety implications (Becks et al., 2023; He et al., 2019). Incorporating conversational AI, as applied in the Gemini-powered dashboard, directly addresses this issue by replacing technical metrics with contextual explanations and proactive suggestions. The AI assistant translates readings from sensors such as ACS712 and SCT-013 into accessible language, allowing users from all age groups to understand consumption trends and make informed decisions. This adaptive communication bridges generational divides and promotes confidence in using smart systems responsibly. As a result, the dashboard supports inclusivity while maintaining technical precision demonstrating how effective interface design, supported by AI-driven interaction, enhances usability for diverse user demographics (Culp-Roche et al., 2020; Stafford et al., 2021; Becks et al., 2023).

2.7.3 Conversational Interfaces as Accessibility Solutions

Conversational interfaces have emerged as an effective means of addressing usability challenges in smart home systems by enabling natural, human-like communication between users and devices. Integrating Natural Language Processing (NLP) and dialog systems allows users to interact with complex data through simple commands or questions, significantly reducing the learning curve associated with traditional graphical dashboards (Gomes et al., 2020; Shirali-Shahreza, 2024). When applied to smart energy systems, these interfaces allow users to query their current energy usage, identify overloaded devices, or request system summaries without needing technical knowledge of power metrics. Research highlights that conversational systems enhance user confidence and engagement by providing immediate, context-aware responses (De Andrade & Tumelero, 2022; Al-Safi et al., 2025). In the present study, this principle underpins the **Gemini-powered cloud dashboard**, where users can ask

questions such as “Which socket consumes the most power?” or “How can I reduce my energy usage today?” and receive AI-generated responses based on real-time sensor data from the ACS712 and SCT-013 modules. This direct conversational model eliminates the complexity of data interpretation, allowing even non-technical individuals to participate actively in energy management.

Beyond convenience, conversational interfaces contribute to inclusivity and long-term system adoption. Studies emphasize that chat-based interaction creates emotional engagement, increasing the likelihood that users will continue using smart systems for energy conservation (De Andrade & Tumelero, 2022; Gomes et al., 2020). The conversational feedback loop built into AI assistants enables users to learn progressively through interaction, reinforcing awareness of power consumption patterns and safe device usage. In this study’s context, the Gemini AI model performs as both a data interpreter and a virtual advisor, generating tailored insights and translating complex parameters into relatable information. This approach aligns with findings that intelligent assistants can foster behavioral change by promoting user education and comprehension rather than simply displaying numeric data (Gomes et al., 2020; Shirali-Shahreza, 2024; Al-Safi et al., 2025). By integrating conversational interaction with predictive energy analytics, the system transforms traditional monitoring into an intuitive, dialogue-driven experience that strengthens accessibility, understanding, and user empowerment.

2.8 Testing Methodologies and Accuracy Evaluation

Evaluating system performance through structured testing is essential to validate accuracy, reliability, and functional consistency in smart energy systems. Testing procedures ensure that sensor readings, communication latency, and AI-driven responses operate within acceptable thresholds for safety and precision. In this study, both material and system-level tests are conducted to assess electrical, communication, and analytical performance. The following subsections

discuss sensor calibration methods, latency and reliability evaluations, and validation frameworks used in related works that guided the design of this research's hybrid outlet-breaker system.

2.8.1 Sensor Calibration and Error Analysis

Accurate calibration is a foundational step in validating sensor performance for IoT-based energy monitoring systems. Studies show that calibration directly impacts the precision of voltage and current readings, which in turn affects total power computation and system safety (Asif et al., 2021; Sudiartama et al., 2023). The process typically involves comparing sensor outputs with reference instruments such as clamp meters or digital multimeters and applying linear correction factors to minimize deviation. The Smart Energy Management and Load Monitoring of Individual Loads experiment (E3S, 2023) employed a Meco clamp meter as a standard for verifying sensor outputs from ACS712, SCT-013, and ZMPT101B, achieving mean errors within $\pm 5\%$ of the reference readings. Similar calibration procedures using controlled load tests and RMS computations were reported by Asif et al. (2021) and Saputri et al. (2024), who demonstrated that proper scaling of current and voltage waveforms significantly improves the reliability of computed power and energy metrics. These findings validate the importance of periodic calibration and confirm that mid-cost sensors can achieve high accuracy when supported by proper compensation algorithms (Asif et al., 2021; Saputri et al., 2024; E3S, 2023).

Noise filtering and temperature compensation further improve sensor stability during continuous operation. Sudiartama et al. (2023) implemented digital filtering methods, including moving average and low-pass filters, to smooth the output of ACS712 and ZMPT101B modules, thereby reducing signal fluctuations caused by environmental interference. Saputri et al. (2024) also emphasized the value of recalibrating sensors periodically, showing that adaptive recalibration every 30 days enhances consistency under varying temperature and

humidity conditions. In the context of this study, each sensor is calibrated individually under controlled load conditions to ensure accurate scaling between measured voltage and current signals. These calibration constants are then applied to the firmware of the ESP32, allowing real-time adjustment and compensation during data acquisition. The error analysis follows the methodologies outlined in the referenced studies, ensuring that measurement accuracy remains within $\pm 5\%$ for both instantaneous and average readings (Asif et al., 2021; Sudiartama et al., 2023; Saputri et al., 2024). This rigorous calibration process ensures that the hybrid outlet-breaker system maintains precise monitoring performance, establishing a reliable foundation for intelligent energy management and predictive overload protection.

2.8.2 Communication Latency and Reliability Tests

Evaluating communication latency and reliability is critical for validating the responsiveness of smart outlet-breaker systems. Transmission speed directly affects how quickly control commands and sensor data propagate between distributed outlets and the central monitoring unit (Asif et al., 2021; Sudiartama et al., 2023). Efficient communication ensures real-time overload protection, minimizes data packet loss, and maintains synchronization between socket-level devices and cloud dashboards. Empirical loopback testing using HC-12 RF modules has shown average round-trip delays of approximately 31 milliseconds, confirming that RF transmission can handle event-driven control signals with minimal latency (Saputri et al., 2024; E3S, 2023). These findings align with hybrid systems described by Asif et al. (2021), where RF is prioritized for rapid local communication and Wi-Fi handles long-range or cloud-bound data. Reliability metrics such as packet delivery ratio, response time, and retransmission rate are key indicators for ensuring stability under dynamic load conditions (Sudiartama et al., 2023). RF-based links generally outperform Wi-Fi when environmental interference is high or when the network lacks consistent internet connectivity, offering a dependable backbone for real-time monitoring.

Latency and reliability assessments typically involve stress testing under varying loads and transmission distances. Experiments conducted in related studies measured data transmission efficiency by recording delays between successive data frames and observing drop rates under interference-prone environments (E3S, 2023; Saputri et al., 2024). Hybrid designs that combine RF for intranet communication and Wi-Fi for internet synchronization provide redundancy if one protocol fails, the other maintains continuity of operation (Asif et al., 2021; Sudiartama et al., 2023). In the proposed system, HC-12 modules facilitate instantaneous socket-to-breaker communication, while the ESP32's Wi-Fi interface ensures cloud integration with the Gemini AI dashboard. This configuration is tested by measuring latency under both connected and offline modes, ensuring consistent data integrity during synchronization. Recorded latency, packet reliability, and reconnection success rates are compared to acceptable standards derived from referenced works, ensuring that communication remains stable even under variable household network conditions. These performance evaluations validate that the hybrid RF–Wi-Fi model offers a reliable and responsive communication framework suitable for intelligent and safety-critical IoT applications.

2.8.3 AI Interface Response and Usability Metrics

Assessing the responsiveness and usability of the AI interface is essential to determine how effectively users interact with smart energy dashboards. Metrics such as response time, accuracy of interpretation, and user satisfaction define the reliability and practicality of conversational systems in real-world environments (Akhtar, 2025; De Andrade & Tumelero, 2022). A well-designed AI interface must generate contextually relevant and timely responses while maintaining minimal delay between query and output. Studies on chatbot-based systems have reported ideal response times within 1–3 seconds, balancing natural dialogue flow with computational efficiency (McDermott & Nicho, 2025). Performance below this threshold ensures a conversational rhythm that users perceive as natural,

reducing frustration and encouraging continued engagement. The Gemini-powered dashboard follows this benchmark by utilizing optimized cloud inference, allowing it to process queries related to socket-level power usage, total load, and predictive overloads with high accuracy. LLM-based assistants such as Vega and LYSIS have demonstrated similar benchmarks, successfully integrating natural language feedback into IoT frameworks while maintaining both fast response rates and low false-positive interpretations (Al-Safi et al., 2025; Cossu et al., 2021).

Usability testing also measures how effectively users can navigate AI interfaces and comprehend system feedback. Research shows that conversational dashboards improve task completion rates and user comprehension compared to traditional text-based interfaces (De Andrade & Tumelero, 2022; Gomes et al., 2020). Systems incorporating NLP and LLM technology, such as Gemini, enable dynamic query handling that adjusts to individual phrasing or context, enhancing personalization and user comfort (Shirali-Shahreza, 2024; Akhtar, 2025). Evaluations in similar AI-assisted monitoring systems used a combination of System Usability Scale (SUS) and task success rate metrics to quantify user performance, with scores above 80% indicating excellent usability (McDermott & Nicho, 2025; Al-Safi et al., 2025). For this research, usability validation includes both quantitative and qualitative measurements: latency between input and AI response, response accuracy against expected output, and subjective user satisfaction gathered through structured feedback. These tests confirm that the Gemini-powered interface offers real-time, context-aware feedback that enhances comprehension and engagement while remaining responsive under diverse conditions. By merging conversational design with performance optimization, the dashboard ensures efficient human AI collaboration, aligning usability with the study's objective of intelligent, accessible energy management.

2.8.4 Validation Frameworks

Validation frameworks establish the scientific basis for verifying whether an IoT-based energy system meets its intended design objectives. In smart monitoring architectures, validation ensures that measured parameters, communication reliability, and AI-driven functionalities operate within acceptable performance tolerances (Asif et al., 2021; Sudiartama et al., 2023). Studies in similar domains often adopt structured validation protocols that combine quantitative performance testing and qualitative usability assessments to evaluate accuracy and efficiency. For instance, the Smart Energy Management and Load Monitoring of Individual Loads project (E3S, 2023) employed a three-phase validation process: sensor accuracy verification, communication integrity testing, and user-interface evaluation to confirm system robustness under simulated household loads. Comparable frameworks (Sudiartama et al., 2023; Saputri et al., 2024) emphasize statistical validation using parameters such as mean absolute percentage error (MAPE) for sensor readings, packet delivery ratio (PDR) for transmission reliability, and overall acceptance rates derived from usability surveys. These studies collectively highlight that validation must address both hardware performance and human interaction metrics to ensure dependable real-world implementation.

In this research, the validation framework synthesizes best practices identified in existing literature while aligning with the system's unique hybrid architecture. Quantitative validation focuses on sensor accuracy within $\pm 5\%$ of reference instruments and RF–Wi-Fi communication latency under 50 ms, ensuring that control decisions are executed with minimal delay (E3S, 2023; Saputri et al., 2024). The qualitative validation phase evaluates AI response accuracy and user comprehension through structured interaction testing with the Gemini-powered dashboard. This two-tier approach follows guidelines outlined in contemporary IoT validation methodologies, where system effectiveness is determined through a balance of functional accuracy and user accessibility (Asif

et al., 2021; McDermott & Nicho, 2025; Al-Safi et al., 2025). Validation outcomes are assessed using defined pass/fail criteria, ensuring reproducibility and scientific credibility in the evaluation process. By applying these validation principles, the hybrid outlet–breaker system demonstrates measurable performance in sensing precision, communication reliability, and AI-assisted usability confirming its alignment with both technical and human-centered design goals.

2.9 Development Methodologies of AI Chatbots for Smart Automation

The development of AI chatbots in smart automation has evolved from rule-based systems to transformer-powered large language models (LLMs). This section outlines the detailed step-by-step methodologies used in key studies covering domain scoping, data preparation, model design, training strategies, and integration with IoT systems.

2.9.1 Prototyping and Integration of Gemini AI Using Google AI Studio

Google AI Studio officially provides a structured and developer-oriented workflow for building conversational agents through prompt configuration rather than model training, making it suitable for direct integration into custom dashboards. According to the Google AI Studio Quickstart, the process begins with Step 1 – Prototype, where the developer defines *system instructions* that establish the assistant’s role and behavior (for example, “You are an AI assistant that helps monitor electrical loads and controls smart sockets”). These instructions are entered in the Chat Prompt interface, which supports multi-turn conversations during testing. In Step 2 – Refine, developers iteratively improve the chatbot by adding clearer behavioral constraints, examples of expected outputs, and adjustments to settings such as model parameters, safety filters, or structured output formatting. This phase also allows enabling function-calling or structured JSON replies, which is essential for IoT control scenarios where the model must return a machine-readable command such as { "socket1": "OFF" }. Finally, in Step 3 – Integrate, once the conversational behavior is validated, AI Studio

provides a “Get Code” export function, which automatically generates ready-to-use API integration code. This exported code can then be placed directly into the backend of a web dashboard, allowing the deployed system to forward the AI-generated command to an ESP32 or microcontroller for execution. Because the export preserves the system instructions and prompt logic, this workflow enables seamless application-level integration without requiring model fine-tuning or Vertex AI deployment, confirming the feasibility of embedding Gemini directly into an IoT dashboard via prompt engineering alone (Google, 2025).

2.9.2 Vega’s Modular LLM Chatbot Development Architecture

According to Al-Safi et al. (2025), presented *Vega*, a modular, large-language-model (LLM)-driven chatbot platform designed for intuitive control of Internet of Things (IoT) devices using natural language commands. The chatbot was architected through a multilayered system composed of three primary modules: (1) a physical circuit board connected to a Raspberry Pi (RPi), (2) the RPi functioning as the control server, and (3) a web application integrating the LLM and multi-agent framework. The development centered around OpenAI’s GPT-3.5 as the core model, accessed through API rather than local fine-tuning, to manage command translation, planning, and image recognition tasks.

The chatbot’s functional flow begins with the user sending a natural-language input through a web-based UI built with React and TypeScript. This input is received by VegaChat (UI) and passed to VegaAi (logic layer), which then initiates three stateless LLM agents: the Chat Agent (for dialogue interpretation and function calling), the Planning Agent (for decomposing multi-step commands into task plans), and the Image Agent (for interpreting image inputs when available). Redis was used as the non-relational database to store conversation logs and session states, while SQLite handled physical device data logging on the RPi side.

Once the LLM interprets the intent, it formulates a function call, typically in JSON format, and transmits this instruction to the control server hosted on the RPi using RESTful APIs. The RPi server, built with Flask, executes the corresponding GPIO operations on the physical circuit, which includes input sensors (e.g., DHT sensor, GPS, ultrasonic) and output actuators (e.g., LEDs, fans, servo motors). Feedback from the hardware is returned to the user via the chatbot UI, completing the loop of interaction. Notably, no dynamic code generation is used at runtime to prevent security risks; instead, pre-defined functions are executed on command.

2.9.3 LLM Parameter Testing, Model Behavior Tuning, and Evaluation

To ensure robustness, Al-Safi et al. (2025) implemented an automated evaluation system involving over 622 test messages across varying complexity levels and model configurations. Each message was assigned a message complexity score ranging from 0 to 1, accounting for syntactic, semantic, and logical complexity. This scoring influenced the behavior of the model's temperature and top-p parameters, which were tuned dynamically to optimize performance under different scenarios.

The evaluation employed a specialized LLM Evaluation Agent to assess task execution success using four key metrics: functional accuracy (40%), logical coherence (30%), completeness (20%), and error handling (10%). Responses scoring above 80% were marked as successful. Heatmaps generated from test data illustrated optimal performance zones at medium complexity (0.4–0.6) and high temperature/top-p ranges (0.7–1.0). Additionally, the system logged errors into five categories: wrong format, incorrect logic, UI runtime errors, API timeouts, and hardware failures, with LLM formatting and logic errors accounting for 59% of all failures. These insights informed further prompt engineering refinements and fallback strategies.

The Vega platform did not fine-tune the GPT-3.5 model but leveraged it directly via API calls. Nonetheless, the authors emphasized that future iterations may explore fine-tuning for improved domain-specific performance. They also tested how different prompt formats and message lengths (minimal, normal, descriptive) influenced success, finding that descriptive inputs yielded the highest performance (84% success rate), indicating that richer context aids LLM interpretation.

In their study, Al-Safi et al. (2025) established systematic calibration and maintenance protocols to enhance sensor reliability within the Vega LLM-driven IoT platform. For temperature sensors, recalibration was conducted every 1000 measurements, while humidity sensors were recalibrated every 500 measurements. The researchers also incorporated outlier detection algorithms that filtered erroneous readings using statistical thresholds of ± 2 standard deviations from rolling means. Furthermore, sensor health monitoring mechanisms were integrated to track response times and accuracy trends, automatically flagging sensors that exhibited performance degradation for maintenance. These strategies ensured consistent data accuracy and reliability throughout the system's operation (Al-Safi et al, 2025).

Through this multi-agent LLM framework with task planning, context tracking, and real-time control execution, the Vega chatbot demonstrated how AI-driven systems can democratize IoT usage for non-technical users by eliminating the need for rigid syntax or programming knowledge. Its modular and scalable design also supports further extension into robotics, education, and industrial applications.

2.9.4 Development of the LYSIS AI Chatbot for Virtual IoT Platforms

According to Cossu, Girau, and Atzori (2021), developed the LYSIS chatbot to facilitate natural-language control of IoT devices within the Lysis

Social IoT (SIoT) platform. The system uses Google DialogFlow for NLU and is built on a serverless Function-as-a-Service (FaaS) architecture, integrating with Telegram as the messaging interface. User commands are received via an API gateway, authenticated, and routed to cloud functions through event topics. DialogFlow classifies intents using a retrieval-based model and triggers corresponding backend functions such as the Message Handler, Action Service, and Send Message.

The DialogFlow agent is configured with domain-specific intents and entities, stored in JSON format, enabling synonym recognition and intent chaining. The chatbot does not use deep learning but relies on rule-based matching and state tracking. Cloud functions operate independently, allowing modular scaling and reduced latency. Performance was optimized by setting a 0.7 confidence threshold and applying reinforcement techniques. Evaluation showed high intent accuracy (0.956 for ideal and 0.844 for paraphrased inputs). The development process included intent modeling, cloud function design, webhook integration, and iterative evaluation using DialogFlow dashboards and logs.

2.9.5 RF Reliability and Microcontroller Integration Studies

Reliable wireless communication is a critical factor in the performance of microcontroller-based embedded systems, particularly those involving remote sensing and control. Tang et al. (2023) developed a communication reliability evaluation framework using an 8051 microcontroller to analyze the success rate of data transmission over wireless links, emphasizing the use of proportion-based metrics such as the Packet Success Rate (PSR) to quantify reliability. Similarly, Yogarayan et al. (2023) compared the performance of common RF transceivers, HC-12, nRF24L01, and XBee, and concluded that the HC-12 achieved higher range and lower packet loss under stable baud-rate configurations. Supporting this, Deshmukh (2020) demonstrated the HC-12 module's reliability in real-time monitoring and control applications, highlighting its effective UART

communication with low error rates when paired with microcontrollers such as Arduino and PIC. Together, these studies provide the empirical basis for evaluating the reliability of the PIC16F88, HC-12 communication link in the present study through PSR measurement and 95% confidence interval estimation.

2.9.6 Logistic Regression Applications in Embedded Reliability Testing

In recent years, logistic regression has emerged as a powerful tool in embedded system reliability assessment, particularly for classification-type outcomes where a system is either operating normally or in failure mode. For example, Amaral et al. (2025) used logistic regression to classify capacitor health states (normal vs. faulty) within a buck converter system through measured electrical parameters, achieving accuracy and F1-scores above 0.99. Their study highlights how low-complexity regression models can be embedded in microcontroller-based platforms and used for real-time decision-making in reliability-critical applications. This underscores the suitability of logistic regression not only in high-level analytics, but also in embedded control systems where hardware logic paths (e.g., microcontroller → actuator) can be treated as binary decision variables.

While the majority of literature focuses on component health diagnosis or system-level fault prediction, the application of logistic regression to validate microcontroller-to-actuator switching or communication control paths remains relatively under-explored. The STMicroelectronics application note AN6223 (2025) provides a framework for integrating regression models into industrial embedded firmware, including data acquisition, pre-processing, model execution, and deployment. This framework supports the extension of logistic regression methods to functionality testing such as verifying GPIO output reliably drives a relay module thus providing a methodological precedent for the present study's integration test of the PIC16F88 and relay module using logistic regression.

2.10 Summary of Related Studies on Smart Energy Systems

A comprehensive understanding of prior research on IoT-based smart energy systems provides the foundation for evaluating this study's novelty and technical direction. Reviewing comparable implementations clarifies the evolution of smart plugs, breaker-level monitoring, AI-assisted dashboards, and communication models. This section summarizes selected studies that have influenced the design and testing methodologies of the proposed system, emphasizing their objectives, architectures, outcomes, and limitations relative to the hybrid outlet–breaker model introduced in this research.

2.10.1 IoT-Based Smart Plug Systems and Architectures

IoT-based smart plug systems have become fundamental components of modern energy-monitoring architectures, enabling per-appliance visibility, automated control, and real-time data collection. These systems typically integrate microcontrollers, current and voltage sensors, and wireless transceivers to form distributed networks that communicate with a central control unit (Asif et al., 2021; Sudiartama et al., 2023). The most common architecture employs ESP32 or NodeMCU microcontrollers paired with sensors such as ACS712, SCT-013, and ZMPT101B for measuring current, voltage, and power factor. Data are transmitted via Wi-Fi or RF to cloud dashboards that visualize consumption patterns and trigger automated responses based on threshold limits (E3S, 2023). Asif et al. (2021) implemented an adaptive current-cutoff feature that disconnects appliances exceeding safety thresholds, achieving real-time protection through firmware-based decision logic. Similarly, Sudiartama et al. (2023) designed an intelligent energy-monitoring system combining these sensors with predictive analytics to optimize renewable-energy utilization. These implementations demonstrate that modular IoT architectures can achieve accurate monitoring and efficient load control while maintaining cost-effectiveness, forming the technological foundation of the present hybrid outlet–breaker design.

Recent studies have expanded smart plug systems to include edge intelligence and collaborative functionality. Neto et al. (2024) introduced EnAPlug, an open-source smart plug built on the ESP32 platform, integrating additional environmental sensors and edge TinyML inference to recognize device usage patterns with 95% classification accuracy. The system also featured local storage and cloud synchronization, balancing privacy with real-time analytics. Munoz et al. (2022) proposed a smart distribution panelboard utilizing machine learning to predict overload risks and optimize circuit behavior, reinforcing the importance of integrating breaker-level intelligence with distributed plug-level monitoring. These findings align directly with the present study's hybrid approach, where outlet devices equipped with ACS712 sensors transmit data to a central control unit that aggregates load information from SCT-013 and ZMPT101B sensors. The combined data feed the Gemini-powered AI dashboard for intelligent analysis, anomaly detection, and user advisory. Collectively, these IoT-based architectures demonstrate the effectiveness of socket-level sensing integrated with centralized logic and AI analytics, validating the system framework adopted in this research (Asif et al., 2021; Munoz et al., 2022; Neto et al., 2024; Sudiartama et al., 2023; E3S, 2023).

2.10.2 Smart Breaker and Load Management Systems

Smart breaker systems extend the functionality of traditional circuit protection by incorporating sensors, microcontrollers, and wireless connectivity for intelligent load management. These systems not only interrupt power during faults but also monitor voltage, current, and power consumption in real time (Padlak et al., 2025; Munoz et al., 2022). Their architecture typically integrates SCT-013 current transformers at the breaker line and microcontrollers such as ESP32 or Arduino Mega, which gather electrical data for analysis. Munoz et al. (2022) developed a smart distribution panelboard capable of machine learning-based overload prediction and adaptive control, enhancing safety through proactive fault detection. Similarly, Padlak et al. (2025) implemented an

IoT-based smart energy metering system using virtual simulation in Proteus 8, confirming the feasibility of real-time monitoring, threshold-based protection, and data visualization. These implementations illustrate the shift from conventional mechanical protection to digitally controlled circuit management, forming the conceptual backbone of this study's intelligent outlet-breaker framework.

Recent developments in smart breaker research emphasize hybrid integration, where breaker-level sensing is combined with localized socket monitoring for comprehensive system visibility (E3S, 2023; Sudiartama et al., 2023). Hybrid load management ensures that while the breaker monitors total circuit load, socket-level devices detect per-appliance consumption, allowing precise identification of fault sources. In this research, this architecture is realized through SCT-013 sensors at the breaker for total load measurement and ACS712 sensors at each socket for granular monitoring. Load data are transmitted to the control unit and processed through if-else logic to execute automatic shutdowns when current thresholds are exceeded. This layered protection mirrors the multi-tier safety architecture proposed by Munoz et al. (2022), combining local and central intelligence for improved reliability. Additionally, the Gemini-powered AI dashboard supports predictive analysis, enabling early detection of anomalies and proactive user advisories. Together, these studies validate the practicality and technical soundness of a hybrid outlet-breaker system, demonstrating that smart breaker integration enhances both safety and operational efficiency (Padlak et al., 2025; Munoz et al., 2022; Sudiartama et al., 2023; E3S, 2023).

2.10.3 AI-Assisted Energy Monitoring and User Advisory Systems

The integration of Artificial Intelligence (AI) into energy monitoring systems has transformed how users understand, manage, and optimize electricity consumption. Modern frameworks combine IoT sensor networks with machine learning and natural language interfaces to generate insights that are both

actionable and user-friendly (Al-Safi et al., 2025; De Andrade & Tumelero, 2022). LLM-powered platforms such as Vega demonstrate that intelligent chatbots can interpret sensor data and execute energy-saving commands while maintaining real-time dialogue with users. By leveraging Transformer-based architectures, these systems process contextual cues, enabling predictive feedback such as load optimization, anomaly detection, and fault anticipation (Gomes et al., 2020; Shirali-Shahreza, 2024). In AI-driven monitoring, user queries such as requests for consumption trends or overload alerts are answered dynamically based on continuous learning from historical data. These principles directly inform the design of the Gemini-powered dashboard, which analyzes socket-level data from ACS712 and SCT-013 sensors to provide users with intelligent recommendations on load balancing, safety, and efficiency. Through its generative capabilities, Gemini translates raw energy metrics into clear and conversational insights, bridging the gap between complex analytics and user comprehension.

AI-assisted dashboards not only improve energy visibility but also foster sustained engagement through personalized advisory systems. Studies have shown that conversational feedback loops where the AI provides explanations and corrective suggestions significantly increase user participation and satisfaction (De Andrade & Tumelero, 2022; Shirali-Shahreza, 2024). For example, the LYSIS Chatbot (Cossu et al., 2021) enabled users to control IoT devices through natural language, achieving faster task completion and improved decision confidence. Similarly, Vega's AI-driven interface demonstrated energy-saving potential through adaptive behavioral analysis, achieving up to 15% reduction in consumption when personalized recommendations were followed (Al-Safi et al., 2025). These studies validate the core advantage of integrating conversational AI into smart energy systems: it transforms passive data reporting into active advisory. The proposed Gemini-powered system extends this concept by combining generative conversation with predictive analytics allowing users not only to monitor but also to understand and act on energy information. This dual

role of Gemini as both advisor and analytic engine establishes a foundation for intuitive, proactive, and intelligent energy management, aligning with the system's goal of enhancing user interaction and operational safety (Al-Safi et al., 2025; Gomes et al., 2020; De Andrade & Tumelero, 2022; Cossu et al., 2021).

Across the reviewed studies, distinct AI chatbot development methodologies emerged. Al-Safi et al. (2025) employed OpenAI's GPT-3.5 via API, using a modular multi-agent architecture with dynamic prompt tuning but without local training, instead optimizing model behavior through parameter calibration and structured function calls. Cossu et al. (2021) built their LYSIS chatbot on Google DialogFlow, configuring intents and entities manually while refining matching accuracy through synonym mapping, confidence thresholds, and reinforcement techniques. Each approach reflects a different AI modeling strategy ranging from rule-based and retrieval-based systems to LLM-driven architectures tailored to the technological context and domain requirements.

2.10.4 Comparative Analysis of Related Studies

Comparing existing IoT- and AI-based energy monitoring systems provides insight into how current research aligns with, and differs from, the proposed hybrid outlet-breaker design. By analyzing system architectures, communication methods, sensor accuracy, and user interaction models, a clear understanding emerges of the research gap addressed by this study. The following comparative discussion highlights the strengths and limitations of previous works in relation to the present system's innovations in hybrid sensing, predictive safety control, and AI-driven user advisories.

2.11 Summary of Gaps and Conclusions

The reviewed literature demonstrates significant progress in the domains of IoT-based energy monitoring, AI-assisted analytics, and user-interactive dashboards. Studies on smart plugs and breaker systems (Asif et al., 2021; Munoz

et al., 2022; Sudiartama et al., 2023; Neto et al., 2024) have proven that distributed sensing using modules like ACS712, SCT-013, and ZMPT101B can achieve efficient real-time load monitoring. Similarly, cloud-connected dashboards (E3S, 2023; Al-Safi et al., 2025) have enabled remote visualization and control, while conversational interfaces (Gomes et al., 2020; De Andrade & Tumelero, 2022) improved accessibility for non-technical users. Despite these advances, most existing studies remain limited to either plug-level or circuit-level monitoring, without integrating both into a unified hybrid framework. Furthermore, prior systems primarily relied on static dashboards with limited analytical depth and lacked predictive feedback mechanisms powered by Generative AI. These gaps reveal the absence of an intelligent system that merges hardware-based protection, multi-layer sensing, and AI-driven advisory in a cohesive and user-friendly architecture.

The proposed study addresses these limitations through a hybrid outlet-breaker system that integrates per-socket ACS712 sensors and breaker-level SCT-013 modules for comprehensive energy visibility, paired with an AI-powered Gemini cloud dashboard for intelligent analysis and user guidance. This approach extends previous works by combining IoT-based sensing with large-language-model analytics capable of generating personalized feedback and predictive overload warnings in natural language. The literature confirms the individual feasibility of each subsystem IoT sensing, wireless control, and AI interaction but lacks evidence of their coordinated implementation in a single household platform. Hence, this research contributes a novel architecture that not only improves measurement accuracy and communication reliability but also enhances user comprehension through conversational AI. Bridging these technological and usability gaps ensures that energy monitoring evolves beyond data collection toward proactive, intelligent, and human-centered energy management (Asif et al., 2021; Munoz et al., 2022; Neto et al., 2024; Al-Safi et al., 2025)

CHAPTER 3

RESEARCH DESIGN AND METHODOLOGY

This chapter presents the research design and methods applied in developing the smart outlet breaker system. It outlines the Agile V-Model framework, story mapping, and system architecture used throughout the study. The chapter also details the three development sprints: appliance-level data transmission, load monitoring with intelligent safety logic, and dashboard integration with Generative AI assistance, together with their respective design, implementation, integration, and testing procedures. Finally, a costing analysis of the system components is presented to evaluate the feasibility of the prototype.

3.1 Research Method

This study adopts the Agile V-Model as its research and development methodology. In the traditional V-Model, development is represented as a single “V” cycle. The left side of the “V” focuses on progressive decomposition: requirements engineering, system architecture, and component design, while the base of the “V” is where implementation occurs. The right side of the “V” represents integration, verification, and validation, ensuring that each component and the system as a whole meet specifications. While structured and rigorous, the traditional V-Model is rigid, as it assumes that requirements are fixed and does not easily adapt to changes during development (Slama et al., 2023).

The Agile V-Model introduces iteration by breaking this single large V-cycle into a series of smaller cycles called sprints. Each sprint can be seen as a mini-V, where planning, design, implementation, verification, and validation occur within a short time frame. The mini-V ensures that each sprint increment is both tested and validated, creating a working version of the system at the end of every cycle. This combination allows the structured rigor of verification and validation

from the V-Model to coexist with the flexibility and adaptability of Agile practices (AIoT Playbook, 2023).

In the adapted model used for this research, each sprint was aligned with one of the study's major objectives. Sprint 1 developed the socket-level current detection and RF communication. Sprint 2 integrated breaker-level monitoring, and intelligent safety logic for overload protection. Sprint 3 finalized the system through the integration and performance evaluation of a cloud-based dashboard equipped with a Generative AI assistant for user interaction. The continuous cycle of integration, testing, and delivery ensured that every increment is both functional and research-valid, while also supporting online operation with limited offline safety fallback features vital for household energy monitoring and protection (Hornos & Quinde, 2024).

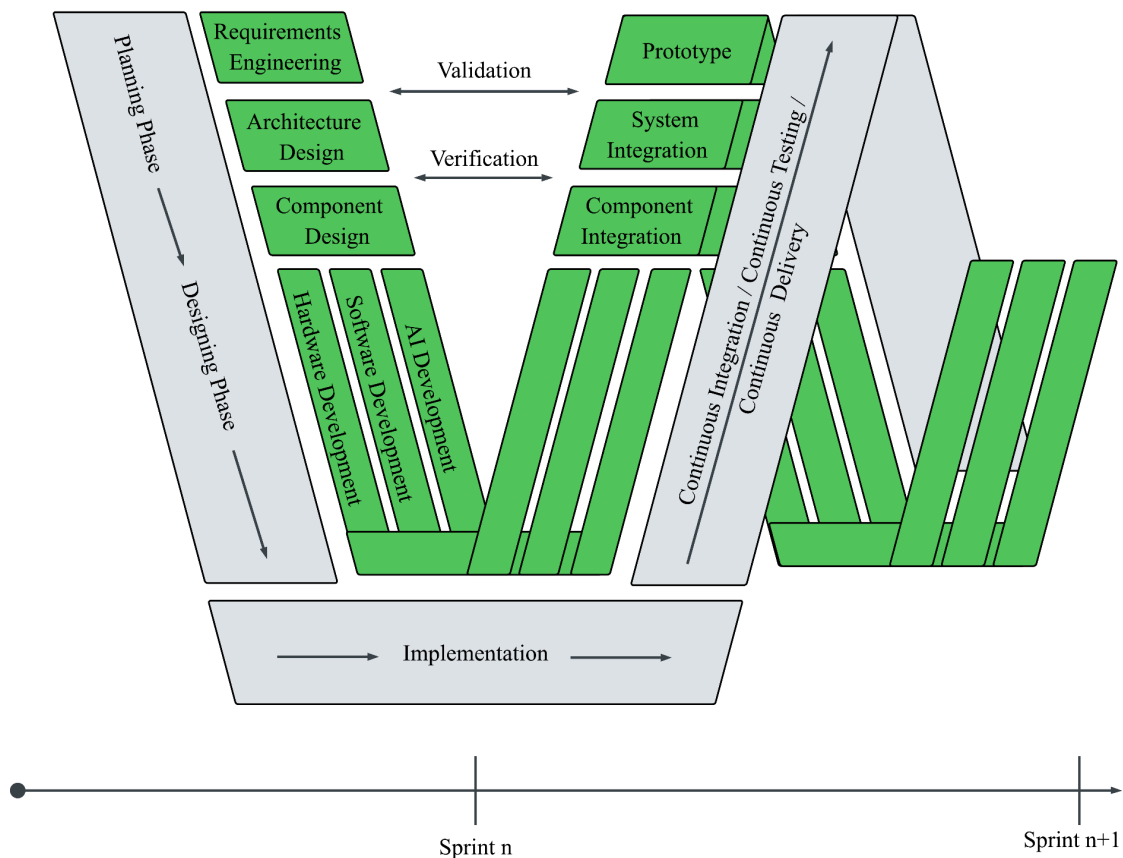


Figure 3.1 Agile V - Model Hybrid Approach

3.2 System Requirements

The development of the smart outlet breaker system relies on both hardware and software components to ensure reliable detection, monitoring, and control of appliances at the household level. The hardware requirements include microcontrollers, sensors, actuators, and power modules that perform the essential physical functions of the system, while the software requirements define the operational logic and communication protocols that integrate each component into a cohesive framework.

3.2.1 Hardware Requirements

The hardware foundation of the smart outlet breaker system is designed to enable socket-level current sensing, circuit load measurement, and automated control. At the breaker side, an ESP32 microcontroller functions as the main unit, coordinating communication, analyzing load data, and applying intelligent safety logic for overload protection. Each outlet is equipped with a PIC16F88 microcontroller, which interprets analog inputs from Hall-effect current sensors integrated into each socket to measure the current draw of connected appliances. It also manages relay actuation for socket switching based on commands from the central control unit. The SCT-013 current sensor is installed at the breaker to provide continuous measurements of overall current flow, serving as input for overload evaluation and system-wide monitoring. Wireless data exchange between outlets and the breaker unit is established through the HC-12 transceiver, offering a long-range serial link for both event reporting and command delivery. Relay modules are integrated into the outlets to handle appliance switching safely, while current sensors ensure precise measurement of power usage per socket. Power to the entire system is supplied by a Hi-Link HLK-5M05 module, which converts AC mains to a stable 5 V DC output suitable for microcontrollers and peripheral modules. Complementary components such as PCBs, wiring connectors, indicator

LEDs, and protective housings complete the assembly, ensuring reliability, ease of testing, and safe operation.

A. ESP32 Microcontroller

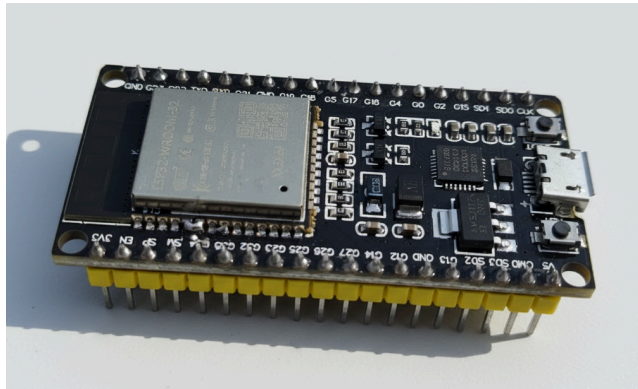


Figure 3.2 ESP32

The ESP32 serves as the primary controller of the smart outlet breaker system. Positioned at the breaker level, it is responsible for coordinating communication with outlet devices, processing current sensor data, and applying local safety logic for overload protection. Its built-in Wi-Fi and Bluetooth interfaces also allow seamless integration with the web-based dashboard, enabling users to monitor real-time data, receive AI-generated insights, and control appliances remotely when internet access is available.

Equipped with a dual-core processor running at 240 MHz and multiple peripheral interfaces such as UART, SPI, and I²C, the ESP32 provides sufficient processing power and flexibility to handle both system-level control and data management tasks. The microcontroller is also optimized for low power consumption, which is advantageous for continuous monitoring applications. Its compatibility with the Arduino IDE and wide community support simplify

firmware development, making it an ideal choice for implementing the central intelligence of the system.

B. PIC16F88 Microcontroller



Figure 3.3 PIC16F88 Microcontroller

The PIC16F88 functions as the outlet-level controller in the smart outlet breaker system. Embedded directly within each outlet device, it is responsible for interpreting analog signals from Hall-effect current sensors to measure the current draw at each socket. It also controls the relay modules that switch appliance sockets ON or OFF in response to commands from the central unit. This localized processing ensures that each outlet can independently handle current measurement and switching functions while maintaining synchronized communication with the breaker-level controller.

This microcontroller was selected due to its compact design, integrated Universal Synchronous Asynchronous Receiver Transmitter (USART) for UART communication, and sufficient program memory to accommodate socket-level control logic. Its proven stability in embedded applications makes it reliable for continuous operation within household outlets. Acting as the intermediary between the physical socket components and the central ESP32, the PIC16F88 encodes socket-level current data into structured RF frames for transmission to the control unit, enabling efficient monitoring and coordination across the system.

C. SCT-013 Current Sensor



Figure 3.4 SCT-013 Current Sensor

The SCT-013 current sensor is utilized at the breaker level to measure the total electrical load passing through the outlet circuit. As a non-invasive current transformer, it clips around a single conductor without requiring direct electrical contact, ensuring safer installation compared to intrusive sensing methods. Its analog signal output provides continuous current readings, which are fed into the ESP32 for monitoring and overload prediction. This allows the system to assess aggregate consumption.

This sensor was chosen because of its practicality, affordability, and compatibility with low-voltage microcontroller inputs when paired with the appropriate burden resistor. It provides sufficient accuracy for household energy monitoring applications, typically within $\pm 3\%$. By serving as the breaker-level reference sensor, the SCT-013 complements the per-socket current sensors, enabling cross-validation of total load against individual socket readings to enhance data reliability and system safety. Its durability and ease of integration make it well-suited for continuous operation in a smart home environment.

D. HC-12 RF Transceiver Module

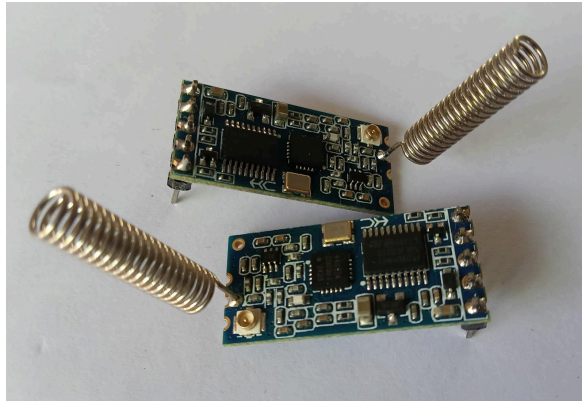


Figure 3.5 HC-12 RF Transceiver Module

The HC-12 module provides the wireless communication link between the outlet devices and the breaker-level controller. Operating in the 433 MHz ISM band, it enables reliable long-range data transmission with minimal power consumption. Through UART interfacing, the PIC16F88 sends socket-level current readings and relay status updates to the ESP32, while also receiving commands for socket switching and load-related safety control in return. This two-way communication ensures that both outlet-level and breaker-level operations remain synchronized.

The module was selected because of its versatility, offering up to 100 communication channels and an effective range of up to one kilometer under open conditions. Compared to short-range wireless modules like Bluetooth, the HC-12 provides stronger signal penetration and stability, making it more suitable for household applications where outlets may be distributed across different rooms and floors. Its compatibility with microcontrollers through simple serial commands also reduces programming complexity. By serving as the system's wireless backbone, the HC-12 supports scalable expansion to multiple outlets without requiring extensive wiring.

E. ACS712 Hall-Effect Current Sensor

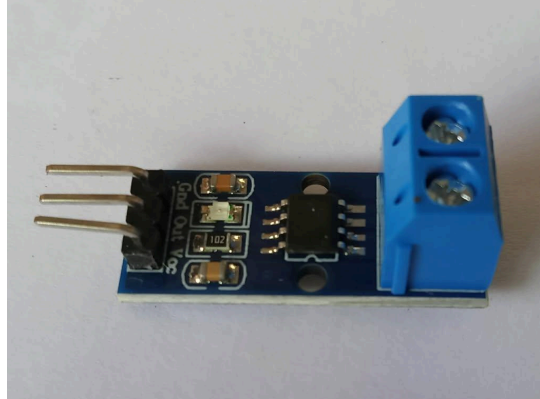


Figure 3.6 ACS712 Current Sensor

The Hall-Effect Current Sensor is integrated into each socket of the smart outlet to measure the electrical current drawn by connected appliances. Acting as a non-intrusive analog sensor, it outputs a voltage proportional to the magnitude and direction of the current flowing through the conductor. This signal is processed by the PIC16F88 microcontroller, which converts the analog readings into digital current values and transmits them via the HC-12 RF transceiver to the ESP32 for centralized monitoring, safety logic execution, and cloud synchronization.

This component was selected for its accuracy, electrical isolation, and durability, offering a reliable solution for continuous household energy monitoring and real-time socket-level current detection. Unlike mechanical detection methods such as microswitches, the Hall-effect sensor provides continuous and precise load monitoring without physical contact or mechanical wear. Its fast response and linear output enable accurate AC current measurement, allowing the system to identify irregular loads, detect faults, and optimize energy usage.

F. Relay Module



Figure 3.7 Relay Module

The relay module is responsible for controlling the flow of electricity to appliances connected to each smart outlet. Driven by the PIC16F88 microcontroller, the relay acts as an electronic switch that can disconnect or reconnect power to the socket based on system commands or safety logic triggered by the breaker-level controller. This enables the system to not only monitor appliance usage but also intervene when necessary, such as cutting off power during unsafe load conditions or allowing users to remotely control devices through the web dashboard and Generative AI assistant.

The relay was selected because of its ability to safely handle standard household voltages and currents, with typical ratings of up to 250 V AC and 10 A. It provides electrical isolation between the low-voltage control circuitry and the high-voltage appliance line, ensuring safe operation. Its proven reliability in automation and embedded systems makes it a practical choice for continuous use in household settings. By integrating the relay into each outlet device, the system extends beyond passive monitoring to active, AI-assisted control, providing users with enhanced safety, convenience, and intelligent energy management.

G. Hi-Link HLK-5M05 Power Module



Figure 3.8 Hi-Link HLK-5M05 Power Module

The Hi-Link HLK-5M05 power module supplies the regulated low-voltage power necessary for the operation of the smart outlet breaker system. It converts the 220 V AC mains input into a stable 5 V DC output that powers the microcontrollers, RF modules, sensors, and relays. By integrating this compact AC-DC converter into each outlet device, the system eliminates the need for external adapters and ensures reliable power delivery to all electronic components.

This module was chosen because of its compact design, electrical isolation, and safety certifications, which make it well-suited for continuous household use. Its 5 W output capacity is sufficient to supply the outlet microcontroller and associated peripherals without overheating or instability. The Hi-Link module also enhances system safety by providing isolation between the high-voltage mains supply and the low-voltage control circuitry. With its robust and efficient design, it serves as the backbone of the system's power infrastructure, ensuring uninterrupted operation of both outlet-level and breaker-level components.

3.3 System Architecture

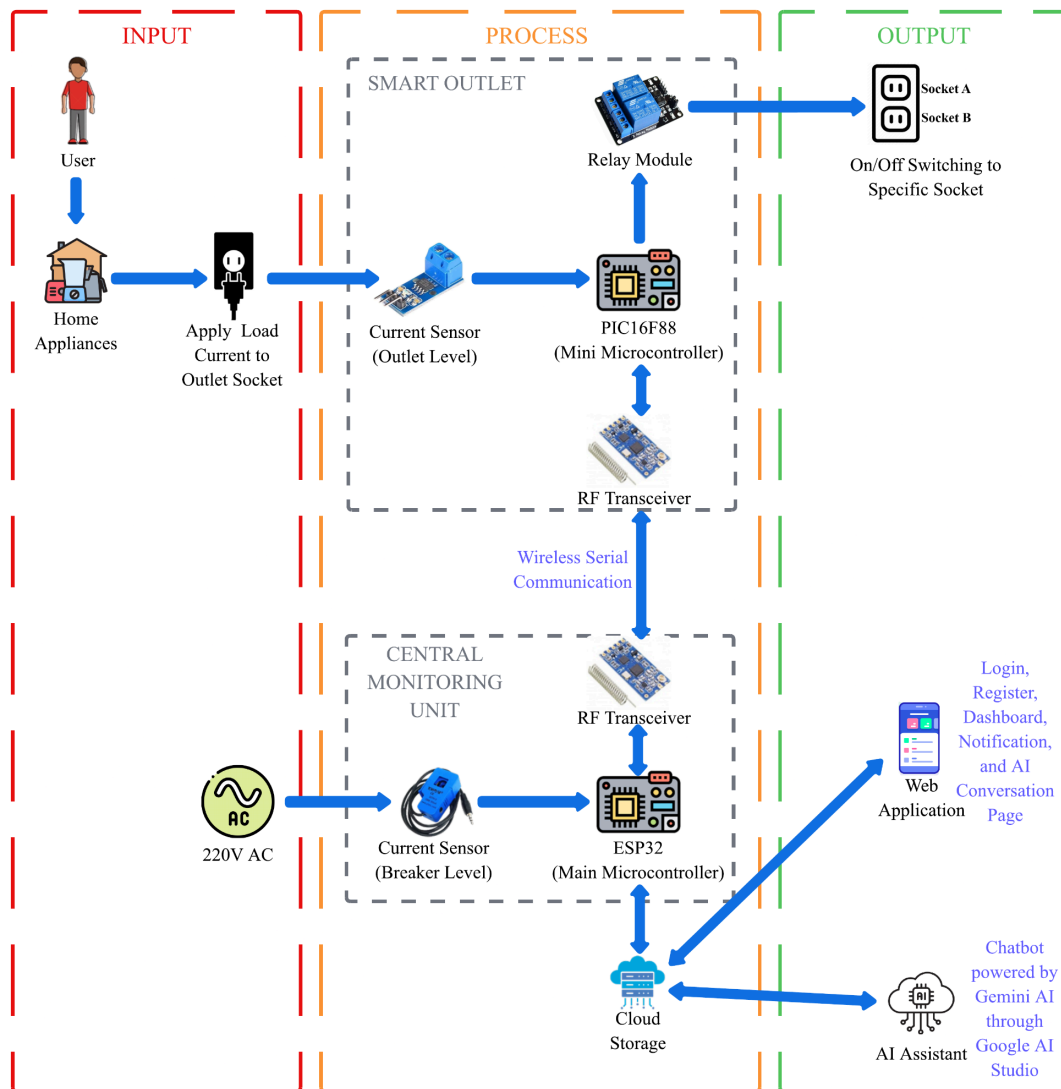


Figure 3.9 System Architecture Diagram

The proposed system architecture is designed to integrate smart outlet devices, a central monitoring unit, and a cloud-based AI-assisted dashboard into one cohesive framework for intelligent load monitoring and appliance-level control. At the device layer, each smart outlet contains a PIC16F88

microcontroller, two Hall-effect current sensors (one per socket), a relay module for switching control, and an HC-12 RF transceiver for communication. The

current sensors continuously measure the load drawn by connected appliances, and the microcontroller encodes these readings with the outlet and socket identifiers before transmitting them via RF to the central monitoring unit. The relays inside the outlet execute commands received from the central unit, such as cutting power to specific sockets during unsafe conditions or toggling sockets based on user control from the web dashboard.

At the control layer, the central monitoring unit is installed at the outlet breaker, functioning as the “brain” of the system. It consists of an ESP32 microcontroller, a breaker-level SCT-013 current sensor, and an HC-12 RF transceiver. The ESP32 aggregates all socket-level current data received from the outlets, evaluates total circuit load, and applies local safety logic to prevent overloads. Whenever internet connectivity is available, the ESP32 synchronizes the aggregated data with the Gemini-powered Generative AI chatbot integrated via Google AI Studio, which analyzes energy trends, interprets activity logs, and provides advisory feedback through the dashboard.

At the application layer, the system connects to a web-based dashboard that allows users to register, log in, and interact with their household energy data in real time. The dashboard displays socket-level current readings, total breaker load, and device status while enabling remote control through manual toggles or AI-assisted commands. The integrated Generative AI chatbot, developed through Google AI Studio, allows users to query the system conversationally for example, asking which outlet consumes the most power or requesting suggestions for reducing energy usage. This integration enhances both accessibility and user understanding, bridging the gap between technical data and actionable insights.

This layered design ensures reliable operation under both offline and online conditions. When offline, the system prioritizes safety through local logic

implemented in the central monitoring unit, maintaining essential monitoring and protection functions. When online, it enhances intelligence and usability through cloud-based analytics and AI-driven interaction, providing an adaptive, user-friendly approach to household energy management.

3.4 Sprint 1: Socket and Breaker-Level Load Detection and RF Communication

3.4.1 Planning Phase

Sprint 1 focuses on developing the foundational hardware and communication framework required for appliance-level data acquisition and wireless transmission. During this phase, the team will design and prototype the smart outlet module that integrates key components PIC16F88 microcontroller, ACS712 current sensors, relay modules, and the HC-12 RF transceiver to establish socket-level detection and control. The goal is to ensure that each outlet can independently sense current changes, toggle relay states, and reliably transmit data to the central control unit. Component datasheets, circuit schematics, and power calculations will guide the configuration and wiring of each submodule to verify proper voltage levels, signal routing, and electrical isolation for safety.

In parallel, the team will define communication protocols, data formats, and testing parameters to validate system responsiveness and transmission reliability. The initial tests will include calibration of current sensors, verification of GPIO switching functions, and RF transmission loopback trials to confirm stable data exchange between the outlet device and the receiver. This phase lays the groundwork for subsequent sprints by ensuring that the essential sensing, control, and communication mechanisms operate accurately and consistently, forming the basis for intelligent load monitoring in later stages of system integration.

3.4.2 Design Phase

Table 3.1

Hardware Components for Sprint 1 Smart Outlet

Component	Description	Key Specifications	Function in System
PIC16F88	8-bit auxiliary microcontroller with UART support.	7 KB Flash, 368 B RAM, 8-pin package, UART-enabled.	Reads microswitch signals and communicates with ESP32 via UART.
ESP32	Dual-core microcontroller for central control and RF data handling.	240 MHz, 12-bit ADC, Wi-Fi / BT, UART, 3.3 V logic.	Aggregates outlet data, reads breaker load, and applies overload safety logic.
HC-12 RF Module	Long-range wireless serial communication module.	433 MHz, UART interface, up to 1 km range, 100 channels.	Handles RF communication between smart outlet and central control.

Table 3.1 Continued

Component	Description	Key Specifications	Function in System
ACS712-5A Current Sensor (2×)	Hall-effect linear current sensor module to detect and measure AC/DC .	Sensitivity ≈ 185 mV/A. Operating voltage 5 V DC, range ± 5 A.	Measures socket-level load current for Sockets A and B.
SCT-013 Current Sensor	Split-core CT for non-invasive breaker-level current sensing.	30 A AC = 1 V out, $\pm 3\%$, <1 ms response, -25 °C to $+70$ °C.	Measures total circuit current for ESP32 load analysis and protection.
5V Relay Module (2-Channel, Active High)	Electromechanical relay for socket control.	Input: 5 V DC, Max load: 250 V AC/10 A, opto-isolated input.	Switches AC socket ON/OFF based on received signals from PIC16F88
HLK-5M05 Power Module	Hi-Link AC-DC converter, compact isolated power supply.	Input: 220 V AC, Output: 5 V DC, 5 W.	Provides regulated 5 V supply to all components.

3.4.2.1 Hardware Design and Implementation

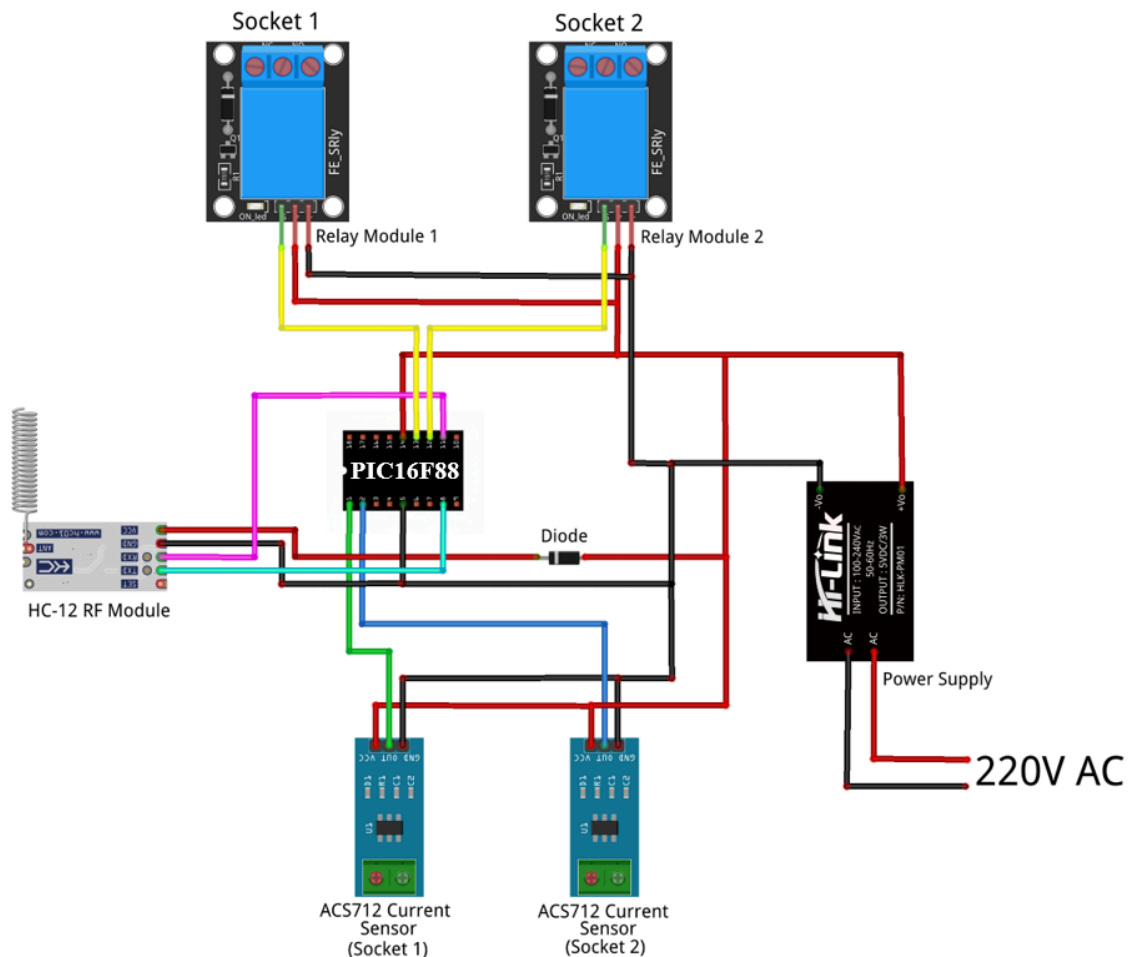


Figure 3.10 Smart Outlet Schematic Diagram

Figure 3.10 presents the schematic diagram of the smart outlet circuit developed in Sprint 1. The design integrates the PIC16F88 microcontroller with two Hall-effect current sensors, a dual-channel relay module, the HC-12 RF transceiver, and a 5 V regulated supply from the HLK-5M05 module. Each sensor monitors socket current while the microcontroller processes and transmits data wirelessly to the central control unit. The layout ensures isolation between high- and low-voltage sections, enabling safe measurement, reliable switching, and stable RF communication within the outlet module.

Table 3.2

Smart Outlet Device — Pin Mapping for PIC16F88 (Figure 3.10)

PIC16F88 Pin No.	Pin Label	Connected To	Function / Description
1	RA2	IN1 of Relay Module (Channel 1)	Digital output to drive Relay 1 (Socket A).
2	RA3	IN2 of Relay Module (Channel 2)	Digital output to drive Relay 2 (Socket B).
5	VSS	GND (HLK-5M05 OUT)	Ground reference for MCU and peripherals.
8	RB2 / RX	TXD of HC-12	UART1 Receive pin; receives serial data from HC-12 RF module.
11	RB5 / TX	RXD of HC-12	UART1 Transmit pin; sends serial data to HC-12 RF module.
14	VDD	+5 V (HLK-5M05 OUT)	Power supply input from regulated 5 V DC output.

Figure 3.11 Central Control Unit Schematic Diagram

Figure 3.11 shows the schematic diagram of the central control unit positioned at the outlet breaker. The ESP32 microcontroller functions as the main controller, receiving socket-level current data from multiple smart outlets through the HC-12 RF transceiver. A non-invasive SCT-013 current sensor measures the total circuit load and sends analog feedback to the ESP32 for evaluation. The Hi-Link HLK-5M05 module supplies isolated 5 V DC power to all low-voltage components, while a voltage-divider and filter network composed of resistors and a capacitor conditions the sensor output. This configuration enables real-time load monitoring, electrical isolation, and synchronized communication between the breaker unit and distributed outlet devices.

Table 3.3

Central Control Unit — Pin Mapping for ESP32 (Figure 3.11)

ESP32 Pin No.	Pin Label	Connected To	Function / Description
1	3.3 V Output	SCT-013 signal conditioning circuit	Provides regulated 3.3 V power to burden resistor–divider network.
14	GND	SCT-013 GND	Common ground reference for sensor and ESP32.
27	GPIO34 (ADC1_6)	Voltage divider output from SCT-013	Samples analog voltage proportional to breaker current.

Table 3.3 Continued

ESP32 Pin No.	Pin Label	Connected To	Function / Description
28	GPIO16 (U1RX)	TXD of HC-12	UART1 receive pin for RF data from Smart Outlet devices.
29	GPIO17 (U1TX)	RXD of HC-12	UART1 transmit pin for sending commands to Smart Outlet devices.

Together, the smart outlet module and the central control unit will form an integrated hardware architecture that will enable precise monitoring, responsive control, and reliable protection within the household electrical system. The smart outlet will provide socket-level sensing and switching, while the central control unit will aggregate all measurements, execute safety logic, and coordinate system-wide communication. Their combined operation will ensure that both local and remote commands are processed accurately, real-time load conditions are evaluated continuously, and protective actions are enforced whenever necessary. This coordinated design will establish a robust and scalable foundation for the overall smart outlet monitoring system, supporting safe energy management and seamless interaction with the AI-assisted dashboard. The architecture will also allow additional outlets or sensors to be integrated in future expansions without requiring major system modifications. Furthermore, the modular structure will simplify maintenance, improve reliability, and ensure that each subsystem can be upgraded independently as newer technologies become available.

3.4.2.2 Software Design and Implementation:

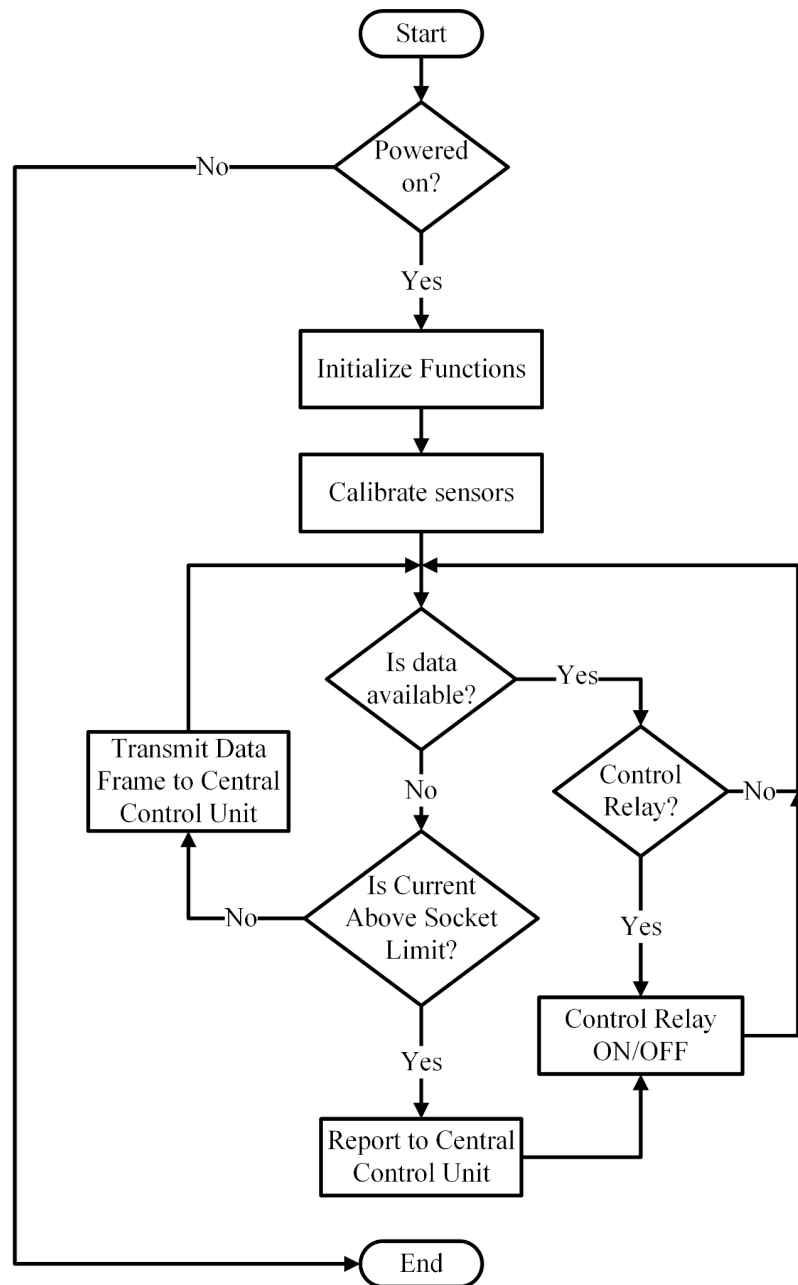


Figure 3.12 Software Flowchart of the Smart Outlet Device

Figure 3.12 illustrates the software flow of the smart outlet device that will operate at the socket level. When powered on, the microcontroller will initialize

its ports, calibrate the current sensors, and load configuration settings, including the transmission interval and socket current limits. The system will then enter a continuous loop where it will measure the current drawn by each socket, evaluate whether the readings exceed their respective limits, and enforce local overload protection if necessary. Regardless of state changes, the outlet will periodically transmit each socket's current data every 50 milliseconds through the HC-12 RF module to the central control unit, while also executing relay actions whenever commands are received. To maintain accuracy, the device will apply filtering routines to reduce noise from sensor readings and ensure that each transmission contains a structured data frame for proper identification. If repeated overload events occur within a short period, the outlet will temporarily disable the affected socket and notify the central unit for logging. This unified flow will support real-time current reporting, responsive switching, and synchronized communication across all smart outlet modules. Additionally, the software logic will be optimized to minimize processing delays and ensure that both sensing and control functions remain highly responsive during peak load periods. This approach will enhance overall system reliability and contribute to stable operation within dynamic household conditions.

Pseudocode Steps (based on Figure 3.12):

1. Start Program
2. Initialize ports, ADC, UART, and relay outputs for Socket A and Socket B.
3. Load Configuration:
 - Calibration constants for each ACS712 sensor.
 - Socket current limits (e.g., 5 A per socket).
 - Inrush and debounce time windows.
 - Set transmission interval = 50 ms.
4. Begin Main Loop

5. Read Sensor Values:
 - Acquire analog current from each ACS712 (Socket A, B).
 - Convert ADC readings to current (A) using calibration factors.
6. Check Socket Load Conditions:
 - If current > limit for T_hold → mark socket = *Overload*.
 - If current < limit and previous state = *Overload* for T_recover → reset status = *Normal*
7. Apply Socket-Level Safety Logic:
 - If *Overload* → cut socket relay OFF, record event, start cool-down timer.
 - After cool-down, auto-retry power ON up to N_retries; if still *Overload* → latch OFF.
 - Ignore short spikes < T_inrush to prevent false trips.
8. Construct and Transmit Data Frame:
 - Include:
[Owner ID, Outlet ID, Socket ID, Current Value]
 - Transmit data every 50 ms continuously, not only on state change.
 - Send frame via HC-12 to Central Control Unit and wait for ACK.
 - Retry transmission if ACK not received.
9. Receive Command Frames:
 - Decode RF commands (ON/OFF requests or reset signals).
 - Execute relay action if socket is not latched by *Overload*.
10. Log Events:
 - Store recent *Overload* events only (limited to 2 records) on EEPROM for diagnostics.
11. Return to Main Loop.

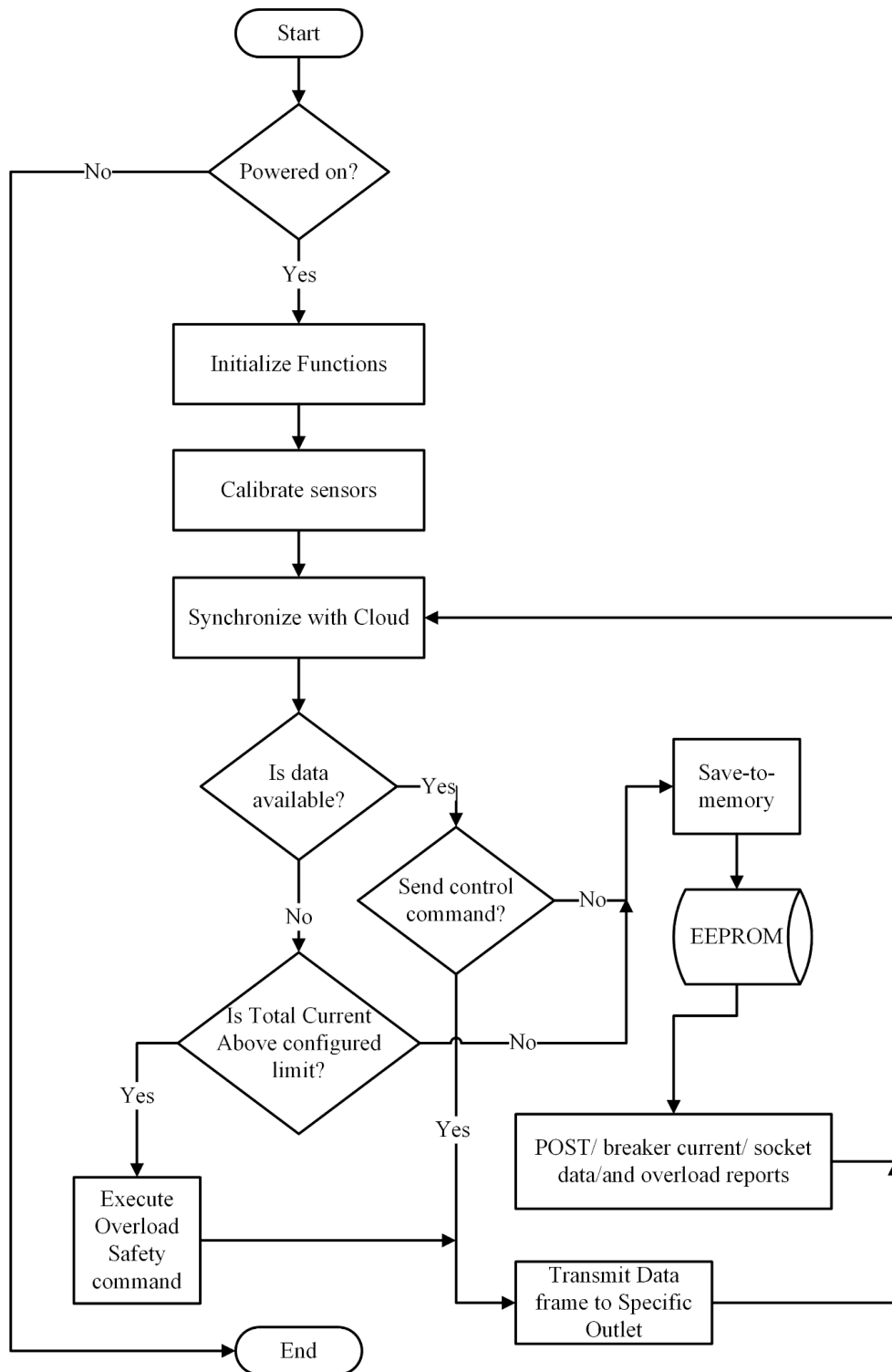


Figure 3.13. General Flowchart of the Central Control Unit (CCU)

Figure 3.13 illustrates the general software flow of the Central Control Unit (CCU) implemented using the ESP32 microcontroller. When powered on, the system initializes its ADC, UART, Wi-Fi, and EEPROM modules, then retrieves the stored breaker limit configuration before synchronizing with the cloud dashboard. During continuous operation, the CCU receives current data frames from multiple smart outlets via RF communication and measures the total breaker current using the SCT-013 sensor. It evaluates whether the total load exceeds the configured limit and, if necessary, executes overload safety commands to disconnect high-load sockets. The CCU also performs periodic synchronization with the cloud using HTTP GET and POST requests to update parameters and upload recorded overload events. Configuration changes and recent overload records are saved to the EEPROM, ensuring that essential settings persist after power loss. This flow ensures coordinated safety control, reliable data communication, and seamless integration between local monitoring and cloud-based management.

Pseudocode Steps (based on Figure 3.13):

1. Start Program
2. Initialize Components:
 - Configure UART for HC-12 RF module communication.
 - Initialize ADC for SCT-013 current sensing.
 - Initialize Wi-Fi connection for dashboard access.
 - Initialize EEPROM for configuration storage.
 - Read stored breaker limit and Owner ID from EEPROM.
 - If no stored value, load default breaker limit (e.g., 10 A).
 - Send “*CCU Active*” broadcast to all outlets.
3. Begin Main Loop
4. Receive Data Frames from Outlets:
 - Continuously read frames via HC-12.

[Owner ID, Outlet ID, Socket ID, Current Value]

- Store the latest readings in an array for each outlet/socket.
 - Send acknowledgment (ACK) to confirm successful reception.
5. Measure Breaker-Level Current:
 - Read analog signal from SCT-013 sensor.
 - Convert ADC reading to RMS current (A) using calibration constants.
 6. Compare Measured Load with Breaker Limit:
 - If total current $\geq$ configured limit $\rightarrow$ mark system = *Overload*.
 - If total current $<$ recovery threshold $\rightarrow$ mark system = *Normal*.
 7. Execute Overload Safety Logic:
 - If *Overload* = TRUE:
 - Identify socket(s) with highest reported current.
 - Send RF command

[Owner ID, Outlet ID, Socket ID, RELAY_OFF].
 - Store overload event record in EEPROM (up to 5 logs).
 - If *Normal* = TRUE:
 - Send RF command(s)

[Owner ID, Outlet ID, Socket ID, RELAY_ON]
 - for restoration.
 8. Transmit Commands to Outlets:
 - Ensure each command frame receives acknowledgment (ACK).
 - Retry transmission if ACK not received.
 9. Synchronize with Cloud Dashboard:
 - POST: Send breaker current, socket data, and overload logs to cloud.
 - GET: Retrieve updated breaker limit or control parameters.
 - If new breaker limit is received, store it to EEPROM.
 - If Wi-Fi offline $\rightarrow$ store data locally and retry when reconnected.

10. Event Logging:

- Maintain up to five latest records in EEPROM or flash.
- Each record includes timestamp, outlet/socket ID, and action taken.

11. Return to Main Loop.

3.4.3 Integration of the Smart Outlet and Central Control Unit

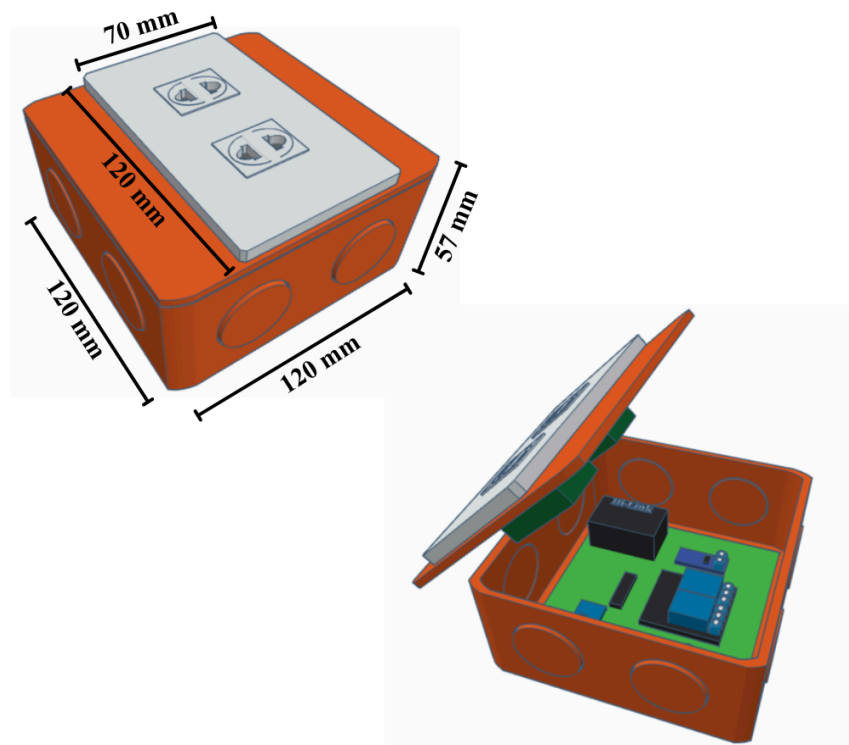


Figure 3.14 3D design of the Smart Outlet device

The smart outlet device integrates the core electronic components responsible for socket-level control and monitoring. Inside the enclosure, each socket is equipped with an ACS712 Hall-effect current sensor for current detection and a relay module for ON/OFF switching. A PIC16F88 microcontroller manages sensor readings, executes switching commands, and communicates data wirelessly via the HC-12 transceiver module to the central control unit at the breaker. The device also contains a Hi-Link HLK-5M05 power module that converts 220 V AC input to a stable 5 V DC supply for all low-voltage circuits.

Together, these internal modules enable accurate current sensing, safe load control, and reliable RF communication within the outlet.

The physical dimensions of the smart outlet measure approximately 12 cm (length) $\times$ 12 cm (width) $\times$ 6 cm (height), designed to fit standard wall boxes while housing the embedded circuitry. A 220 V AC cable entry is positioned at the side of the unit for secure wiring connection. During operation, the device continuously senses the load current at each socket; when an appliance is plugged in, the microcontroller reads the sensor output, processes the data, and transmits it to the central unit. Commands from the web dashboard or AI assistant are relayed back to toggle socket power through the relay module. This integration allows both autonomous and user-controlled operation, forming the foundation of the intelligent load monitoring and control system.

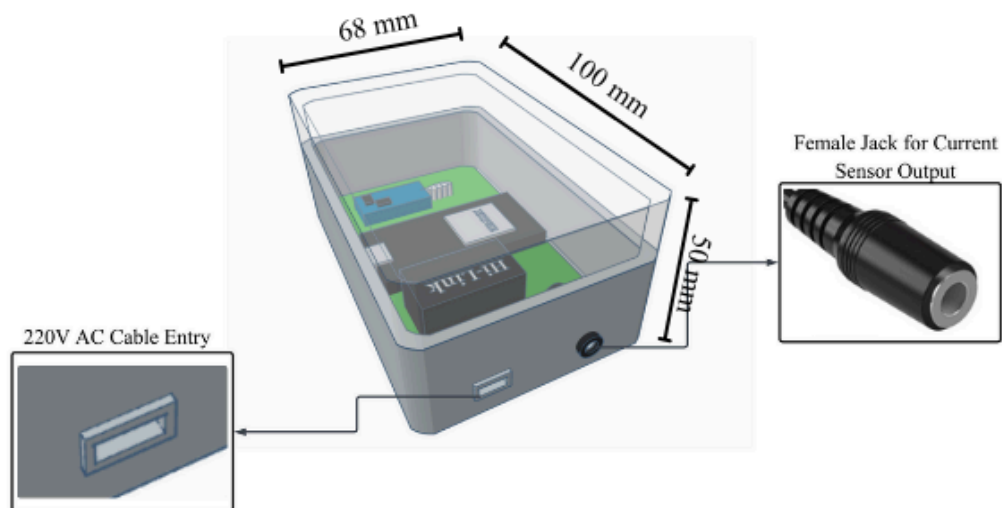


Figure 3.15. 3D Model of the Central Control Unit for Outlet Breaker

Figure 3.15 presents the 3D model of the Central Control Unit (CCU) designed for installation at the outlet breaker. The enclosure measures approximately 100 mm $\times$ 68 mm $\times$ 50 mm, providing a compact and secure

housing for the ESP32 microcontroller, HLK-5M05 isolated power supply, and HC-12 RF transceiver. A 220 V AC cable entry is positioned on one side of the case to deliver power to the internal modules, while the design ensures adequate spacing and insulation for safe high-voltage operation. The layout maintains clear separation between low-voltage control circuits and the AC input to minimize electrical interference and ensure compliance with safety standards.

The model also features a female jack for the current sensor output, allowing external connection to the SCT-013 non-invasive current transformer mounted on the breaker line. This configuration enables the CCU to monitor total circuit current while keeping all control electronics safely enclosed. Internally, the board arrangement provides organized cable routing and accessibility for maintenance or upgrades. Overall, the enclosure design combines compactness, electrical safety, and modularity, supporting reliable operation and easy integration with the smart outlet network.

3.4.3.1 Material Testing

This phase focuses on verifying the functionality, accuracy, and efficiency of each hardware component before system integration. Material testing ensures that every module, microcontrollers, sensors, transceivers, and power supplies operates within its specified performance range under controlled conditions. Each test is designed to validate that the components deliver reliable readings, consistent outputs, and stable performance aligned with their datasheet parameters. The evaluation covers both measurement accuracy and operational efficiency. Accuracy testing determines how closely each component's readings match reference instruments, while efficiency testing assesses stability and responsiveness during prolonged or repeated operation. By analyzing parameters such as voltage consistency, response time, transmission delay, and current measurement accuracy, this process establishes the reliability of all modules and

provides a validated foundation for subsequent integration and system-level testing.

Table 3.4

PIC16F88 GPIO Pins Testing

Test no.	RA2			RA3			RB2			RB5			RA0			RA1		
	V	I	P	V	I	P	V	I	P	V	I	P	V	I	P	V	I	P
1																		
2																		
3																		
4																		
.....																		
10																		

Table 3.4 presents the recorded voltage, current, and power measurements from the GPIO pins of the PIC16F88 microcontroller to assess their output accuracy and consistency. Each of the six pins, RA2, RA3, RB2, RB5, RA0, and RA1 will be tested over ten trials using the multimeter to capture real-time voltage readings while configured in a logic HIGH state. The recorded data will be statistically analyzed using a one-way Analysis of Variance (ANOVA) to determine whether significant differences existed among the mean output voltages of the tested pins. A nominal reference of 5 V will be used based on the microcontroller's typical operating voltage, with an acceptable tolerance of $\pm 5\%$ (4.75–5.25 V) as specified in the PIC16F88 datasheet. The ANOVA results provide a comparative assessment of GPIO output performance, verifying that all tested pins maintain stable voltage levels within the expected tolerance range for reliable digital operation in the smart outlet circuit.

Table 3.5

Relay operate time

Test No.	Measured Time (ms)
1	
2	
3	
4	
5	
6	.
7	.
8	
9	
10	

Table 3.5 presents the recorded operate times of the relay during multiple activation cycles. Each measurement represents the elapsed time between the control signal sent by the microcontroller and the relay's response, as recorded through the Arduino Serial Monitor. The data will be gathered from ten consecutive trials under consistent load and supply conditions to evaluate the relay's switching speed and timing stability. The recorded values will be analyze using a one-sample t-test to determine whether the average operate time is statistically lower than the 10 ms reference value specified in the relay's datasheet, following the performance guidelines of IEC 61810-7 for electromechanical relays.

Table 3.6

HC-12 RF Wireless Round-Trip Delay Test Plan (10-Byte Frame, 9600 Baud)

Distance (m)	Measured RTT (ms)	Error (ms)
2		
4		
6		
8		
10		
12		
14		

Table 3.6 shows the recorded round-trip time (RTT) values of the HC-12 RF module measured at different distances to evaluate its transmission efficiency. The test will be performed using the Arduino Serial Monitor under default settings (FU3 mode, 9600 bps, 433.4 MHz). For each distance, several RTT readings will be taken to observe how transmission speed changes as range increases. The collected data will be analyze using linear regression to determine whether there is a significant change or error in milliseconds as the communication distance increases. This method provides a simple measure of how efficiently the HC-12 maintains transmission speed across varying distances relative to its typical 31 ms reference delay stated in the datasheet.

Table 3.7

Output Voltage Test for HLK-5M05 Power Module

Test no.	Minute Duration	Measured Output (V)	Error (V)
1			
2			
3			
4			
...	...	...	...
10			

Table 3.7 records the measured output voltage of the HLK-5M05 power module at hourly intervals to evaluate long-term stability. Linear regression will be applied to determine whether the output voltage changes significantly over time. The resulting slope (β_1) indicates the rate of drift in volts per hour, while the coefficient of determination (R^2) assesses consistency. A slope value near zero and a mean voltage within the $5.00 \text{ V} \pm 5 \%$ range confirm stable performance, consistent with the specifications in the HLK-5M05 datasheet (Hi-Link, 2021).

3.4.3.2 System Testing

Table 3.8

Calibration Test for ACS712 Socket Current Sensor with PIC16F88 ADC
(Compared to Current tester)

Trial No.	Richmeter	ACS712	Error (%)
	Current tester (A)	Sensor Output (A)	
1			
2			
3			
4			
5			
...	...	...	...
10			
MAPE%			

Table 3.8 presents the calibration results of the ACS712 current sensor compared to a reference digital ammeter (Richmeter). Each trial records simultaneous readings from both instruments at increasing load levels. Linear regression analysis will be applied to determine the relationship between the

reference current (A) and the sensor output (A). The resulting regression parameters slope, intercept, and coefficient of determination (R^2) will be use to evaluate the sensor's accuracy and linearity. A slope close to 1 and an R^2 value near 1 indicate that the ACS712 provides highly accurate and consistent current measurements suitable for socket-level monitoring in the smart outlet system.

Table 3.9

RF Integration and Reliability Test for PIC16F88–HC-12 and ESP32–HC-12 Communication Links

Distance (m)	Frames Sent (n)	Frames Received (x)	PSR (%)	Mean RTT (ms)	95% Confidence Interval (PSR)
2					
4					
6					
8					
10					
12					

This table presents the results of the RF integration and reliability tests conducted for both the PIC16F88 + HC-12 and ESP32 + HC-12 communication links. The test aimed to verify the successful integration of the RF transceiver with each microcontroller and to evaluate transmission reliability under identical operating conditions. An Arduino transmitter sent 500 data frames at 9600 bps (FU3 mode) to each receiver, which validated the frames through UART communication and returned an acknowledgment (ACK). The Packet Success Rate (PSR) will be computed and its 95 % confidence interval will be estimated using the normal-approximation formula to quantify link reliability. Both setups will be considered reliable when the lower bound of the confidence interval will be at least 0.95, indicating consistent data reception and acknowledgment between the microcontrollers and the HC-12 modules. This methodology follows the wireless communication reliability evaluation approaches proposed by Tang et al. (2023) and Yogarayan et al. (2023), ensuring that the same analytical framework is applied across both microcontroller platforms.

Table 3.10

GPIO Control Response Test Between PIC16F88 and Relay Module

Trial No.	GPIO Signal	Relay State	Output (1 = Triggered, 0 = No Response)
1			
2			

Table 3.10 Continued

Trial No.	GPIO Signal	Relay State	Output (1 = Triggered, 0 = No Response)
3			
4			
...	...	...	...
10			

This table presents the results of the GPIO control response test conducted to verify the integration between the PIC16F88 microcontroller and the relay module. The test evaluated whether the relay consistently responded to digital HIGH and LOW signals generated by the PIC's GPIO pin. Each trial recorded the relay's switching outcome as a binary response "1" for successful activation or deactivation, and "0" for failure to respond. A logistic regression model will be applied to determine how strongly the GPIO signal influenced the probability of correct relay switching. A statistically significant coefficient ($p < 0.05$) and high odds ratio confirm that the relay module reliably responded to the control logic of the PIC16F88, consistent with empirical reliability evaluation frameworks used in embedded systems (Amaral et al., 2025; AN6223, STMicroelectronics, 2025).

Table 3.11

Calibration Test for SCT-013 Breaker-Level Current Sensor with ESP32 ADC
(Compared to Clamp Ammeter)

Trial No.	Clamp Ammeter	<i>SCT-013 Sensor</i>	Error (%)
	Reading (A)	<i>Output (A)</i>	
1			
2			
3			
4			
5			
...	...	...	...
10			
MAPE%			

Table 3.11 presents the calibration results of the SCT-013 current sensor integrated with the ESP32 ADC, compared against a reference clamp ammeter. Each trial records simultaneous readings from both instruments at increasing load levels to verify the sensor's precision and stability. Linear regression analysis will

be applied to determine the relationship between the reference current (A) and the sensor output (A). The resulting regression parameters slope, intercept, and coefficient of determination (R^2) will be used to evaluate the sensor's accuracy and linearity.

A slope close to 1 and an R^2 value near 1 indicate strong correlation and high measurement reliability, while a $\text{MAPE} \leq 5\%$ confirms that the SCT-013 sensor provides acceptable calibration accuracy for breaker-level current monitoring in the system.

3.4.3.3 Verification and Validation

Table 3.12

Verification and Validation Summary for Sprint 1

Component (Table Ref.)	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
PIC16F88 GPIO Pins (Table 3.4)	One-Way ANOVA	$p \geq 0.05 \rightarrow$ no significant difference between pin outputs	F-statis tic, p-value	ISO 3534-1 (Statistical Methods); Voltage tolerance ± 5 % of 5 V	Output Uniformity and Consistency

Table 3.12 Continued

Component (Table Ref.)	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
Relay Operate Time (Table 3.5)	One-Sample t-Test	$p \geq 0.05$ → meets datasheet timing spec; $p <$ 0.05 → deviation	t-value, p-value	Datasheet (operate $\approx$ 10–15 ms, release $\approx$ 5–10 ms); Cohen (1992)	Switching Accuracy and Reliability
HC-12 RF Wireless RTT (Table 3.6)	Linear Regression	$R^2 \rightarrow 1$; $MAPE \leq$ 5 %	R^2 , MAPE (%)	$R^2 \geq 0.95 =$ strong correlation (Schober et al., 2018); $MAPE \leq 5 \%$ (IEEE 1057-2017)	Transmissio n Accuracy and Stability
HLK-5M05 Power Module Voltage (Table 3.7)	Linear Regression	$R^2 \rightarrow 1$; $MAPE \leq$ 5 %	R^2 , MAPE (%)	$R^2 \geq 0.95$; $MAPE \leq 5 \%$ (IEEE 1057-2017)	Voltage Regulation Accuracy

Table 3.12 Continued

Component (Table Ref.)	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
ACS712 Socket Current Sensor (Table 3.8)	Linear Regression	$R^2 \geq 0.98$; $MAPE \leq 5\%$	R^2 , MAPE (%)	IEEE 1057-2017 (sensor error limit)	Measureme nt Accuracy and Reliability
RF Integration and Communicatio n Links (Table 3.9)	Proportion Reliability (95 % Confidence Interval)	Lower bound of $CI \geq 0.95$ → reliable link	PSR (%), 95 % Confiden ce Interval	Tang et al. (2023); Yogarayan et al. (2023) — $CI \geq 0.95 =$ reliable communicati on	Transmissi on Reliability and Link Consistenc y
GPIO Control Response (PIC16F88 ↔ Relay) (Table 3.10)	Logistic Regression	$AUC \geq 0.9$ → outstandin g; Nagelkerk $e R^2 \rightarrow 1$	AUC, Nagelker ke R^2	Hosmer et al. (2013) — $AUC \geq 0.9 =$ excellent fit	Control Accuracy and Integration Reliability

Table 3.12 Continued

Component (Table Ref.)	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
SCT-013 Breaker-Level Sensor (Table 3.11)	Linear Regression	$R^2 \geq 0.95$; $MAPE \leq$ 5 %	R^2 , MAPE (%)	IEEE 1057-2017 (sensor error limit)	Breaker-Le vel Accuracy and Reliability

The verification and validation results summarized in Table 3.11 confirm that all material-testing procedures in Sprint 1 satisfy the system's performance requirements for measurement accuracy, reliability, and hardware consistency. Statistical analyses demonstrate that the PIC16F88 GPIO pins maintain uniform output characteristics, the relay operates within datasheet timing limits, and all sensors including ACS712 and SCT-013 exhibit high correlation and low error percentages relative to reference instruments. Wireless communication through the HC-12 transceivers achieved reliability above the 95 % confidence threshold, while logistic regression confirmed dependable GPIO-to-relay control response. Collectively, these findings verify that the core electronic modules developed in Sprint 1 perform accurately and reliably, meeting Objective 1 of the study.

3.4.3.4 Data Analysis and Procedure

A. One-Way ANOVA

The one-way ANOVA (Analysis of Variance) test was used to determine whether the mean output characteristics of the PIC16F88 microcontroller's GPIO pins differ significantly from one another under identical supply conditions. This

analysis verifies the uniformity of the device's voltage, current, and power delivery among its digital-output pins. All computations were performed with an online ANOVA calculator to ensure precision and reproducibility.

$$F = \frac{MS_B}{MS_W}$$

$$MS_B = \frac{SS_B}{k - 1}, \quad MS_W = \frac{SS_W}{N - k} \quad (3.1)$$

Where:

- F = F-statistic (ratio of between-group to within-group variance)
- SS_b = sum of squares between groups
- MS_B = mean square between groups
-
- k = number of groups (GPIO pins tested)
- SS_W = sum of squares within groups
- MS_W = mean square within groups
- N = total observations

Interpretation Criteria:

- $p \geq 0.05 \rightarrow$ no significant difference among group means; GPIO outputs are statistically uniform (ISO 3534-1).
- $p < 0.05 \rightarrow$ significant difference; at least one pin shows output deviation.
- Acceptable electrical tolerance: $\pm 5 \%$ from the nominal 5 V reference.

A.1 PIC16F88 GPIO Pins Testing (Table 3.4)

The test compared the mean output voltage, current, and power measured across all GPIO pins of the PIC16F88 to verify uniform signal performance.

Procedure:

1. Power the PIC16F88 at 5 V DC and configure all GPIO pins as outputs under identical logic conditions.
2. Measure each pin's voltage, current, and power using a calibrated multimeter for multiple trials.
3. Enter grouped readings by pin number into an online one-way ANOVA tool.
4. Compute the F-statistic and p-value to evaluate whether mean differences are statistically significant.
5. Interpret results at a 95 % confidence level: $p \geq 0.05$ indicates uniform pin performance; $p < 0.05$ suggests non-uniform output requiring calibration.

Consistent p-values above 0.05 and voltage readings within $\pm 5\%$ of 5.00 V confirm that all PIC16F88 GPIO pins operate uniformly and deliver stable outputs suitable for reliable peripheral control.

B. One-Sample t-Test

The one-sample t-test was used to determine whether the relay module's average operate time differs significantly from the benchmark value stated in its datasheet. This statistical tool compares a sample mean ($\bar{X}$) with a hypothesized population mean (μ_0) to verify compliance with design specifications.

Calculations were performed using an online t-test calculator to ensure precision and reproducibility.

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \quad (3.2)$$

Where:

t = t-statistic (test value)

$\bar{X}$ = sample mean

μ_0 = benchmark mean (datasheet value)

s = sample standard deviation

n = number of observations

Interpretation Criteria:

- $p \geq 0.05 \rightarrow$ no significant difference; the relay meets its specified operate time.
- $p < 0.05 \rightarrow$ significant difference; measured times deviate from specification.
- Benchmark reference: datasheet operate time $\approx 10\text{--}15$ ms, release time $\approx 5\text{--}10$ ms.

B.1 Relay Operate Time Test (Table 3.5)

This analysis verifies whether the relay's switching performance conforms to manufacturer timing specifications when driven by the PIC16F88 control signal.

Procedure:

1. Connect the relay module to the PIC16F88 output pin and supply 5 V DC.
2. Trigger the relay ten times while measuring each operate and release delay using an oscilloscope or serial-timed logging.
3. Record all delay readings (ms) and compute the sample mean ($\bar{X}$) and standard deviation (s).
4. Compare the sample mean to the datasheet benchmark (μ_0) using the one-sample t-test formula.
5. Use an online t-test calculator to obtain the t-statistic and p-value.
6. Interpret results at a 95 % confidence level: $p \geq 0.05$ confirms the relay's operate and release times are within the acceptable range; $p < 0.05$ indicates deviation requiring recalibration or component replacement.

A non-significant result ($p \geq 0.05$) demonstrates that the relay module's dynamic response aligns with manufacturer specifications, confirming its reliability for system switching operations.

C. Linear Regression

Linear regression was used to analyze the linear relationship between measured and reference values for all analog-based components in the system. This method quantifies how closely a device's measured output corresponds to its reference or theoretical value, ensuring that each sensor or module performs within its specified accuracy range. Calculations were performed using an online regression calculator to derive the regression equation, coefficient of determination (R^2), and Mean Absolute Percentage Error (MAPE).

$$\begin{aligned}
\hat{Y} &= b_0 + b_1 X \\
R^2 &= 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2} \\
MAPE &= \frac{100}{n} \sum \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right|
\end{aligned} \tag{3.3}$$

Where:

$\hat{Y}$ = predicted value of the dependent variable

b_0 = intercept

b_1 = slope (change in Y for every 1-unit change in X)

X = reference reading independent variable

Y_i = observed value of Y

$\bar{Y}$ = mean of observed Y

Interpretation Criteria:

- R^2 close to 1 $\rightarrow$ strong linear relationship between X and Y (Schober et al., 2018).
- $MAPE \leq 5\%$ $\rightarrow$ acceptable measurement error for electronic instrumentation (IEEE 1057-2017).
- $b_1 \approx 1$ $\rightarrow$ system readings increase proportionally with reference values, indicating proper calibration.

C.1 HC-12 RF Wireless Round-Trip Delay Test (Table 3.6)

This analysis verified the consistency of the HC-12 wireless transceiver by correlating transmission distance with measured round-trip delay (RTT).

Procedure:

1. Configure two HC-12 modules at 9600 baud, sending 10-byte frames across distances of 1–50 m.
2. Record RTT values for each trial using the serial monitor.
3. Compare measured RTTs with the expected reference (≈ 31 ms).
4. Compute R^2 and MAPE to evaluate transmission stability.
5. Results with $R^2 \geq 0.95$ and $\text{MAPE} \leq 5\%$ confirm consistent wireless performance.

C.2 HLK-5M05 Power Module Output Voltage Test (Table 3.7)

This test assessed the voltage regulation stability of the HLK-5M05 AC-to-DC converter.

Procedure:

1. Power the module with 220 V AC and measure output voltage at fixed intervals for 10 trials.
2. Compare each reading to the $5.00\text{ V} \pm 5\%$ reference.
3. Calculate Error % for each trial and determine R^2 and MAPE.
4. Compliance is verified when $\text{MAPE} \leq 5\%$ and $R^2 \rightarrow 1$, confirming stable DC output suitable for microcontroller supply.

C.3 ACS712 Socket-Level Current Sensor Calibration (Table 3.8)

This test evaluated the calibration accuracy of the ACS712 Hall-effect sensor connected to the PIC16F88 ADC.

Procedure:

1. Apply varying current loads (0.5–5 A) and record both ACS712 output and reference ammeter readings.
2. Compute Error % and analyze correlation via regression.
3. $R^2 \geq 0.98$ and $\text{MAPE} \leq 5\%$ indicate accurate current measurement suitable for per-socket load sensing.

C.4 SCT-013 Breaker-Level Sensor Calibration (Table 3.11)

This analysis validated breaker-level current readings of the SCT-013 sensor processed by the ESP32 ADC.

Procedure:

1. Place SCT-013 and a reference clamp ammeter on the same line.
2. Increase load incrementally, recording both sensor and reference readings.
3. Compute Error % for each sample and evaluate R^2 and MAPE.
4. $R^2 \geq 0.95$ and $\text{MAPE} \leq 5\%$ confirm accurate current detection at the circuit-breaker level.

D. Logistic Regression

Logistic regression was employed to evaluate the switching reliability between the PIC16F88 microcontroller and the relay module, ensuring that control signals from the GPIO pins consistently activate or deactivate the relay coil as intended. This method models the probability of a binary outcome—successful

switching (1) or failure (0)—based on the logic level of the control signal. Calculations were performed using an online logistic-regression calculator to obtain regression coefficients (b_0 , b_1), classification accuracy, and model-fit indicators (Nagelkerke R^2 and AUC).

$$P(Y = 1) = \frac{1}{1 + e^{-(b_0 + b_1 X)}} \quad (3.4)$$

Where:

p = probability that the dependent variable = 1

b_0, b_1 = regression coefficients determined from observed data

X = control input condition from PIC16F88 (GPIO state = High or Low)

e = Euler's number (≈ 2.71828)

Y = relay output (1 = ON, 0 = OFF)

Interpretation Criteria:

- $AUC \geq 0.9 \rightarrow$ Outstanding classification performance (Hosmer et al., 2013)
- Nagelkerke $R^2 \rightarrow 1 \rightarrow$ High reliability of control prediction
- A model achieving both criteria indicates strong one-to-one response between the microcontroller signal and relay activation.

D.1 GPIO Control Response Test Between PIC16F88 and Relay Module (Table 3.10)

This test validated whether each logic-high signal from the PIC16F88 GPIO reliably energized the relay coil and each logic-low signal properly released it.

Procedure:

1. Connect the relay module to the PIC16F88 GPIO output and power the circuit at 5 V DC.
2. Apply alternating logic-high and logic-low signals from the microcontroller over 20 trials.
3. Record the relay output state (1 = energized, 0 = de-energized) for each control command.
4. Classify outcomes in a binary table (Control Signal, Relay Output) and analyze using an online logistic-regression calculator.
5. Obtain coefficients (b_0, b_1), Nagelkerke R^2 , and AUC values.
6. Interpret results: $AUC \geq 0.9$ and R^2 close to 1 indicate that relay responses are highly predictable and synchronized with GPIO control logic.

High AUC and R^2 values confirm that the PIC16F88 microcontroller consistently triggers the relay as designed, demonstrating reliable hardware integration for socket-level control operations.

E. Proportion Reliability Analysis

This analysis evaluates the wireless communication reliability between the PIC16F88 + HC-12 and ESP32 + HC-12 modules by calculating the Packet Success Rate (PSR) and its 95 % confidence interval. The test quantifies the probability that transmitted frames are correctly received and acknowledged, thereby verifying link stability under normal operating conditions.

$$\hat{p} = \frac{x}{n}, \quad CI = \hat{p} \pm z \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \quad (3.5)$$

Where:

- $\hat{p}$ = PSR (proportion of successful receptions)
- x = frames received
- n = frames sent
- z = 1.96 for 95 % confidence

Interpretation Criteria:

- Communication is considered reliable when the lower bound of the confidence interval ≥ 0.95 .
- This criterion ensures ≥ 95 % of transmitted frames are successfully acknowledged.

Procedure:

1. Configure each microcontroller pair (PIC16F88 + HC-12 and ESP32 + HC-12) at 9600 bps in FU3 mode.
2. Transmit 500 frames at each distance increment (2–12 m).
3. Record frames sent (n), frames received (x), and compute PSR = x / n.
4. Calculate the 95 % confidence interval using the normal-approximation formula.

5. Compare results: if the lower bound ≥ 0.95 , the RF link is deemed reliable.

Following the methodology of Tang et al. (2023) and Yogarayan et al. (2023), this analysis validates that both microcontroller + HC-12 configurations maintain consistent and reliable wireless communication within the tested range.

3.5 Sprint 2: Load Monitoring & AI-Assisted Dashboard Application

3.5.1 Planning Phase

Sprint 2 addresses Objective 2: To develop a web dashboard that monitors circuit load, alerts users of safety actions, and integrates an AI assistant for control, inquiry, and advisories. While Sprint 1 focused on per-socket current detection and bidirectional RF communication between the smart outlets and the central unit, Sprint 2 extends the system to a web application that collects, stores, alerts users, and visualizes data from the central unit. This sprint integrates the Generative AI assistant for conversational user interaction and advisory.

This sprint aims to design and implement a web-based user application featuring an AI-assisted chatbot dashboard capable of real-time load monitoring, user alerts, and intelligent interaction with the smart outlet system. This phase centers on the development of the dashboard interface, including both user interface and user experience design, and the integration of real-time data transmitted from the central control unit (ESP32) via Wi-Fi. It also encompasses the creation of a structured database for logging load measurements, safety-logic events, and user account information, alongside the incorporation of a Gemini-powered Generative AI chatbot assistant for contextual feedback and command interpretation. The dashboard visualizes breaker-level and per-socket loads with dynamic alerts that reflect the system's safety-logic status generated by the ESP32. The scope of Sprint 2 excludes any hardware modification to the central monitoring unit or smart outlet modules, as well as long-term cloud

deployment or multi-household scalability testing. Testing activities in this sprint include functional testing to validate dashboard data presentation, usability testing to assess chatbot responsiveness and interaction flow, system testing to verify database logging and data synchronization, and verification and validation to compare measured dashboard accuracy, chatbot response time, and data integrity against the established Sprint 2 acceptance criteria.

3.5.2 Design Phase

Software Components

The system's software stack integrates front-end, back-end, and cloud technologies to ensure real-time monitoring, secure data handling, and intelligent user interaction.

Table 3.13

Software Components and Platforms Utilized in the System.

Component	Description / Function
HTML / CSS / JavaScript	Used to design the dashboard interface for real-time load visualization, safety status display, and user controls.
Python Backend (Django or Flask Framework)	Manages data processing, API communication with the central unit, and database synchronization.

Table 3.13 Continued

Component	Description / Function
SQLite3 and PostgreSQL (via Supabase)	SQLite3 handles local data logging, while PostgreSQL in Supabase stores synchronized cloud records for remote access.
Google AI Studio (Generative AI Integration)	Integrates the AI assistant for conversational queries, data retrieval, and intelligent user feedback.

Software Design and Implementation

The software design for Sprint 2 establishes the full workflow connecting the central unit's data to the AI-assisted web application. The system is organized into four layers: front end, back end, database, and AI integration. Data transmitted from the central unit are received through Python-based API routes, processed by the backend, and logged into a local SQLite3 database. These records are synchronized with a PostgreSQL cloud server in Supabase to ensure continuous data availability and remote access.

The web interface, developed using HTML, CSS, and JavaScript, provides real-time visualization of total load, specific socket loads, and user alerts. The backend manages authentication and database communication, while Google AI Studio integrates the Generative AI assistant within the dashboard. This assistant enables natural-language interaction, allowing users to check current load, alerts users for safety actions, or issue control commands all within a unified, responsive, and intelligent interface.

Dashboard Implementation

The dashboard implementation focuses on creating a web-based interface that enables real-time monitoring and control of the smart outlet system. It presents continuous readings of total circuit load and individual socket activity from the central control unit, giving users immediate visibility into household power usage. The dashboard will also display safety-related events such as socket activations, disconnections, overload responses, and breaker status changes, along with notifications for load updates and system alerts. Users will be able to manually toggle sockets ON or OFF and review visual summaries of historical energy consumption to support informed decision-making. To enhance usability, the interface integrates a Generative AI assistant that allows conversational queries, provides data summaries, executes socket control commands, and offers context-aware energy-use recommendations.

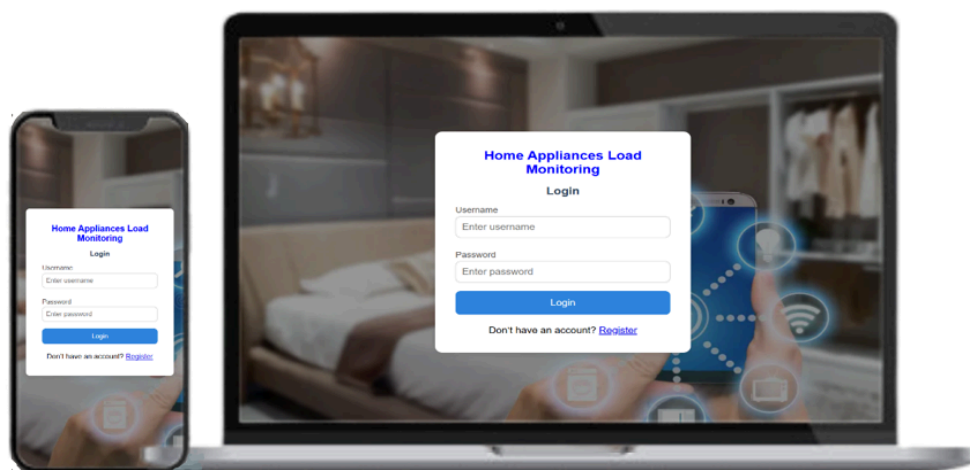


Figure 3.16 Login Page

Figure 3.16 shows the login page of the web-based AI-assisted load monitoring dashboard, presented in both mobile and desktop views. The interface provides fields for entering a username and password to securely access the

system, along with a registration link for new users who do not yet have an account. During registration, users are required to provide a username, password, country, and barangay or village information for system identification. This interface serves as the entry point to the AI-assisted dashboard, ensuring secure and authenticated user access.

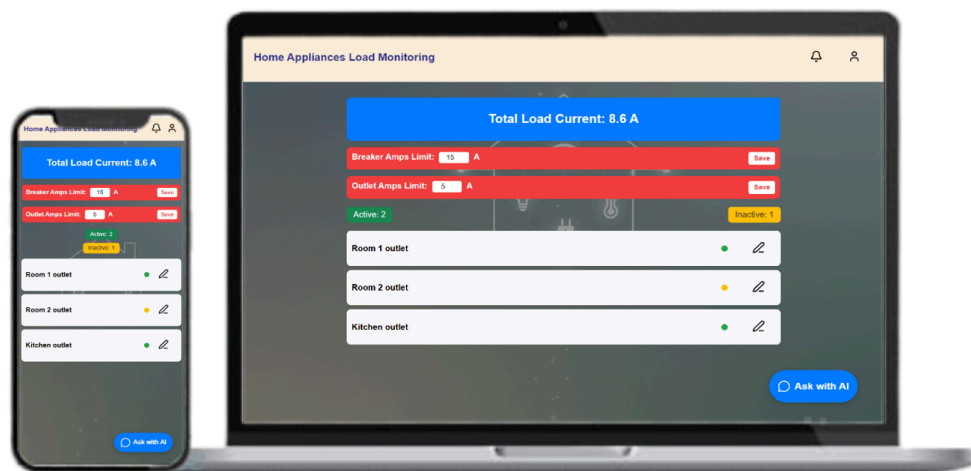


Figure 3.17 Home Page

Figure 3.17 shows the homepage of the web-based AI-assisted load monitoring dashboard, which serves as the main interface for both mobile and desktop users. The layout reflects the team's effort to design a responsive and user-friendly interface that adapts to any device. The navigation header contains the system title, a Notification icon for safety alerts such as breaker trips or socket disconnections, and a Profile icon for account access or logout. Below it, the dashboard displays the Total Load Current, the Breaker Amps Limit panel, per Outlet Amps Panel and indicators showing the number of Active and Inactive sockets. Each outlet Room 1, Room 2, and Kitchen is shown as a status card with a color-coded activity dot and an edit icon for configuration. A floating "Ask with AI" button at the bottom-right connects users to the Generative AI assistant, enabling control commands, inquiries, and real-time energy advisories. Overall,

the homepage combines real-time monitoring, safety alerts, and AI-driven interaction, highlighting the system's intelligent and accessible design for effective energy management.

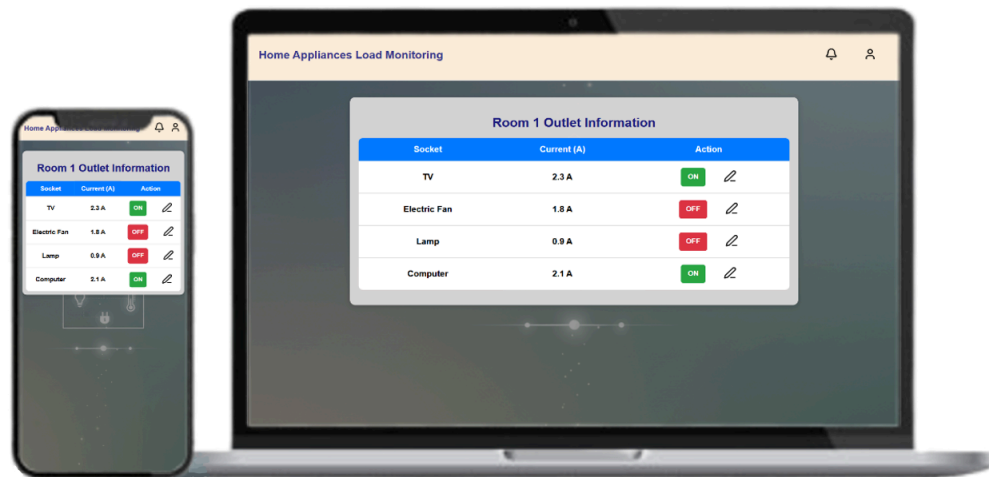


Figure 3.18 Specific Outlet Page

In this figure, the Specific Outlet Page presents the manual control interface for a selected outlet in both mobile and desktop views. It follows the same theme and visual hierarchy as the main dashboard to maintain design consistency and improve user familiarity. The interface displays a structured table listing each socket's assigned appliance name, its corresponding real-time current draw, and ON/OFF control buttons, complemented by an edit icon for renaming or reassigning connected devices.

Beyond manual toggling, the page provides visual feedback whenever a switching action is executed, ensuring users can verify socket status changes instantly. Each update is synchronized with the system database and reflected across all connected devices in real time. This interface allows users to manually manage appliances, monitor individual energy consumption patterns, and maintain accurate socket labeling. By combining usability, feedback responsiveness, and

multi-platform accessibility, the Specific Outlet Page reinforces the system's goal of intelligent, user-driven energy management and transparent device control within a smart-home environment.

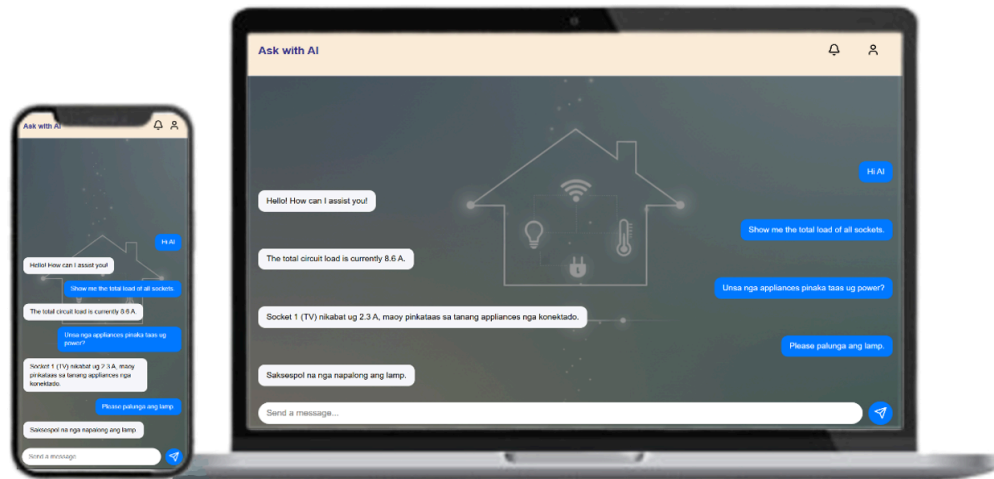


Figure 3.19 Conversation Page with AI Assistant

In this figure, the Conversation Page displays the interactive interface between the user and the system's Generative AI assistant, shown in both mobile and desktop views. Within the page, the chat area presents an example conversation where the AI responds to user commands and inquiries, such as requesting total circuit load, identifying appliances with the highest power consumption using natural-language, or performing control actions like turning off a device in a using natural-language . User messages appear on the right side in blue, while AI responses are aligned on the left in white for visual clarity. At the bottom of the page is a message input bar and a send button that allows users to prompt or interact naturally with the AI assistant. Overall, this page demonstrates the system's capability for real-time, conversational interaction, enabling users to monitor, inquire, and control household devices through an intuitive and language-based interface.

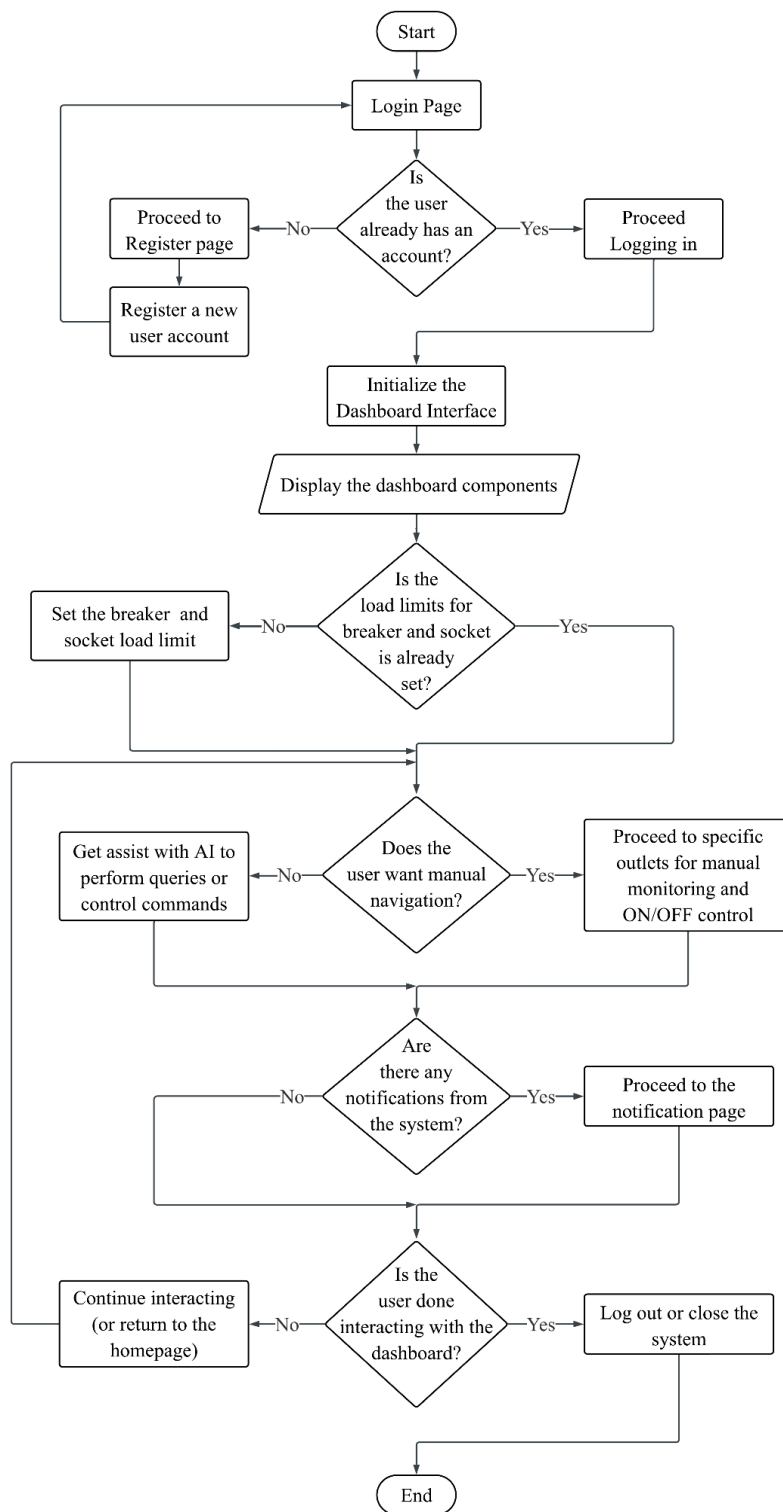


Figure 3.20 System Dashboard Flowchart

Figure 3.20 illustrates the overall flow of the system's Dashboard operation, showing how users interact with the application from login to logout. After a new user successfully registers, the system automatically redirects them to the Login Page to verify their credentials before accessing the main dashboard, ensuring security and authenticated access. The flowchart highlights the user's journey through different functionalities, including load monitoring, outlet navigation, notification access, and AI-assisted control.

Pseudocode Steps (based on Figure 3.20):

1. Start
2. Display Login Page
 - If the user already has an account → proceed to login.
 - If the user is new → register a new user account then redirect to login again.
3. Initialize the Dashboard Interface.
4. Display the dashboard components
5. Set Load Limits
 - If the load limit for breaker and socket is already set → If yes, proceed to the next step.
 - If not → set the breaker and outlet load limit.
6. Select Interaction Mode
 - Manual navigation → proceed to specific outlets for manual monitoring and ON/OFF control.
 - Assist with AI → open the “Ask with AI” page to perform queries or control commands.
7. Check Notifications
 - Are there any notifications from the system? → If yes, proceed to the notification page.
 - If not → continue.
8. User Interaction Check

- Is the user done interacting with the dashboard? → If yes, proceed to log out or close the system.
- If not → continue interacting (or return to the homepage).

9. End Process

Database Implementation

The database implementation is designed to record, organize, and manage all essential system data that supports user account authentication, dashboard access, and personalized load monitoring. It functions as the backbone of the system's data management layer, handling both real-time and historical information transmitted from the central control unit. The database stores continuous breaker-level load measurements and per-socket current readings, as well as safety-logic event logs that include socket disconnections, activations, overload detections, and breaker status updates. These records are timestamped and categorized to ensure that data can be efficiently retrieved, visualized, and analyzed through the dashboard interface.

To enable responsive and intelligent data utilization, the structure of the database will be composed of multiple tables linked by relational keys, covering user profiles, device information, socket status, load readings, and safety events. This relational design ensures scalability, integrity, and traceability across all data points in the system. The database will also support Generative AI integration, allowing the large language model (LLM) assistant to query stored records, generate summarized insights, and provide personalized advisory feedback to users. Through structured query execution, the AI assistant will be able to retrieve recent measurements, identify peak load periods, and respond to user prompts. Furthermore, the database ensures cloud synchronization between the local database and the cloud server, maintaining data availability and consistency even after periods of offline operation.

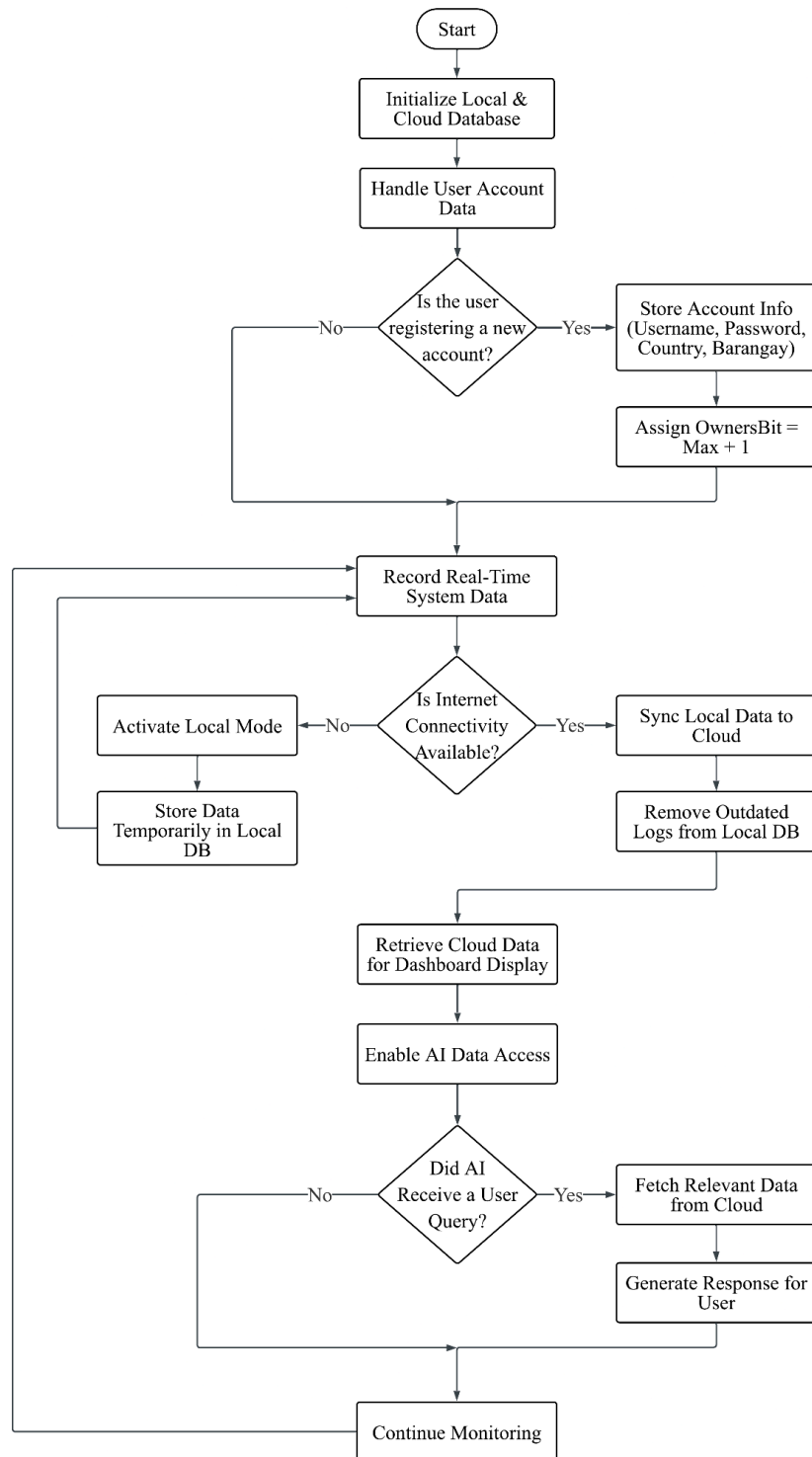


Figure 3.21 Database Synchronization Flowchart

Figure 3.21 illustrates how the system will manage data processing and synchronization between the local database and the cloud database. The flowchart shows how each user will interact with their data—from account registration and real-time data recording to the synchronization of records with the cloud database for dashboard visualization and Generative AI access. This process will ensure consistent data flow, accessibility, and reliable integration between local and cloud storage. Through this mechanism, both the dashboard and AI assistant will always operate using the most recent and accurate data available.

Pseudocode Steps (based on Figure 3.21):

1. Start
2. Initialize the local database and the cloud database
3. Handle user account data
 - If the user register a new account → store account details (username, password hash, country, and barangay/village) in a cloud database. → Assign OwnersBit max + 1 and record it in the cloud database
 - If not → continue
4. Record real-time system data (breaker load readings, socket current data and safety-logic events from the central control unit)
 - Is internet connectivity available? → If yes, synchronized all the data from local storage to cloud storage → Remove outdated logs beyond the data retention period on local DB
 - If offline → activate local-only mode and store all data temporarily
5. Retrieve cloud data for dashboard display
6. Enable AI data access from cloud database
 - Is the AI assistant receiving a user query? → If yes, retrieve stored data from the cloud database → continue monitoring
 - If no → continue monitoring

Generative AI Implementation

The Generative AI implementation integrates a cloud-based AI assistant developed through Google AI Studio into the dashboard to enable intelligent and conversational user interaction. It processes user queries and control commands through natural language understanding (NLU), allowing intuitive management of outlets and simplified access to system information. The AI assistant retrieves relevant data from the cloud database in response to user prompts and generates context-aware responses, explanations, and recommendations such as describing load status, advising on efficient energy usage, or confirming socket control actions. The assistant operates on a large language model (LLM) framework referenced from Google's Gemini architecture, enabling it to interpret complex instructions and maintain contextual continuity across interactions.

To ensure traceability and improvement of AI performance, the system will maintain detailed event logs of all AI interactions, including user-issued commands, generated responses, and decision outcomes. These logs will be reviewed for monitoring, validation, and further model tuning to ensure that the assistant continues to respond accurately as the dataset grows. The AI module will be synchronized with the cloud database to ensure consistent query processing and reliable access to the latest sensor readings and safety events. Continuous cloud connectivity will sustain uninterrupted AI operation and prevent data desynchronization, ensuring that user requests are always met with accurate and up-to-date information. Additionally, the logging and synchronization framework will support future enhancements, such as adaptive prompting or model refinement based on user behavior patterns. The structured integration of AI logic and cloud data flow will further strengthen system reliability and ensure that conversational interactions remain coherent, precise, and aligned with real-time household conditions.

A. Development and Configuration of the Generative AI Model via Google AI Studio

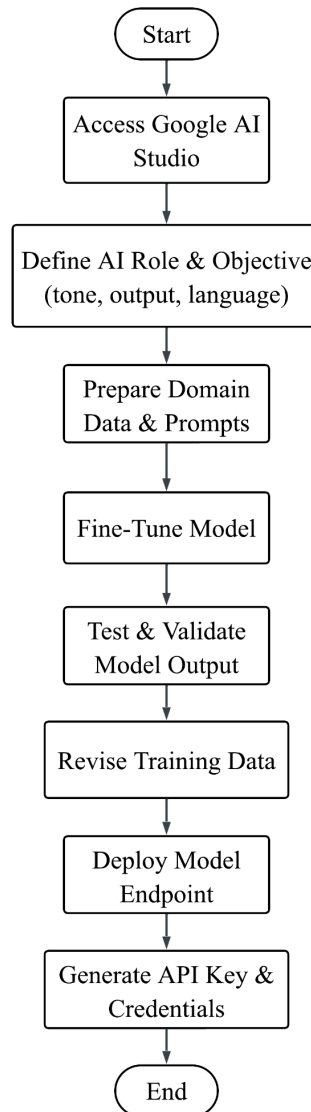


Figure 3.22 Development of Generative AI in Google AI Studio Flowchart

In Figure 3.22, the flowchart presents the one-way process showing how the Generative AI model is developed and configured in Google AI Studio before integration into the system, ensuring the feasibility and reliability of the AI assistant in understanding domain-specific commands and generating accurate,

context-aware responses for the smart outlet monitoring environment. The process begins by defining the model's role and objective, which includes setting its tone, response style, and language parameters. Domain-specific prompts and datasets are then prepared and used for fine-tuning to align the model's behavior with the project's functional requirements. After fine-tuning, the model undergoes testing and validation to evaluate its accuracy, coherence, and responsiveness. The training data may be revised iteratively until the desired performance metrics are achieved. Finally, the model is deployed through an endpoint, and an API key with authentication credentials is generated to enable secure communication and integration with the web-based dashboard.

Pseudocode Steps (based on Figure 3.22):

1. Start.
2. Access Google AI Studio platform.
3. Define AI Role and Objective (Specify tone formal or concise, text response or JSON command, and supported languages English and Filipino).
4. Prepare Domain Data and Prompts.
5. Fine-Tune the Model (Upload few-shot examples to teach the model how to respond to domain-specific prompts).
6. Test and Validate the AI Model (Send test prompts and test responses).
7. Revise training data until expected output behavior is achieved.
8. Deploy Model Endpoint (Save and publish the customized Gemini API endpoint).
9. Generate API key and access credentials for system integration.
10. End Process

B. Integration of Generative AI to our System Dashboard

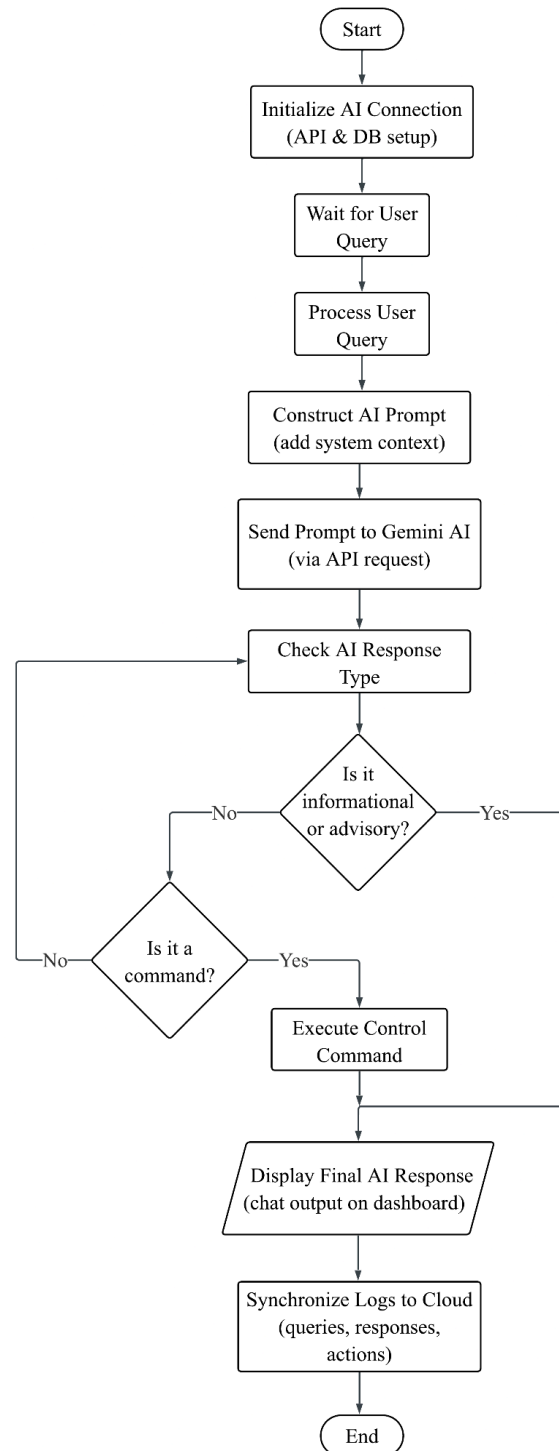


Figure 3.23 Integration of Generative AI to our System Dashboard

In Figure 3.23, the flowchart illustrates how the Generative AI component is integrated into the system's dashboard. It shows the interaction sequence from establishing the AI connection, processing user queries, and sending prompts to the Gemini model, to interpreting responses and executing control commands. The process ensures that the AI assistant seamlessly manages both informational and command-based queries while synchronizing all responses and actions with the cloud database for real-time monitoring and record management. This ensures the feasibility and reliability of AI-driven interaction within the smart outlet monitoring environment, enhancing user accessibility and intelligent system performance.

Pseudocode Steps (based on Figure 3.23):

1. Start
2. Initialize AI Connection
 - Import Gemini API credentials from Google AI Studio.
 - Connect the backend server (Flask/FastAPI) with the deployed model endpoint.
 - Link the cloud database (Supabase/PostgreSQL) for real-time data retrieval.
3. Wait for User Query
4. Process User Query
 - Retrieve latest breaker and socket data (current, voltage, status) from the cloud database.
5. Construct AI Prompt
 - Combine system context and user query into a single structured prompt.
6. Send Prompt to Gemini AI
 - Transmit prompt for Natural Language Understanding (NLU) and response generation.
7. Check Type of AI Response

- If response is informational or advisory → display answer on the dashboard → return to Step 3.
 - If response contains an action command (e.g., “Turn off socket 2”) → proceed to Step 8.
8. Execute Control Command
 - Parse command (socket ID + desired state).
 - Backend sends command to ESP32 central control unit via HTTP/MQTT.
 9. Display Final Response
 - Present Gemini’s confirmation or advisory message (e.g., “Socket 2 has been switched off. Total load = 0.97 A.”).
 10. Synchronize Logs
 - Save all user queries, AI responses, and executed commands to the cloud database for record and analysis.
 11. End Process

3.5.3 Testing Phase

The testing phase of Sprint 2 validates the complete web-based monitoring system and its integration with the Generative AI assistant to meet Objective 2. Sprint 2 emphasizes the connection between the central control unit, the cloud database, and the AI-assisted dashboard. Testing ensures that real-time data from the ESP32 is accurately recorded in the local and cloud databases, displayed correctly on the dashboard, and retrievable through AI queries and user interactions. Testing covers two levels: (1) System Testing to validate dashboard functionalities, user login and registration flow, cloud synchronization, AI query responses, and control command execution. (2) Verification & Validation to confirm that the web dashboard meets the acceptance criteria.

System Testing

System testing in Sprint 2 evaluated the integrated performance of the AI-assisted web dashboard, cloud database, and central control unit against the defined acceptance criteria. The accuracy of real-time current data displayed on the dashboard will be validated by comparing the recorded values from the ESP32 with reference readings measured by a clamp ammeter, ensuring deviations remained within $\pm 5\%$. Cloud synchronization will be tested by monitoring response time of synchronization between the local SQLite3 storage and the Supabase cloud database, requiring at least 2 seconds of successful record transmission. The responsiveness of the Generative AI assistant will be assessed by measuring the time from user prompt to system reply, with a target latency of ≤ 3 seconds. Response accuracy is also verified by comparing the AI's output with expected system context and control actions to ensure alignment with actual load data and device states. Finally, overall system integrity will be verified through the successful execution of user commands and data retrieval operations, confirming the dashboard's reliability, accuracy, and interaction efficiency.

Table 3.14

Dashboard Current Display Accuracy Test (Breaker-Level Load vs Clamp Ammeter Reference)

Trial No.	Clamp Ammeter (A)	Total Load Current Dashboard Display (A)	Error (%)
1			
2			
3			

Table 3.14 Continued

Trial No.	Clamp Ammeter (A)	Total Load Current Dashboard Display (A)	Error (%)
4			
...	...	...	...
20			

This test evaluates the accuracy of the dashboard in displaying breaker-level current readings. The clamp ammeter serves as the reference instrument, and its readings are treated as the ground truth.

Table 3.15

Dashboard Current Display Accuracy Test (Per-Socket Load vs Per Socket Clamp Ammeter Reference)

Trial No.	Per Socket Clamp Ammeter Test (A)	Per Socket Dashboard Display (A)	Error (%)
1			
2			
3			
4			
...	...	...	...
20			

This test measures the accuracy of the dashboard in reporting per-socket current readings obtained from the central control unit. The clamp ammeter provides the reference measurement, and its values are treated as the true current draw per socket.

Table 3.16

Data Synchronization to Cloud Database Response Time Testing

Trial No.	Time of Upload in Local DB	Time of Upload in Cloud DB	Synchronization Response Time (ms)
1			
2			
3			
4			
5			
.....			
20			

This test measures the response time of data synchronization from the local SQLite3 database to the Supabase cloud database. Each trial records the time taken for a new data entry stored locally to appear in the cloud. The dataset could include user credentials, sensor readings, socket status updates, or safety-logic events.

Table 3.17

Generative AI Assistant Responsiveness and Accuracy Test

Test No.	User Prompt	Ping (ms)	Action Executed Correctly (1/0)	Cross Trigger Detected (1/0)	Response Time (s)
1					
2					
3					
4					
....					
20					

This presents the evaluation of the Generative AI assistant's responsiveness and accuracy. The test measures the latency between a user prompt and the system's reply, with an acceptance threshold of ≤ 3 seconds. Accuracy is determined by comparing the AI-generated output with the expected system response, requiring at least 95% alignment with actual load data and control actions.

Verification and Validation

Verification will be carried out by comparing measured results from system testing against each specific acceptance criteria. Validation confirmed that the integrated system achieved Objective 2 by demonstrating dashboard, database and Generative AI functionality.

Table 3.18

Verification and Validation Summary for Sprint 2

Component	Test Type	Criteria	Metric	Acceptable Range	Testing Outcome
Breaker-Level Current Measurement (SCT-013 ESP32)	Accuracy +	Deviation measured vs clamp ammeter reference.	of Percentage current (%)	$\leq \pm 5$ % error or ≥ 95 % accuracy vs reference meter (IEC class).	
Per-Socket Current Display (Dashboard vs Clamp Reference)	Accuracy	Consistency dashboard readings per-socket data.	of Percentage (%) with clamp	≥ 95 % alignment with reference values.	
Cloud Database Synchronization	Reliability	Response time successful synchronization.	of Time (s) data	≤ 2 seconds per record transmission.	

Table 3.18 Continued

Component	Test Type	Criteria	Metric	Acceptable Range	Testing Outcome
Generative Assistant Response Latency	AI Efficiency	Time between user prompt and AI system reply.	Time (s)	≤ 3 seconds maximum latency.	
Generative Response Accuracy	AI Accuracy	Alignment of AI-generated output with expected system context and control intent.	Percentage (%)	≥ 95 % match with expected response.	
Dashboard Command Execution Integrity	Reliability	Success rate of ON/OFF command execution and data retrieval.	Percentage (%)	100 % successful execution of issued commands.	

This table summarizes Sprint 2 system testing by comparing the proportion of successful trials against acceptance criteria. Verification indicates whether design requirements will be met, while validation confirms if Objective 2 will be achieved under controlled conditions.

Data Analysis and Procedure

Data analysis for Sprint 2 focuses on evaluating the system introduced in this sprint. Metrics include dashboard accuracy, processing latency, response time, and dashboard data logging. These procedures extend the methods applied in Sprint 1.

Dashboard Accuracy.

The dashboard display values will be compared with clamp ammeter readings, treated as reference. Accuracy will be computed as:

$$Accuracy(\%) = \left(1 - \frac{|Dashboard - Reference|}{Reference} \right) \times 100 \quad (3.7)$$

A trial will be considered valid if accuracy $\geq 95\%$, consistent with standard calibration practices (Hughes & Drury, 2019).

Data Synchronization Time.

Synchronization performance between the local SQLite3 database and the Supabase cloud database will be determined by recording the timestamps of local storage and cloud upload events. The response time will be computed as:

$$Sync\ Time(ms) = Cloud\ Upload\ Time - Local\ Storage\ Time \quad (3.8)$$

Each trial passed if synchronization occurred within ≤ 2000 ms (2 s) per record, reflecting efficient cloud communication for real-time data updates.

Generative AI Response Latency

Latency will be measured from the moment the user entered a prompt to the time the AI system returned its response on the dashboard. The latency will be

calculated as:

$$\text{Latency(ms)} = \text{AI Response Time} - \text{User Prompt Time} \quad (3.9)$$

A trial passed if latency ≤ 3000 ms (3 s), consistent with acceptable response times for conversational AI systems to maintain natural interaction rhythm (McDermott & Nicho, 2025).

Generative AI Response Accuracy

The AI assistant's output will be compared with the expected system response to determine contextual accuracy. Accuracy will be computed as:

$$\text{Response Accuracy(\%)} = \frac{\text{Number of Correct AI Responses}}{\text{Total Responses Tested}} \times 100 \quad (3.10)$$

Each trial passed if Response Accuracy ≥ 95 %, indicating that the AI correctly interpreted the user's intent and aligned with actual system context or control state.

3.6 Sprint 3: System Reliability, Accuracy, and Intelligent Dashboard Evaluation

Sprint 3 addresses our Objective 3: To test the system's accuracy and reliability through different load variations, safety logic evaluation, and AI assistant response.

3.6.1 System Integration

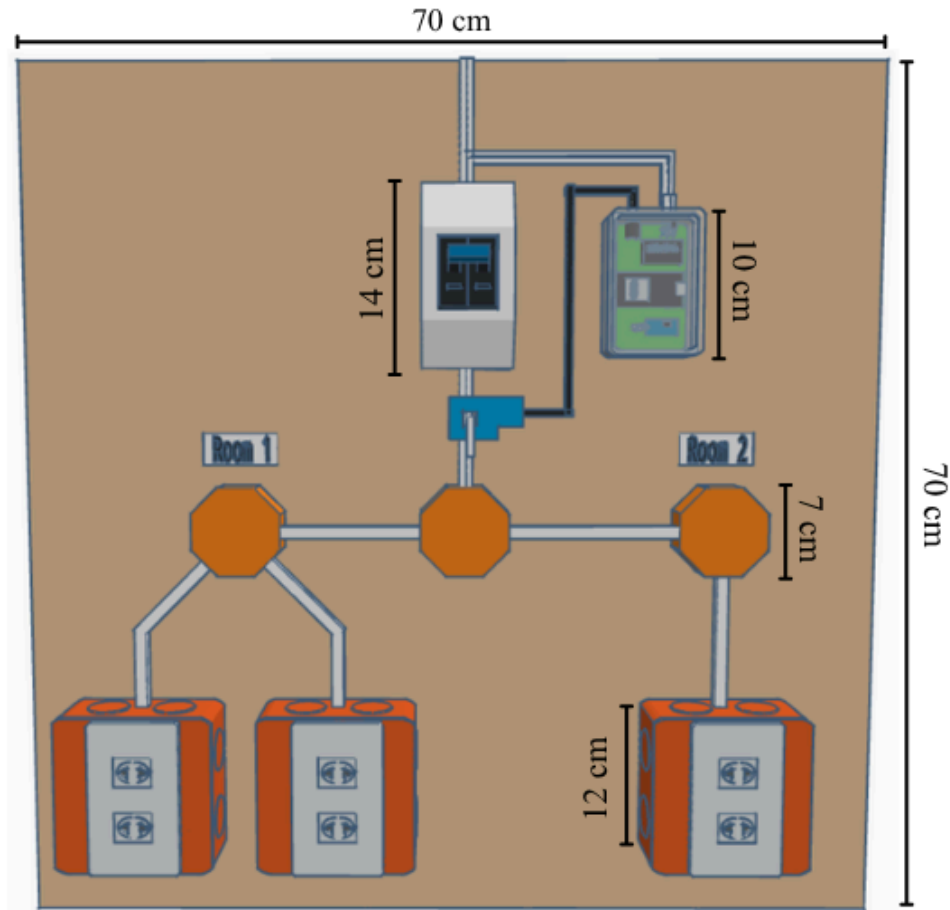


Figure 3.24 Implementation of the Smart Outlet and Central Monitoring Unit

Figure 3.24 illustrates the complete physical integration of the smart outlet devices and the central monitoring unit installed at the outlet breaker. The layout demonstrates how the system interconnects multiple outlet nodes positioned across two test rooms (Room 1 and Room 2) through a structured wiring network. Each room contains two smart outlets, each equipped with ACS712 current sensors and controlled by the PIC16F88 microcontroller, while the main control line connects to the ESP32-based central monitoring unit located near the breaker panel. The wiring distances between components are clearly indicated to ensure uniform testing conditions and consistent RF signal coverage across the network.

This setup replicates a scaled household configuration, allowing evaluation of both local outlet performance and centralized breaker-level monitoring.

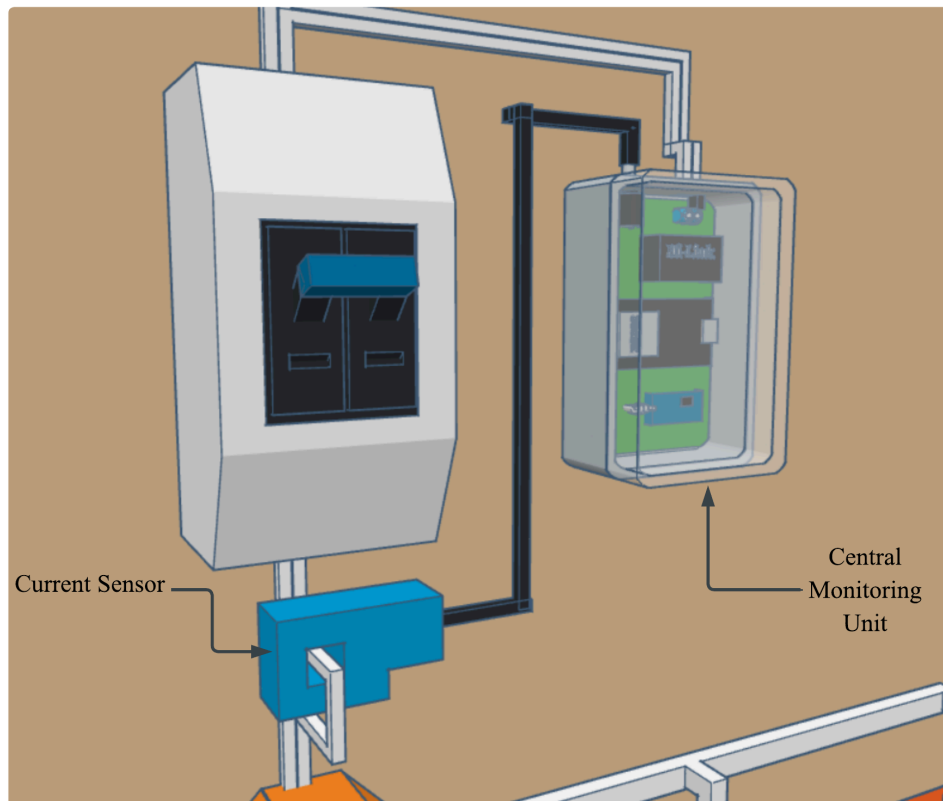


Figure 3.25 Close-Up View of the Central Monitoring Unit and Current Sensor

Figure 3.25 shows an enlarged view of the central monitoring section, highlighting the connection between the SCT-013 current sensor, outlet breaker, and the ESP32-based Central Monitoring Unit (CMU). The current sensor is clamped onto the main supply line to detect the total circuit load in real time, while the CMU receives this signal through an isolated interface and performs data aggregation, safety logic, and wireless communication via the HC-12 transceiver. This compact assembly demonstrates the hardware integration of sensing, protection, and communication modules within the breaker panel

enclosure, forming the foundation for intelligent load monitoring and centralized control in the smart outlet system.

3.6.2 System Testing

Table 3.19

Dashboard Load Monitoring Calibration Test for Outlet breaker and sockets.

Test no.	Dashboard Reading (mA)	Reference Instrument (mA)	Error%
1			
2			
3			
.....			
20			
MAPE%			

This table presents the results of the calibration test conducted to verify the accuracy of load readings displayed on the dashboard compared to a reference measurement instrument. The purpose of this test is to ensure that both outlet- and

breaker-level current readings are consistent with the actual measured values, confirming the reliability of the dashboard's monitoring function.

Table 3.20

Smart Outlet Overcurrent Trip Test (Socket-Level)

Test no.	Dashboard Setup Value for AC Outlet (mA)	Socket A-B per Outlet	
		Increasing Load (mA) > Set value	Relay Output (5.00V $\pm$ 5 %)
1			
2			
3			
.....			
20			

This table shows the results of the socket-level safety logic evaluation performed on the smart outlet devices. Each test verifies whether the socket's relay activates when the applied current exceeds its configured set value. The purpose of this test is to confirm that every outlet socket can independently detect overcurrent conditions and trigger protection at the correct threshold.

Table 3.21

Breaker-Level Overcurrent Trip Test (Central Control Unit)

Test no.	Set Value for Outlet breaker (A)	Outlet Breaker	
		Increasing Load (mA) > Set value	Socket Identified as Highest Load (1/0)
1			
2			
3			
.....			
20			

This table presents the results of the breaker-level safety logic evaluation managed by the central control unit. The test determines whether the system accurately identifies the socket contributing the highest load and initiates a trip action when the total circuit current exceeds the configured breaker limit. The purpose of this test is to confirm the central unit's ability to detect overload conditions at the breaker level and apply selective protection by isolating the correct socket.

Table 3.22

Dashboard Safety Notification Accuracy Test for Outlet breaker and Sockets

Test no.	Trip Source	Overlimit Margin (A)	Notification Correctly Displayed (1/0)	Notification Delay (s)
1	e.g “Outlet 1 Socket A”	0.4	1	1.5
2				
3				
....				
20				

This table presents the results of the dashboard notification test conducted to verify whether protection events are correctly displayed when triggered by either the outlet sockets or the breaker. The test records each trip source, the extent of overload, and whether the dashboard accurately identified the event and displayed it with minimal delay. The purpose of this test is to ensure that the system provides users with timely and accurate safety alerts reflecting real-time protection actions.

Table 3.23

Manual Socket Control Accuracy and Cross-Trigger Test

Test No.	Selected Control Button	Ping (ms)	Dashboard Status Updated (1/0)	Toggled the correct socket (1/0)	Cross Trigger Detected (1/0)	Response Time (s)
1	eg. Outlet 1 Socket A → ON	35	1	1	0	1.4
2	eg. Outlet 2 Socket B → OFF	40	1	1	1	1.9
3	eg. Outlet 3 Socket A → ON	32	1	1	0	1.7
....						

20

This table presents the results of the manual control evaluation conducted through the web dashboard to verify the reliability of direct user operations. Each

test records whether the selected socket will be correctly toggled, whether the dashboard interface accurately reflects the new state, and whether any unintended sockets will be activated. The purpose of this test is to confirm that manual commands execute with high accuracy, display timely feedback, and maintain isolation between controlled outlets without cross-triggering events.

Table 3.24

Outlier Filtering and Data Quality Evaluation

Load Level (W)	Total Samples Collected	Outliers Removed (± 2 SD)	Outlier Rate (%)	Drift Flag (Yes/No)
Variable Load 1	480	...	...	
Variable Load 2	480	...	...	
Variable Load 3	480	...	...	
Variable Load 4	480	...	...	
Variable Load 5	480	...	...	
Variable Load 6	480	...	...	

Each outlet load level will be sampled at defined time intervals representing typical household usage conditions until 480 total samples per load are obtained. The collected data will be filtered using a ± 2 SD rolling-mean algorithm. An outlier rate of $\leq 5\%$ per load will be considered acceptable, while a persistent rate above 5% for three consecutive sampling periods will trigger recalibration.

Table 3.25

AI Retrieval Accuracy and Response Latency

Load Level (W)	Queries Executed	Correct Responses	Accuracy (%)	Mean Response Time (s)	SD (s)
Variable Load 1	10	...	...	...	...
Variable Load 2	10	...	...	...	...
Variable Load 3	10	...	...	...	...
Variable Load 4	10	...	...	...	...
Variable Load 5	10	...	...	...	...
Variable Load 6	10	...	...	...	...

The chatbot will be queried with factual requests (e.g., “Which socket has the highest load?” “Show breaker current.”). Responses will be cross-checked against database values and reference readings. A retrieval accuracy of at least 95% will be required, with a mean response time of ≤ 3 seconds and a standard deviation of ≤ 1 second.

Table 3.26

Intent and Entity Recognition Accuracy

Load Level (W)	Commands Issued	Intent Accuracy (%)	Entity Accuracy (%)
Variable Load 1	50	...	...
Variable Load 2	50	...	...
Variable Load 3	50	...	...
Variable Load 4	50	...	...
Variable Load 5	50	...	...
Variable Load 6	50	...	...

Fifty commands per load will be tested across variants (“turn off socket 1,” “pataya ang socket 1,” etc.). Jamovi confusion matrices and per-class precision metrics will be used to evaluate intent and entity recognition. $\geq 98\%$ accuracy for both intent and entity extraction under all load conditions.

3.6.3 Data Analysis and Procedure

A. Linear Regression

Linear regression will be utilized to determine the accuracy and correlation between the system’s measured current values and those from the reference instrument. This statistical tool establishes the best-fit line that predicts a dependent variable (Y) based on an independent variable (X), showing how closely the system readings follow the actual measured values. It is particularly useful for quantifying both the strength of the relationship and the accuracy of prediction between two continuous variables. All computations will be performed using an online statistical calculator to ensure precision and consistency when analyzing multiple datasets.

The coefficient of determination (R^2) measures how much of the variation in the dependent variable (Y) is explained by the independent variable (X). A higher R^2 indicates a stronger correlation and better reliability of the model fit. The Mean Absolute Percentage Error (MAPE), on the other hand, expresses the average error between the measured and predicted values as a percentage, providing a straightforward indicator of accuracy. Together, these two metrics allow evaluation of how well the dashboard’s readings match the reference measurements.

$$\begin{aligned}
\hat{Y} &= b_0 + b_1 X \\
R^2 &= 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2} \\
MAPE &= \frac{100}{n} \sum \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right|
\end{aligned} \tag{3.11}$$

Where:

$\hat{Y}$ = predicted value of the dependent variable

b_0 = intercept

b_1 = slope (change in Y for every 1-unit change in X)

X = independent variable

Y_i = observed value of Y

$\bar{Y}$ = mean of observed Y

Interpretation Criteria:

- R^2 close to 1 $\rightarrow$ strong linear relationship between X and Y (Schober et al., 2018).
- $MAPE \leq 5\%$ $\rightarrow$ acceptable measurement error for electronic instrumentation (IEEE 1057-2017).
- $b_1 \approx 1$ $\rightarrow$ system readings increase proportionally with reference values, indicating proper calibration.

A.1 Dashboard Load Monitoring Calibration

This subsection applies the general linear regression model to the Dashboard Load Monitoring Calibration test, where the independent variable (X) represents the reference instrument reading measured in amperes (A) or milliamperes (mA), the dependent variable (Y) represents the dashboard's displayed reading, and the Error % corresponds to the computed percentage difference between the dashboard and reference measurements.

Procedure:

1. The system will be powered on and connected to a calibrated reference ammeter to obtain baseline readings for comparison.
2. The load connected to each outlet or breaker will be gradually increased in predefined increments (e.g., 0.5 A, 1 A, 1.5 A, etc.) while recording the corresponding current displayed on the dashboard.
3. For every test point, the reference instrument reading (X) and the dashboard reading (Y) will be recorded simultaneously to minimize timing bias.
4. The Error % for each trial will be computed using the formula:

$$\text{Error}\% = \left| \frac{Y - \hat{Y}}{Y} \right| \times 100 \quad (3.12)$$

Where:

Y = Reference Instrument Reading (Actual)

$\hat{Y}$ = Dashboard Reading (Measured)

5. All recorded data will be entered into an online statistical calculator to compute the linear regression equation, coefficient of determination (R^2),

and Mean Absolute Percentage Error (MAPE).

6. The obtained R^2 and MAPE values will be compared against the benchmark criteria (IEEE 1057-2017) to determine if the dashboard readings met the accuracy requirement of $\text{MAPE} \leq 5\%$.

B. Logistic Regression

Logistic regression will be employed to analyze tests involving binary outcomes, such as whether a socket tripped (1 = Triggered, 0 = Not Triggered) or whether the system correctly identified a specific condition. Unlike linear regression, which predicts continuous values, logistic regression estimates the probability that a categorical outcome occurs based on one or more independent variables. It is particularly suited for modeling decision-based system responses such as overload detection, tripping logic, or safety notifications. All computations will be performed using an online statistical calculator to obtain regression coefficients (b_0 , b_1) and model performance metrics, including the Nagelkerke R^2 and the Area Under the Curve (AUC).

$$p = \frac{1}{1 + e^{-(b_0 + b_1 X)}} \quad (3.13)$$

Where:

p = probability that the dependent variable = 1

b_0 = intercept

b_1 = coefficient (slope) for X

X = independent variable

e = Euler's number (≈ 2.71828)

Interpretation Criteria:

- Nagelkerke R^2 quantifies how much variation in the outcome is explained by the model. Values close to 1 indicate strong reliability and explanatory power.
- AUC (Area Under the ROC Curve) measures classification accuracy. $AUC \geq 0.8$ indicates a strong model, and $AUC \geq 0.9$ indicates outstanding predictive performance (Hosmer et al., 2013).
- The model's overall quality is determined by both metrics, where a high AUC confirms precise classification and a high Nagelkerke R^2 confirms consistent reliability in predicting binary events.

B.1 Smart Outlet Overcurrent Trip Test (Socket-Level)

This subsection applies the general logistic regression model to the Smart Outlet Overcurrent Trip Test (Socket-Level). The analysis determines whether each socket's built-in protection logic reliably activates when the measured load exceeds the user-defined set value configured through the dashboard. Here, the independent variable (X) represents the increasing load current (mA) applied to the socket, while the dependent variable (Y) represents the relay output condition, classified as 1 = Triggered ($5.00 \text{ V} \pm 5 \%$) and 0 = Not Triggered. The model estimates the probability that a socket will trip based on the applied load, assessing both reliability and accuracy of the protection response.

Procedure:

1. Each smart outlet will be configured through the dashboard with a predefined current limit (e.g., 1000 mA, 1500 mA, 2000 mA, etc.).

2. A variable load will be gradually increased beyond the set value while the corresponding relay output voltage will be monitored for each socket (A–B).
3. For every trial, the measured increasing load (mA) and the resulting relay output (V) will be recorded.
4. Each record will be classified using the binary condition:

$$Y = \begin{cases} 1, & \text{if } I_{\text{load}} > I_{\text{set}} \text{ and Relay Output} = 5.00V \pm 5\% \\ 0, & \text{otherwise} \end{cases} \quad (3.14)$$

where I_{load} is the measured socket current and I_{set} is the dashboard-configured threshold.

5. The collected data will be analysed using an online logistic regression calculator to compute regression coefficients (b_0, b_1), the Nagelkerke R^2 , and the Area Under the Curve (AUC) to evaluate predictive reliability and classification accuracy.
6. The results will be interpreted using accepted criteria: $\text{AUC} \geq 0.9 =$ Outstanding, $0.8\text{--}0.9 =$ Excellent, $0.7\text{--}0.8 =$ Acceptable, and Nagelkerke R^2 close to 1 = high model reliability (Hosmer et al., 2013).

B.2 Breaker-Level Safety Logic

This subsection applies the logistic regression model to the Breaker-Level Overcurrent Trip Test (Central Control Unit). The analysis determines whether the central control unit can reliably identify and deactivate the socket with the highest

load when the total circuit current exceeds the user-configured breaker threshold. The independent variable (X) represents the over-limit margin the difference between the total measured load and the breaker's set value while the dependent variable (Y) represents whether the system correctly identified the highest-load socket (1 = Correct, 0 = Incorrect). This test validates the reliability of the central control unit's selective tripping algorithm under overload conditions.

Procedure:

1. The outlet breaker threshold will be set through the dashboard (e.g., 10 A, 12 A, or 15 A).
2. Loads will be gradually applied across multiple sockets until the total current approaches and then exceeds the breaker's limit, while individual socket currents will be simultaneously recorded.
3. The over-limit margin (X) will be computed for each trial using the formula

$$X = I_{total} - I_{limit} \quad (3.15)$$

where I_{total} is the measured total circuit current and I_{limit} is the configured breaker current limit.

4. For each test, the system's selected socket (the one automatically turned off) will be compared against the true highest-load socket to determine accuracy, defined by:

$$Y = \begin{cases} 1, & \text{if } X > 0 \text{ and System-Selected Socket} = \text{True Highest Load} \\ 0, & \text{otherwise} \end{cases} \quad (3.16)$$

5. All collected data will be processed using an online logistic regression calculator to obtain coefficients (b_0, b_1) and model performance indicators Nagelkerke R^2 and AUC.
6. Results will be interpreted using the criteria: $AUC \geq 0.8$ = strong classification, $AUC \geq 0.9$ = outstanding, and Nagelkerke R^2 approaching 1 = highly reliable detection (Hosmer et al., 2013).

B.3 Dashboard Safety Notification Accuracy Test

This subsection applies the logistic regression model to the Dashboard Safety Notification Accuracy Test for Outlet Breaker and Sockets (Table 3.21). The objective is to determine whether the system's dashboard correctly identifies and displays the trip source when a protective event is triggered, either at the socket or breaker level. The independent variable (X) represents the over-limit margin (A) the amount by which the measured current exceeded the configured limit and the dependent variable (Y) represents whether the notification will be correctly displayed (1 = Correct, 0 = Incorrect). This test evaluates the reliability of dashboard notifications in accurately communicating which component initiated the trip event.

Procedure:

1. The system's safety thresholds will be configured for both the sockets and the outlet breaker through the dashboard interface.
2. Loads will incrementally increase until an overcurrent event will be triggered at either the socket or breaker level.

3. For each trial, the trip source (e.g., *Outlet 1 Socket A*, *Outlet 2 Socket B*, or *Circuit Breaker*) and the over-limit margin (A) will be recorded.
4. The dashboard display will be observed to determine if the correct trip source will be identified and shown to the user.
5. Each record will be classified into binary outcomes according to the condition:

$$Y = \begin{cases} 1, & \text{if Dashboard Notification matches Actual Trip Source} \\ 0, & \text{otherwise} \end{cases}$$

(3.17)

6. The resulting data will be analysed using an online logistic regression calculator to determine coefficients (b_0 , b_1), Nagelkerke R^2 , and AUC.
7. Results will be interpreted using the established criteria: $AUC \geq 0.8$ = strong classification, $AUC \geq 0.9$ = outstanding, and Nagelkerke $R^2 \rightarrow 1$ = high reliability, indicating accurate and consistent dashboard notifications (Hosmer et al., 2013).

B.4 Manual Socket Control and Cross-Trigger Test

This subsection applies the logistic regression model to the Manual Socket Control Accuracy and Cross-Trigger Test. The analysis determines the reliability of the dashboard's manual control interface by evaluating whether the correct socket will be toggled, whether the dashboard status updated accurately, and whether unintended cross-triggering occurred. The dataset includes three dependent variables Dashboard Status Updated (1/0), Toggled the Correct Socket (1/0), and Cross Trigger Detected (1/0) each representing a separate binary

outcome analyzed individually. The independent variables are the Selected Control Button, which identifies the specific outlet or socket targeted, and Ping (ms), representing the network latency between the dashboard and the outlet.

Procedure:

1. Each outlet socket will be individually toggled through the web dashboard interface using the corresponding Selected Control Button (e.g., *Outlet 1 Socket A* $\rightarrow$ *ON*, *Outlet 2 Socket B* $\rightarrow$ *OFF*).
2. For each manual operation, the Ping (ms) delay between the dashboard command and the smart outlet response will be recorded.
3. The physical relay states and dashboard indicators will be monitored simultaneously to verify command execution and feedback synchronization.
4. Each trial produced three binary outcomes, defined as:

$$\begin{aligned}
 Y_1 &= \begin{cases} 1, & \text{if Dashboard status updated correctly after the toggle command} \\ 0, & \text{otherwise} \end{cases} \\
 Y_2 &= \begin{cases} 1, & \text{if the correct socket was toggled according to the selected control button} \\ 0, & \text{otherwise} \end{cases} \\
 Y_3 &= \begin{cases} 1, & \text{if Cross-triggering was detected (unintended socket activation occurred)} \\ 0, & \text{if no cross-triggering occurred} \end{cases}
 \end{aligned}
 \tag{3.18}$$

5. The dataset will be analyzed using an online logistic regression calculator, with separate regression models computed for each dependent variable (Y_1 , Y_2 , Y_3) to obtain coefficients (b_0 , b_1 , b_2), Nagelkerke R^2 , and AUC.
6. The resulting values will be interpreted using the following benchmarks: $AUC \geq 0.8$ = strong, $AUC \geq 0.9$ = outstanding, and Nagelkerke R^2 approaching 1 = highly reliable performance (Hosmer et al., 2013). A high

reliability for Y_1 and Y_2 combined with a low frequency of $Y_3 = 1$ indicates that the manual control interface executes commands accurately and maintains isolation between sockets.

B.5 Outlier Filtering and Data Quality Evaluation

This test aims to determine the reliability and consistency of raw sensor readings transmitted to the dashboard AI assistant. The ± 2 SD rolling-mean filter will be applied to remove statistically inconsistent samples and identify drift in the current-sensing modules.

Procedure:

1. For each load level, the system will collect a total of 480 samples under realistic household operating conditions. Sampling will be performed at defined intervals aligned with common household load activity periods to capture variations in appliance usage.
2. The mean $\bar{x}$ and standard deviation (SD) will be computed for each sampling window.
3. Any reading x_i outside the interval

$$\bar{x} \pm 2(SD) \tag{3.19}$$

will be classified as an outlier and automatically removed.

4. The total number of removed samples will be divided by the total collected samples to obtain the Outlier Rate (%):

$$\text{Outlier Rate (\%)} = \frac{N_{\text{outliers}}}{N_{\text{total}}} \times 100 \tag{3.20}$$

Where:

- $N_{outliers}$ = number of samples beyond ± 2 SD
- N_{Total} = total samples collected per load

Outlier rates $\leq 5\%$ will indicate stable and reliable sensor data, while 5–10 % will suggest minor drift requiring recalibration. Rates above 10 % will signify unstable readings or interference, and a drift flag will be triggered if the outlier rate exceeds 5 % for three consecutive sampling periods.

B.6 AI Retrieval Accuracy and Response Latency

To evaluate the chatbot's ability to fetch correct information from the cloud database and respond within an acceptable time window, ensuring smooth user interaction through the dashboard.

Procedure:

1. Ten queries will be executed per load level (total = 60 queries).
2. For each query, the time of submission (T_1) and time of complete response (T_2) will be logged.
3. Response Time for each query will be computed as:

$$\text{Response Time (s)} = T_2 - T_1 \quad (3.21)$$

4. The Mean Response Time and Standard Deviation will be then obtained:

$$\bar{t} = \frac{\sum t_i}{n}, \quad SD_t = \sqrt{\frac{\sum (t_i - \bar{t})^2}{n - 1}} \quad (3.22)$$

5. Each AI reply will be verified against the corresponding database or sensor value. The Retrieval Accuracy (%) will be calculated using:

$$\text{Accuracy (\%)} = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100 \quad (3.23)$$

Where:

- t_i = individual response time per query
- n = total queries tested
- N_{correct} = number of correct responses
- N_{total} = total queries executed

A mean response time of 3 seconds or less with a standard deviation of 1 second or below will indicate that the chatbot responds quickly and consistently during interactions. A retrieval accuracy of at least 95% confirms that the AI reliably fetches correct data from the database, while values beyond these limits suggest latency issues or inefficiencies in query processing.

B.7 Intent and Entity Recognition Accuracy

To assess the AI language model's comprehension of user commands by measuring its success in correctly identifying the intended action (Intent) and the targeted device or socket (Entity) across varying loads.

Procedure:

1. Fifty commands per load level (total = 300 commands) will be issued using different phrasing, languages, and abbreviations (e.g., "turn off socket 1," "pataya ang socket 1," "disable S1").

2. Each output from the AI will be compared with the expected system action.
3. Results will be tabulated into a confusion matrix to identify true positives (TP), false positives (FP), and false negatives (FN).
4. The following formulas will be applied:

$$\bullet \text{ Precision (\%)} = \frac{TP}{TP + FP} \times 100 \quad (3.24)$$

$$\bullet \text{ Recall (\%)} = \frac{TP}{TP + FN} \times 100 \quad (3.25)$$

$$\bullet \text{ Intent Accuracy (\%)} = \frac{TP_{intent}}{N_{intent}} \times 100 \quad (3.26)$$

$$\bullet \text{ Entity Accuracy (\%)} = \frac{TP_{entity}}{N_{entity}} \times 100 \quad (3.27)$$

Where:

- TP = correctly classified commands
- FP = incorrectly classified as a different intent/entity
- FN = missed intents/entities
- N_{intent}, N_{entity} = total tests per category

An intent and entity accuracy of 98% or higher indicates excellent language understanding and precise command mapping by the AI language model. Likewise, precision and recall values close to 100% confirm that the AI consistently interprets user commands correctly, even when phrased differently or tested under varying load conditions.

C. System Response Time Evaluation (One-Sample T-Test)

The one-sample t-test will be used to evaluate the overall responsiveness of the smart outlet monitoring system by combining all recorded response-time data from the Dashboard Notification, Manual Control, AI-Controlled Socket Command, and AI Load Log Retrieval tests. This unified analysis determines whether the system's average response time ($\bar{X}$) meets the modern usability benchmark for interactive and intelligent systems. Recent studies have shown that user satisfaction and perceived intelligence decline sharply when delays exceed 2 seconds (Følstad et al. 2019; Huang et al., 2023; Google UX Research, 2022; Nielsen Norman Group, 2024). Therefore, the hypothesized population mean will be set to $\mu_0 = 2s$ as the performance benchmark for real-time operation.

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \quad (3.28)$$

Where:

t = t-statistic (test value)

$\bar{X}$ = sample mean response time

μ_0 = benchmark mean (2.0 s based on modern UX standards)

s = sample standard deviation

n = total number of recorded trials across all system modules

Procedure:

1. All response-time values (s) from the four modules Dashboard Notification, Manual Socket Control, AI-Controlled Command, and AI Load Log Retrieval will be compiled into a single dataset.

2. Each record represents the duration between a user action (manual command or AI prompt) and the system's feedback (relay activation or dashboard update).
3. The sample mean ($\bar{X}$), standard deviation (s), and sample size (n) will be calculated from the pooled dataset.
4. The one-sample t-test will be computed using the formula above to compare the overall mean response time against the 2.0 s benchmark.
5. Statistical computation will be performed using an online t-test calculator to obtain the t-statistic and p-value with 95% confidence.

Interpretation:

- $p \geq 0.05 \rightarrow$ The overall mean response time does not significantly differ from the 2.0 s benchmark, indicating that the system responds within acceptable real-time limits.
- $p < 0.05 \rightarrow$ The average response time significantly exceeds the benchmark, suggesting slower performance and reduced real-time interaction quality.

Based on modern interaction design research, maintaining system responses under 2 seconds ensures a smooth, real-time user experience in IoT and AI-integrated applications. This threshold reflects the typical human perception of instantaneous feedback, where longer delays may disrupt task flow and reduce user satisfaction. Achieving response times within this range demonstrates that the system effectively manages data transmission, cloud synchronization, and AI processing without noticeable lag (Følstad & Brandtzaeg, 2021; Huang et al., 2023; Google UX Research, 2022; Nielsen Norman Group, 2024).

3.6.4 Verification and Validation

Table 3.27

Verification and Validation Summary for Sprint 3

Component	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
Dashboard Load Monitoring Calibration	Linear Regression	R^2 close to 1; $MAPE \leq$ 5 %	R^2 , MAPE (%)	$R^2 \approx 1$ = strong relationship (Schober et al., 2018); $MAPE \leq 5\%$ = accurate (IEEE 1057-2017)	Accuracy and Reliability
Smart Outlet Socket-Level Safety Logic	Logistic Regression	$AUC \geq$ 0.9; Nagelker ke R^2 close to 1	AUC, Nagelker ke R^2	$AUC \geq 0.9$ = outstanding (Hosmer et al., 2013); $R^2 \rightarrow 1$ = high reliability	Protection Accuracy and Reliability

Table 3.27 Continued

Component	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
Central Unit Breaker-Level Safety Logic	Logistic Regression	$AUC \geq 0.9$; Nagelkerke R^2 close to 1	AUC, Nagelkerke R^2	$AUC \geq 0.9$ = outstanding (Hosmer et al., 2013); R^2 $\rightarrow 1$ = strong predictive fit	Overload Detection Reliability
Dashboard Safety Notification	Logistic Regression	$AUC \geq 0.8$ = strong; $AUC \geq 0.9$ = outstanding	AUC, Nagelkerke R^2	Hosmer et al. (2013) – AUC classification accuracy; R^2 $\rightarrow 1$ = model reliability	Notificati on Accuracy and Consisten cy
Manual Socket Control and Cross-Trigg er	Logistic Regression	$AUC \geq 0.8$ = strong; $AUC \geq 0.9$ = outstanding	AUC, Nagelkerke R^2	Hosmer et al. (2013); AUC ≥ 0.9 = outstanding classification	Control Accuracy and System Isolation

Table 3.27 Continued

Component	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
Outlier Filtering and Data Quality Evaluation	Descriptive Statistical Analysis	Outlier Rate $\leq 5\%$; drift flag triggered $> 5\%$ for 3 consecutive hours	Outlier Rate (%)	$\leq 5\%$ = statistically stable data (ISO 5725-1: Accuracy of Measurements)	Data Stability and Sensor Consistency
AI Retrieval Accuracy and Response Latency	Descriptive and Inferential Analysis	Accuracy $\geq 95\%$; Mean Response Time ≤ 3 s; SD ≤ 1 s	Accuracy (%), Mean Response Time (s), SD	< 3 s = optimal chatbot latency (McDermott & Nicho, 2025; Google UX Research, 2022)	Query Accuracy and Response Efficiency
Intent and Entity Recognition Accuracy	Confusion Matrix Analysis	Intent and Entity Accuracy $\geq 98\%$; Precision and Recall $\approx 100\%$	Accuracy (%), Precision, Recall	$\geq 98\%$ = excellent command mapping (Gomes et al., 2020; Shirali-Shahr eza, 2024)	Language Understanding and Command Reliability

Table 3.27 Continued

Component	Test Type	Criteria	Metric	Benchmark (Reference)	Testing Outcome
System Response Time Evaluation	One-Sam ple t-Test	$p \geq 0.05 =$ within benchmark ; $p < 0.05$ = significant difference	p-value, Mean Delay (s)	< 2 s benchmark (Følstad & Brandtzaeg, 2021; Huang et al., 2023; Google UX Research, 2022; Nielsen Norman Group, 2024)	Overall Responsi veness and Consisten cy

The results summarized in Table 3.26 confirm that all Sprint 3 statistical evaluations validated the system's accuracy, reliability, and consistency across hardware and software components. Linear regression verified dashboard measurement accuracy within sensor error limits, while logistic regression confirmed stable overload detection and control through high AUC and Nagelkerke R^2 values. The one-sample t-test further showed that the overall response time remained below the 2-second usability benchmark, demonstrating that the system meets its design thresholds for safe, responsive, and user-friendly operation.

3.7 Costing

The costing analysis ensures the feasibility of the proposed smart outlet breaker system by accounting for both structural materials required for assembly and hardware components necessary for system functionality. This breakdown

highlights the affordability of the prototype while reflecting its scalability for practical household deployment.

Table 3.28

Cost of Structural and Installation Materials

Item			Quantity	Unit Price (₱)	Total Cost (₱)
Plastic enclosure (ABS)/ Junction Box			2	100.00	200.00
Wiring set			1	300.00	300.00
PCB board			2	250.00	500.00
Heat shrink tubes			1 pack	200.00	200.00
Plywood			1	300.00	300.00
Outlet			3	50.00	150.00
Miscellaneous materials			—	2,145.00	2,145.00
Total					₱3,795.00

Table 3.29

Cost of Hardware and Electronic Components

Component	Quantity	Unit Price (₱)	Total Cost (₱)
ESP32 microcontroller	1	450.00	450.00
PIC16F88 microcontroller	3	300.00	900.00
HC-12 RF modules	3	250.00	750.00
Relay module 2 channel	3	90.00	270.00
ACS712 current sensor	6	60.00	360.00
SCT-013 current sensor	1	150.00	150.00
Hi-Link HLK-5M05 power	4	35.00	140.00
Total			₱ 3,010.00

The combined cost of structural materials (₱3,795) and hardware components (₱3,010) amounts to a total of ₱6,815.00 for the prototype system. This budget aligns with the project's objective of developing a cost-effective and scalable household energy monitoring solution, remaining well within the range of affordable IoT-based academic prototypes. Costing analysis also highlights that the majority of expenses come from structural fabrication, while the electronic components remain inexpensive, further reinforcing the system's feasibility for replication and deployment.

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APPENDIX A

REQUEST LETTER TO THE ADVISER

Ranian J. Rulona
Jhade Japhet P. Bangquiao
Emmazen Faye A. Tampus
BSCpE – 4th Year
Jose Rizal Memorial State University
September 15, 2025

Engr. Troy Lasco
College of Engineering
Jose Rizal Memorial State University

Dear Engr. Lasco,

Good day!

We are students currently enrolled in Project and Design 1, and we respectfully write to formally request your mentorship and guidance as our Project Design Adviser for our research study entitled, “IoT and Edge AI-Based Smart Outlet Breaker for Predictive Load Monitoring”, which we shall continue to undertake in Project and Design 2.

Our study focuses on developing a smart outlet breaker system equipped with IoT and edge AI capabilities that will not only provide per-outlet monitoring and control but also predict potential overload conditions to ensure safety, efficiency, and reliability in power usage. With your expertise in programming, embedded systems, computer engineering, and IoT technologies, we firmly believe that your insights and professional guidance will significantly contribute to the successful completion of our project.

We are eager to learn from your experience and knowledge, and we are confident that your mentorship will help us enhance the technical, research, and professional aspects of our study.

We sincerely hope for your favorable response to this request. Thank you very much for considering our invitation, and we look forward to working with you throughout the course of Project and Design 1 and Project and Design 2.

Respectfully yours,

Researchers:

Ranian J. Rulona
Jhade Japhet P. Bangquiao
Emmazen Faye A. Tampus

Acknowledged by:

I hereby accept the request to serve as Adviser for Project Design 1 and 2.


ENGR. TROY LASCO
Project Design Adviser

APPENDIX B

MINUTES OF THE PROPOSAL DEFENSE



JRMSU-COE-027

Republic of the Philippines
JOSE RIZAL MEMORIAL STATE UNIVERSITY
 The Premiere University in Zamboanga del Norte
COLLEGE OF ENGINEERING



Minutes of the BSCPE Project Design Final Defense/ Oral Examination of following:
 Ranian J. Rulona, Jhade Japhet P. Banquiao, Emmazen Faye A. Tampus

Adviser: Troy C. Lasco

Panel Of Examiners

Chair: Dr. Romie D. Laranjo

Member: Engr. Richie L. Lacaya

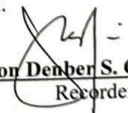
Member: Engr. Raynon Falconete

Member: Engr. French Aleca Recamara

Title as Corrected: “IoT and Edge AI-Based Smart Outlet Breaker for Predictive Load Monitoring”

Summary of Comments and Suggestions:

- Follow strictly the JRMSU – COE institutional format of research.
- Add preliminaries. Use the format for the approval sheet, table of contents etc.
- Chapter 1...Revise Objective 1. Determine the current from the plug socket.
 - Objective 2. Add operation for control of socket to overload.
 - Objective 3. Test the current comparing 6 different loads.
- Chapter 2... Add RRL to support the location of the sensors as presented in the design. Present a study that supports on the method presented. What is the basis of the method used?
- Chapter 2... Provide details about edge AI operation on the proposed system.
- Chapter 3... Improve the system architecture presentation. Base the architecture on similar studies as presented in the RRL.
- Chapter 3... Include a testing procedure to show the correctness of the edge AI model.
- Chapter 3... Revise system flowchart. Use only relevant data connection in the system flowchart.
-
- Chapter 3... Consider using actual picture of components in the system requirements.
- Chapter 3... Use test variation of loads to test the accuracy and consistency of the system.
- Use APA 7 format in writing your paper (specially the citations and references). Cited authors should be listed in the references.
- Check the manuscript for plagiarism with 15% similarity index.


Maricon Denber S. Gahisan
 Recorder



Gov. Guading Adasa St., Sta. Cruz, Dapitan City 7101
 main.coe@jrmsu.edu.ph
 09175991352





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COLLEGE OF ENGINEERING
ORAL EXAMINATION PANEL REPORT

September 26, 2025
 Date of Examination

The panel members of the Oral Examination have examined the manuscript of the student's Kanton J. Rulona, Jade Japhet P. Baraguias, Immanuel Faye A. Tampus

_____, on the basis of the student's proposal defense of their Project Design (THESIS), we submit the following result:

- ☐ Passed the proposal defense without modification of the manuscript
☐ Passed the proposal defense provided that the attached list of suggestions/changes be reflected
☐ Failed the proposal defense because of the following reasons:

Remarks (if any):

Engr. <u>ROMIE LARANJO</u>	<u>11/19/25</u>
Panel Chairperson	Date
ENGR. <u>RICHEL LUCENA</u>	<u>11/11/25</u>
Member	Date
ENGR. <u>RAYMOND FALCONETE</u>	<u>11/06/25</u>
Member	Date
ENGR. <u>FRENCH ALEA RECAMARA</u>	<u>11/10/25</u>
Member	Date
ENGR. <u>MARCON GATHISAN</u>	<u>11/21/25</u>
COE Research Chairperson	Date



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COLLEGE OF ENGINEERING

PROJECT DESIGN (THESIS) ORAL EXAMINATION REPORT

Name of Students: RAVIAN J. RULONA, SHADE JAPHET P. BANGQUINO,
EMMAZEN FAYE A. TAMPAU

Title: IoT and Edge AI-Based Smart Outlet Breaker for Predictive
Load monitoring

Please rate each item either passed or failed.

- | | | |
|--|---|-------|
| 1. Thesis Presentation | | |
| a. Organization | - | _____ |
| b. Presentation | - | _____ |
| c. Content | - | _____ |
| 2. Mastery of subject matter | - | _____ |
| 3. Ability to defend ideas | - | _____ |
| 4. Receptiveness towards suggestions given | - | _____ |

Recommendations: Follow the suggestions of the panel members.

Overall Evaluation: Passed _____
 Failed _____
 Conditional _____

ENGR. ROMIE D. DARANJO
 Examiner (Signature over Printed Name)



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Name of Students: RAVIAN J. RULONA, JHADE JAPHET P. BAUGQUIAO,
EMMAZEN PAVE A. TAMPAUS

Title: IoT and Edge AI-Based Smart Outlet Breaker for Predictive
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| 4. Receptiveness towards suggestions given | - | _____ |

Recommendations:

Overall Evaluation:

Passed
 Failed
 Conditional

✓

ENGR. RAYMON L. FALCONETE
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Name of Students: RANIAN I. KULOWA, SHADE JAPNET P. BANGQUAD,
EMMAZEN FAYE A. TAMPOS

Title: IoT and Edge AI-Based Smart Outlet Breaker for Predictive
Load Monitoring

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| c. Content | - | _____ |
| 2. Mastery of subject matter | - | _____ |
| 3. Ability to defend ideas | - | _____ |
| 4. Receptiveness towards suggestions given | - | _____ |

Recommendations:

1. Integrate individual sensors for each line.
2. Consider using local nodes from sensors.
3. Define the use of Edge AI

Overall Evaluation: Passed _____
 Failed _____
 Conditional _____

ENGR. FRENCH ALECA RECAMARA
 Examiner (Signature over Printed Name)



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| c. Content | - | _____ |
| 2. Mastery of subject matter | - | _____ |
| 3. Ability to defend ideas | - | _____ |
| 4. Receptiveness towards suggestions given | - | _____ |

Recommendations:

Outlet-Switch + contact sensor
How AI being used in the system?

Overall Evaluation: Passed _____
 Failed _____
 Conditional _____

ENGR. RICHIE L. KACANA
 Examiner (Signature over Printed Name)

APPENDIX C

PLAGIARISM CHECK



Digital Receipt

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Page count: 191
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IoT-Enabled Smart Outlet Breaker with AI-Assisted Dashboard for Socket-Level Control and Load Monitoring

by Ranian Rulona, Jhade Japhet Bangquiao, Emmazen Tampus

Submission date: 04-Nov-2025 04:05PM (UTC+0300)

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